Aerodynamic Response of Rectangular Cylinders in Steady and Unsteady Inflow: An All-Theme Quantitative Evidence Review

PETW Wind-Tunnel Review Team

August 7, 2026, internal technical review draft

Abstract

This review targets three planned cross-sections of 60×60, 60×180, and 60×300 mm, integrating 55 studies on steady and unsteady aerodynamics of rectangular cylinders and placing table-by-table and figure-by-figure extracted numerical values in prominent positions within the main text. Newly added quantitative extractions cover 15 sources; together with the complete 15,518-point baseline dataset of Yang and Mason (2019), a total of 16 deep-extraction sources are formed. The remaining sources enter their respective themes through condition tables, original figures, direct quotations, and numerical-mechanism records. The review separately discusses $B/D = 1/3$, 1, 3, and 5, distinguishing steady uniform flow, steady turbulent flow, monotonic acceleration, acceleration and deceleration, downburst-like transient, streamwise sinusoidal inflow, vertical sinusoidal inflow, and numerical acceleration studies, while maintaining separation among surface pressure, pressure-integrated force, independent balance, PIV, hot-wire, and numerical channels. Physical evidence shows that C_D and St of square cylinders in accelerating inflow generally decrease, while the 5:1 rectangular cylinder exhibits different responses depending on the inflow family and reattachment state, including increases in C_D, increases in St, or enhanced pressure fluctuations. The existing physical evidence does not yet support a unified transient correction relationship across cross-sections. The test mapping herein serves as the basis for the next measurement and analysis steps and does not constitute facility, safety, or hardware approval.

Keywords: rectangular cylinder; accelerating inflow; sinusoidal gust; surface pressure; aerodynamic force; PIV; reattachment; cross-literature quantitative comparison

Contents

1 Introduction and literature scope **5**

2 Geometry and condition space **5**

2.1 Cross-sectional width-to-depth ratio coverage 5

2.2 Reynolds number range 6

2.3 Inflow velocity history families 6

3 Key experimental studies under accelerating and gust inflow **6**

3.1 Brusco et al. (2022): Multi-fan active-control wind tunnel experiment on a square cylinder in accelerating inflow 7

3.1.1 Precise condition statement . . . 7
3.1.2 Response and mechanisms . . . 8
3.1.3 Condition table . . . 9
3.1.4 Limitations . . . 9
3.1.5 Nearest-neighbor comparison . . . 9
3.2 Morello et al. (2025a): DICI wind tunnel experiment on square and 5:1 rectangular cylinders in accelerating inflow . . . 10
3.2.1 Precise condition statement . . . 10
3.2.2 Response and mechanisms . . . 11
3.2.3 Condition table . . . 12
3.2.4 Limitations . . . 12
3.2.5 Nearest-neighbor comparison . . . 13
3.3 Brusco et al. (2024): Drag and pressure response of a square cylinder in accelerating inflow in the TKU-MFWT . . . 13
3.3.1 Precise condition statement . . . 13
3.3.2 Response and mechanisms . . . 14
3.3.3 Condition table . . . 15
3.3.4 Limitations . . . 15
3.3.5 Nearest-neighbor comparison . . . 15
3.4 Wu et al. (2021): Multi-fan wind tunnel experiment on a 5:1 rectangular cylinder in sinusoidal inflow . . . 16
3.4.1 Precise condition statement . . . 16
3.4.2 Response and mechanisms . . . 17
3.4.3 Condition table . . . 17
3.4.4 Limitations . . . 18
3.4.5 Nearest-neighbor comparison . . . 18
3.5 Yang and Mason (2019): Accelerating-inflow experiment on rectangular cylinders of three aspect ratios . . . 19
3.5.1 Precise condition statement . . . 19
3.5.2 Response and mechanisms . . . 19
3.5.3 Condition table . . . 20
3.5.4 Limitations . . . 20
3.5.5 Nearest-neighbor comparison . . . 21
3.6 M. Li et al. (2023): Square cylinder experiment in streamwise and vertical sinusoidal gusts . 21
3.6.1 Precise condition statement . . . 21
3.6.2 Response and mechanisms . . . 22
3.6.3 Condition table . . . 22
3.6.4 Limitations . . . 22
3.6.5 Nearest-neighbor comparison . . . 23

4 Steady-State and Mechanism Anchor Sources **23**
4.1 Experimental Data for Three Cross-Sections Under Steady Inflow . . . 23
4.2 Geometry-Effect Anchoring Under Steady Uniform Inflow . . . 27
4.3 Vortex Shedding Branch Anchoring for Flat-Plate Cross-Sections . . . 27
4.4 High-Reynolds-Number and PIV Anchoring for the 5:1 Rectangular Cylinder Under Steady and Accelerating Inflow . . . 28
4.5 Numerical Simulation Anchoring . . . 29

5 Transient Family Comparison **29**
5.1 Monotonic Acceleration, Acceleration and Deceleration, and Gust-Type Transients . . . 29
5.2 Streamwise Sinusoidal Inflow . . . 36
5.3 Vertical Sinusoidal Inflow . . . 38
5.4 Multi-Axis Sinusoidal Inflow . . . 38

6 Measurement Channel Normalization and Measurement System **38**
6.1 Pressure Channels . . . 38
6.2 Force Channels . . . 38
6.3 Numerical Channels . . . 39

7 Evidence and Gaps in PIV Measurements **39**
7.1 Steady PIV Evidence for Planned Cross-Sections . . . 40
7.2 PIV Evidence under Unsteady Inflow . . . 40
7.3 Adjacent-Geometry and Multi-Section PIV Evidence . . . 40
7.4 Accelerating-Flow PIV Gaps and Planned Measurements . . . 40

8 Separation of Physical and Numerical Evidence **40**

9 Bounded Experimental Implications and Planned Cross-Section Mapping **41**
9.1 Bounded Implications of Current Evidence . . . 41
9.2 Planned Cross-Section Mapping . . . 42

10 Synthesis and Discussion **42**

11 Conditions and Quantitative Evidence of Key Transient Experiments **43**
11.1 Brusco et al. (2022), Time Cells and Frequency Jumps during Monotonic Acceleration and Deceleration . . . 44
11.2 Morello et al. (2025a), Time Cells, Pressure-Fluctuation Peak Location, and Reattachment under Monotonic Acceleration of Square and 5:1 Rectangular Cylinders . . . 46
11.3 Brusco et al. (2024), Local Pressure and Pressure-Integrated Drag in the Acceleration and Deceleration Branches of a Downburst-Like Transient . . . 53
11.4 Wu et al. (2021), Pressure-Integrated Forces, Component Decomposition, and Two-Dimensional Aerodynamic Admittance of a 5:1 Rectangular Cylinder under Streamwise Sinusoidal Inflow . . . 57

11.5 Yang and Mason (2019), Monotonic Acceleration Comparison of B/D=1 and B/D=3 in the Same Facility . . . 65
11.6 M. Li et al. (2023), Mean and Fluctuating Force Response of a Square Cylinder in Independent Streamwise and Vertical Sinusoidal Gusts . . . 74

12 Comparison Matrix of Aspect Ratio, Inflow Family, and Measurement Channel 76

13 Source-text verification of steady-state and mechanism-anchoring studies 92
13.1 Source Da Matha Sant'Anna et al. (1988) . . . 92
13.2 Source Robertson et al. (1978) . . . 95
13.3 Source Vickery (1966) . . . 97
13.4 Source Yu et al. (2013) . . . 98
13.5 Source Nakamura et al. (1991) . . . 98
13.6 Source Zhang and Liu (2017) . . . 100
13.7 Source Ohya et al. (1992) . . . 100
13.8 Source Han et al. (2025) . . . 100
13.9 Source Guissart et al. (2022) . . . 101
13.10 Source Li et al. (2021a) . . . 101
13.11 Source Zhou et al. (2025) . . . 101
13.12 Source Chiarini and Quadrio (2021) . . . 103

14 Numerical assumptions, physical gaps, and experimental research directions 103

15 Concluding remarks 104

A Direct-quotation and page-number index 105

1 Introduction and literature scope

The steady aerodynamic characteristics of rectangular cylinders have established an experimental and numerical foundation spanning Reynolds number, angle of attack, turbulence intensity, and aspect ratio, yet unsteady-inflow research still comprises multiple mutually non-equivalent velocity histories. Monotonic acceleration, acceleration and deceleration, downburst-type velocity pulses, streamwise sinusoidal, and vertical sinusoidal inflows each modify vortex shedding frequency, separation bubble, reattachment location, pressure fluctuations, and force coefficients in distinct ways. Merging these results into a single "transient effect" would obscure the differences arising from aspect ratio, inflow direction, reference velocity, and measurement channel.

This review adopts a table-first organizational approach. Steady-state baselines are first compared by B/D and by physical or numerical source, followed by comparisons of transient coefficient variations, frequency response, and flow mechanisms by inflow family. Source figures are used to verify axes, series, and spatial topology; cross-paper figures plot only source-reported values, point-by-point digitized values, or explicitly labeled project-converted values. Three sources still in isolation status are retained only for methods or local context and do not enter controlled quantitative conclusions. The overshoot results of Li et al. (2025) continue to serve as a numerical hypothesis, not as primary physical evidence.

The scientific question is how different aspect ratios alter C_D, C_L, St, C_p, and the reattachment process under the same or similar $U(t)$, and which existing measurements can directly support comparisons for the planned experiments. The strongest existing evidence covers square-cylinder pressure and pressure-integrated force, 5:1 cylinder steady-state and sinusoidal inflow, and a small number of monotonic-acceleration pressure experiments; missing evidence is concentrated in transient physical measurements for $B/D = 3$, same-protocol pressure plus PIV for the three planned cross-sections, simultaneous comparison of independent balance and pressure-integrated force, and multi-axis gusts. The measurement that would most reduce uncertainty in the next step is synchronous acquisition of pressure and 300 Hz PIV under the same measured $U(t)$, using consistent coordinate, dynamic pressure, and time references for all three cross-sections.

2 Geometry and condition space

2.1 Cross-sectional width-to-depth ratio coverage

The width-to-depth ratios B/D of rectangular cross-sections covered in this review span a broad range. Among high-completeness transient sources, Brusco et al. (2022), Morello et al. (2025a) (square cylinder portion), Brusco et al. (2024), and M. Li et al. (2023) tested $B/D = 1$ square cross-sections; Morello et al. (2025a) (rectangular cylinder portion), Wu et al. (2021), and Zhou et al. (2025) tested $B/D = 5$; Yang and Mason (2019) tested three cross-sections with $B/D = 1/3$, $B/D = 1$, and $B/D = 3$. Among anchor sources, Da Matha Sant'Anna et al. (1988) tested seven width-to-depth ratios in the range $0.5 \leq H/D \leq 3.0$, Yu et al. (2013) numerically simulated $H/D = 0.3$ to 7.0, Nakamura et al. (1991) and Ohya et al. (1992) covered flat-plate cross-sections with $c/t = 3$ to 16 ($c/t = 9$), Zhang and Liu (2017) tested blunt-nosed flat plates with $L/D = 3$, 6, and 9, Han et al. (2025) tested the $B/D = 5$ cross-section, Guissart et al. (2022) and Chiarini and Quadrio (2021) both tested the $B/D = 5$ cross-section, and Li et al. (2021a) tested the $B/D = 5$ cross-section. Li et al. (2025) (numerical hypothesis line, $B/D = \mathrm{N/A}$) numerically simulated $R = 3$ to 16, but B/D values were not explicitly labeled ($B/D = \mathrm{N/A}$). The master source table presents aspect ratio and evidence-level coverage, and the master source table presents aspect ratio and inflow-family coverage;

Table 13, Table 14, and Table 15 provide condition-level details for $B/D = 1$, 3, and 5, respectively.

2.2 Reynolds number range

The Reynolds number definitions across sources differ substantially. Brusco et al. (2022) and Brusco et al. (2024) define $Re = Ub/\nu$ based on the square-cylinder side length b, with a range of 1.72×10^4 to 6.34×10^4. Morello et al. (2025a) defines $Re = u_\infty D/\nu$ based on the square-cylinder side length $D = 6$ cm, with a range of 1.72×10^4 to 6.54×10^4 (standard acceleration interval), extending to 1.09×10^5. Wu et al. (2021) defines $Re_D = 16080$ based on depth $D = 0.03$ m, a single Reynolds number condition. Yang and Mason (2019) defines $Re = UD/\nu$ based on depth $D = 60$ mm, with a range of 9.3×10^3 to 1.23×10^5. M. Li et al. (2023) defines $Re = 2.1 \times 10^4$ based on the square-cylinder side length $D = 50$ mm. Han et al. (2025) defines $Re = UD/\nu$ based on depth D, with a range of 3.12×10^5 to 1.01×10^6, the highest Reynolds number among all sources. Chiarini and Quadrio (2021) is a DNS at $Re = U_\infty D/\nu = 3000$. Ohya et al. (1992) numerically simulated at $Re = 10^3$. Yu et al. (2013) numerically simulated at $Re = 10^5$. The above differences in Reynolds number definition (choice of characteristic length, assumed kinematic viscosity, and choice of reference velocity) directly affect the cross-source comparison ceiling.

2.3 Inflow velocity history families

The inflow velocity histories $U(t)$ covered in this review can be classified into the following families. Steady uniform inflow: Brusco et al. (2022) conditions SF1 through SF12, Morello et al. (2025a) steady reference, Brusco et al. (2024) SF1 through SF12, Yang and Mason (2019) steady conditions, Han et al. (2025), Guissart et al. (2022), Chiarini and Quadrio (2021) steady baseline, Morello et al. (2025b) steady baseline, and steady tests from multiple anchor sources. Steady turbulent inflow: Robertson et al. (1978) with 10.4% turbulence intensity grid inflow, Vickery (1966) with 10% turbulence intensity inflow. Monotonic acceleration: Brusco et al. (2022) UF1 and UF6 (containing acceleration and deceleration segments), Morello et al. (2025a) WTG (Gaussian-type) and WTC (constant-magnitude-type), Brusco et al. (2024) UF1 through UF4 and UF10 through UF13, Yang and Mason (2019) Case 1 through Case 8, Zhou et al. (2025) nine conditions (ACS/DCS), Lunghi et al. (2024) Case A/B/C, Morello et al. (2025b) Case A/B/C. Streamwise sinusoidal inflow: Wu et al. (2021) SF(f_u, $\Delta u/U$) conditions, Li et al. (2021a) SSOF I through IV, M. Li et al. (2023) Streamwise gust A/B/C. Vertical sinusoidal inflow (transverse/vertical sinusoidal): M. Li et al. (2023) Transverse gust A/B/C. Synchronous multi-axis sinusoidal inflow (multiaxis sinusoidal): no source in the present corpus simultaneously applied streamwise and vertical sinusoidal forcing in a single condition; the two gust types in M. Li et al. (2023) are independent single-axis conditions and cannot be classified as multi-axis evidence. Downburst-like transient: Yang and Mason (2019) explicitly referenced the gust parameter range of Mason and Yang (2016). The master source table presents inflow-family source counts, and the master source table presents the channel matrix of transient sources. Table 19 through Table 22 cover the detailed conditions of the four transient families, respectively.

3 Key experimental studies under accelerating and gust inflow

This section provides individual condition-level reconstructions of six Class-A sources that passed high-completeness transient post-hoc verification and condition-level independent review. Each subsection contains a precise condition statement (geometry, B/D, Reynolds number definition and range, angle of attack, $U(t)$ family and parameters, measurement channels, sampling and synchronization, force source and normalization), references to original cropped figures and controlled overlay figures, a condition table (with explicit

NR entries), a discussion of limitations, and a nearest-neighbor comparison paragraph within the declared comparison ceiling.

3.1 Brusco et al. (2022): Multi-fan active-control wind tunnel experiment on a square cylinder in accelerating inflow

3.1.1 Precise condition statement

The experiment of Brusco et al. (2022) was conducted in the Tamkang University Multi-Fan Wind Tunnel (TKU-MFWT). The wind tunnel is equipped with 72 independently controlled fans arranged in a 12×6 matrix, with a test section of $1.32\,\text{m} \times 1.32\,\text{m}$, an aspect ratio of approximately 1:2, and a maximum uniform freestream velocity of approximately 16 m/s (PDF p. 2). The total length of the wind tunnel is 10.43 m, with relevant dimensions from the fans to the test section of 2.38, 3.66, and 4.39 m, the specific units of which are not explicitly labeled (not reported-14). The mean freestream turbulence intensity is approximately 2.5% (PDF p. 2), with a non-uniformity deviation of 4% to 5% across the cross-section.

The test model is a square-cross-section cylinder with side length $b = 6\,\text{cm}$, equipped with 94 pressure taps, of which 46 are located at the mid-span cross-section. The tap spacing is 5 mm, with the four taps on the upper and lower surfaces nearest the leading edge spaced at 2.5 mm. Circular end plates with a diameter of 40 cm (approximately $7b$) are installed at both ends with a spacing of 600 mm, yielding an aspect ratio of 10. The blockage ratio is less than 5%, and the first bending frequency of the model is estimated at 120 Hz. The test angle of attack is $0°$ (the precision of angle-of-attack adjustment is not reported in the NR).

Steady inflow comprises 12 conditions SF1 through SF12, with reference wind speed U ranging from 4.3 m/s to 15.9 m/s, each condition repeated at least 25 times, with each repetition containing 2^{16} sampling points over a duration of 218.4 s. The Reynolds number is defined as $Re = Ub/\nu$, where ν is assumed to be $1.5 \times 10^{-5}\,\text{m}^2/\text{s}$ (temperature conditions not reported, not reported-08), yielding a Re range of 1.72×10^4 to 6.34×10^4. Transient inflow comprises 13 conditions, each repeated at least 30 times, with each repetition also containing 2^{16} sampling points. Of these, UF1 (the maximum acceleration condition) was repeated 90 times, and UF6 (a lower acceleration condition) was repeated 30 times. The sampling frequency was 300 Hz, using two Scanivalve ZOC 33 units (64 channels each) with a range of 2500 Pa and an accuracy of $\pm 0.15\%$ (i.e., ± 3.75 Pa). The connecting tubing consists of 1.5 m PVC tubes with an inner diameter of 1.5 mm, and the tubing transfer function was treated following the method of Irwin et al. (1979).

The dynamic pressure P_{dyn} is defined as the difference between the measurement from pressure tap C06 and the mean static pressure from reference pressure sensors P2 and P3. P2 and P3 are located approximately 20 cm upstream of the model. The lift difference coefficient c'_{PL} is defined as the pressure difference between taps C18 and C40 divided by the dynamic pressure (steady normalization, Eq.3) or the instantaneous dynamic pressure (transient normalization, Eq.4); this is a two-tap pressure difference coefficient, not a cross-sectional lift coefficient (F3). The CWT analysis employs a Morlet wavelet with center frequency $\omega_0 = 6\pi$ over a frequency range of 5.5 Hz to 55.1 Hz. The Hilbert transform is used for instantaneous frequency extraction with a threshold of 50% of the mean. The identification thresholds for acceleration and deceleration are $\langle a \rangle_W = \pm 0.25\,\text{m/s}^2$.

Figure 15 shows the TKU-MFWT facility and model layout (direct source figure excerpt, for internal academic review only, copyright permission required before public publication; source Brusco et al. (2022), original Figure 1, PDF p. 2; permitted interpretation: facility geometry and model installation position; prohibited interpretation: quantitative assessment of inflow uniformity; public release status: separate permission required before public publication). Figure 16 shows the velocity history and acceleration time history for

condition UF1 (direct source figure excerpt, for internal academic review only, copyright permission required before public publication; source Brusco et al. (2022), original Figure related panels, PDF p. 6; permitted interpretation: qualitative shape of the acceleration time history; prohibited interpretation: precise numerical reading in place of tabulated values; public release status: separate permission required before public publication). Figure 17 shows the steady-state Strouhal number and comparison table with the literature (direct source figure excerpt, for internal academic review only, copyright permission required before public publication; source Brusco et al. (2022), original Table 1, PDF p. 5; permitted interpretation: source-reported Strouhal number range; prohibited interpretation: generalizing this value as a universal benchmark; public release status: separate permission required before public publication). Figure 18 shows the variation of transient Strouhal number with acceleration (direct source figure excerpt, for internal academic review only, copyright permission required before public publication; source Brusco et al. (2022), original Figure 16 related panels, PDF p. 14; permitted interpretation: trend effect of acceleration on Strouhal number; prohibited interpretation: quantitative claims on precise peak locations beyond tabulated reported values; public release status: separate permission required before public publication).

3.1.2 Response and mechanisms

Brusco et al. (2022) reported a mean Strouhal number of $S_{W,m} = 0.129$ across all steady-inflow tests, ranging from 0.125 to 0.132 (PDF p. 5). Under transient inflow condition UF1, the estimated maximum acceleration in the acceleration segment is 7.32 m/s^2, and the minimum in the deceleration segment is -4.93 m/s^2 (PDF p. 6). Table 2 (PDF p. 7) provides more detailed statistics: for UF1, the maximum $\langle a \rangle_W$ in the acceleration segment is 7.21 m/s^2 (mean 3.80), with -5.28 m/s^2 (mean -2.75) in the deceleration segment; the corresponding values for UF6 are 3.05, -2.39, 1.93, and -1.55 m/s^2.

During the acceleration segment, the vortex shedding frequency exhibits piecewise lock-in behavior: the frequency locks at approximately 20 Hz for $\tau < 13.8$ s, then abruptly jumps to approximately 25.7 Hz for $\tau > 13.8$ s (PDF p. 10). Table 12 (PDF p. 12) records representative instantaneous states, including entry (4) with $n_{\mathrm{VS,W}} = 19.9$ Hz, $U_W(\tau) = 11.30$ m/s, $a_W(\tau) = 7.14$ m/s^2, $\tau = 13.75$ s, and entry (5) with $n_{\mathrm{VS,W}} = 25.7$ Hz, $U_W(\tau) = 11.54$ m/s, $a_W(\tau) = 7.06$ m/s^2, $\tau = 13.78$ s. In the deceleration segment, Table 13 records a decrease from $n_{\mathrm{VS,W}} = 22.4$ Hz at entry (4) to 18.4 Hz at entry (5). The authors also reported multiple frequency jump instants during the acceleration segment: $\tau = 12.45$ s (9.7 to 11.4 Hz), $\tau = 14.15$ s (27.6 to 30.8 Hz), and $\tau = 14.98$ s (31.1 to 35.5 Hz) (PDF p. 12).

Regarding the systematic reduction of Strouhal number, the authors reported a maximum reduction of approximately 8% in the UF1 acceleration segment, occurring at the U_6 upcrossing instant and corresponding to an acceleration of $\langle a \rangle_W = 7.11$ m/s^2; the maximum reduction in the deceleration segment is approximately 5%, occurring at the U_7 downcrossing instant and corresponding to $\langle a \rangle_W = -5.11$ m/s^2 (PDF p. 14). For UF6, the maximum reductions are 4% (acceleration segment) and 3% (deceleration segment), respectively. The 30-repetition and 90-repetition results yield the same trend, indicating that 30 repetitions provide sufficient repeatability for this test series (PDF p. 14).

The dimensionless acceleration parameter $K_{a,W} = (a_W / U_W^2)\, b$ ranges from 0.007 to -0.005 in UF1 and from 0.0036 to -0.003 in UF6 (PDF p. 7). The maximum acceleration in full-scale events is 1.78 m/s^2, with a mean of the 15 largest values of 0.87 m/s^2 (PDF p. 7); the a/U^2 ratio ranges from 0.05 to 0.18 m^{-1} across the 15 full-scale events.

Figure 19 and Figure 20 are controlled formalized extraction overlay figures for Brusco et al. (2022). Figure 19 overlays the steady-state Strouhal number and the transient Strouhal number under condition UF1 as a function of Reynolds number, with each condition maintaining separate coordinate axes (L4 same-source

comparison). Figure 20 overlays the CWT spectrogram time-frequency structure of c'_{PL} in the acceleration and deceleration segments.

3.1.3 Condition table

Table 7 is the high-completeness transient condition table for Brusco et al. (2022), containing the following fields. Facility: TKU-MFWT, 1.32×1.32 m cross-section. Model: square cross-section, $b = 60$ mm, end plate diameter 400 mm, spacing 600 mm, aspect ratio 10. B/D: 1. Reynolds number definition: $Re = Ub/\nu$, $\nu = 1.5 \times 10^{-5}$ m^2/s. Reynolds number range: 1.72×10^4 to 6.34×10^4. Angle of attack: 0°. $U(t)$ family: steady (SF1 through SF12) and transient (13 conditions, including UF1 through UF6 and others). Turbulence intensity: approximately 2.5%. Blockage ratio: $< 5\%$. Blockage correction: not reported. Sampling frequency: 300 Hz. Record length: 2^{16} points (218.4 s). Number of repetitions: steady ≥ 25, transient ≥ 30 (90 for UF1). Pressure sensors: ZOC 33, ± 3.75 Pa. Force channel: not reported (no independent balance, no pressure-integrated force reported). c'_{PL} definition: two-tap (C18 and C40) pressure difference coefficient (not a cross-sectional lift coefficient). PIV: not reported. Hot-wire: not reported. Numerical simulation: not reported. Individual wind speed values (U_1 through U_{12}) for the 12 steady conditions: not reported (only crossing values for U_6 and U_7 are reported). Detailed conditions for UF2 through UF5 and UF7 through UF13: not reported. Data availability status for each condition: not reported (the authors stated that data should be obtained upon request).

3.1.4 Limitations

The authors of Brusco et al. (2022) stated several limitations. The freestream turbulence intensity of approximately 2.5% exceeds the recommended values for conventional general-purpose wind tunnels, and the test results may be affected by freestream turbulence (PDF p. 2). The inflow non-uniformity deviation of 4% to 5% exceeds the typically recommended range for conventional wind tunnels (PDF p. 2). The number and timing of constant-frequency time cells are not entirely regular (PDF p. 9), and the choice to exclude the lower-energy peak of a bimodal distribution may be debatable (PDF p. 10). The mechanism underlying the transient Strouhal number reduction remains unclear, and the vortex shedding configuration under acceleration may differ from that under steady conditions (PDF p. 16). The analysis of low-acceleration, high-velocity segments may be subject to certain biases (PDF p. 8). The content of the related doctoral thesis Brusco (2021) did not undergo this review's pre-verification procedure (not reported-19).

3.1.5 Nearest-neighbor comparison

Within the declared comparison ceiling, Brusco et al. (2022) is compared with the physical source Brusco et al. (2024). Both use the same facility (TKU-MFWT), the same model geometry ($b = 60$ mm square cylinder), the same Reynolds number definition and range, and the same sampling frequency (300 Hz). Matching conditions include facility, model dimensions, B/D, inflow type (including steady and monotonic acceleration), and Reynolds number range. Condition differences include: Brusco et al. (2024) explicitly reports the pressure-integrated drag coefficient c_D and the base pressure coefficient $c_{P,b}$, whereas Brusco et al. (2022) uses the two-tap pressure difference coefficient c'_{PL} as the primary response quantity. Brusco et al. (2024) spans a larger parameter space for UF conditions (varying U_{low} and T_b) and reports the percentage drag reduction due to acceleration (maximum 8% in the acceleration segment, 4% in the deceleration segment). The comparison ceiling is L4 (same-facility, same-model, same-source-group condition comparison), but because the response quantity definitions differ (c'_{PL} vs. c_D), quantitative comparison is limited to the universally appli-

cable Strouhal number and acceleration parameters. Brusco et al. (2022) reports a steady-state $S_{W,m} = 0.129$; Brusco et al. (2024) does not explicitly report the Strouhal number (the primary reported quantity is the c_D reduction); the acceleration parameter ranges of the two sources are compatible ($K_{a,W}$ ranges are similar). A residual limitation is that Brusco et al. (2022) does not report c_D and Brusco et al. (2024) does not report c'_{PL}, so the force responses of the two sources cannot be directly overlaid. Experimental implication: the two sources together indicate that, under this facility and model condition, accelerating inflow causes a systematic reduction of Strouhal number and deviations in force coefficients, but the differences in response-quantity definitions require future experiments to report multiple channels simultaneously.

Brusco et al. (2022) is also compared with the numerical source Lunghi et al. (2024). Lunghi et al. (2024) used the LES method in Nek5000 to simulate a square cylinder ($B/D = 1$) in accelerating inflow, with a Reynolds number range of 1.720×10^4 to 6.536×10^4 that fully overlaps that of Brusco et al. (2022). However, the inflow conditions differ fundamentally: Lunghi et al. (2024) uses a turbulence-free smooth inflow, while the freestream turbulence intensity of Brusco et al. (2022) is approximately 2.5%. Lunghi et al. (2024) reports a numerical steady-state $C_{D,\mathrm{sim}} = 2.19$, compared with the experimental value $C_{D,\mathrm{exp}} = 2.06$ (from same-facility data of Brusco et al. (2024)), with the difference attributed to the turbulence intensity difference. The comparison ceiling is L1 (physical and numerical results do not share coordinate axes, qualitative adjacency only). The Strouhal number of Lunghi et al. (2024) remains in the range 0.10 to 0.14 during the acceleration process, which is qualitatively adjacent to but cannot be merged with the 0.125 to 0.132 range (steady state) of Brusco et al. (2022). Residual limitations include the turbulence intensity difference, normalization definition differences (Lunghi et al. (2024) normalizes by the instantaneous $u_\infty(t)$, while Brusco et al. (2022) normalizes by the CWT-extracted U_W), and the three-dimensional domain span of Lunghi et al. (2024) being only $5D$. Experimental implication: the numerical and physical sources separately report piecewise Strouhal number behavior, but the inflow turbulence intensity, domain size, and normalization are not aligned, so the results cannot be described as consistent or as validation.

3.2 Morello et al. (2025a): DICI wind tunnel experiment on square and 5:1 rectangular cylinders in accelerating inflow

3.2.1 Precise condition statement

The experiment of Morello et al. (2025a) was conducted in the DICI wind tunnel, which has a circular test section with a diameter of 1.1 m and a length of 1.42 m, and a freestream turbulence intensity of 0.9% (PDF p. 3). The test models include a square-cross-section cylinder (side length $D = 6$ cm) and a 5:1 rectangular-cross-section cylinder (streamwise length 20 cm, transverse width 4 cm), both with a span of 80 cm. The model edges were machined with a corner radius of 0.002 cm, measured by an RS PRO digital microscope. The blockage ratio is 5% for the square cylinder and 3.37% for the 5:1 rectangular cylinder (PDF p. 3). The square cylinder is equipped with 30 pressure taps (7 on each of the front and rear faces, 8 on each side face), and the 5:1 rectangular cylinder with 72 taps (9 on each of the front and rear faces, 27 on each side face). The angle of attack is $0°$.

The inflow velocity histories comprise two types of acceleration: WTG (Gaussian-type, with acceleration amplitude following a Gaussian distribution in time) and WTC (constant-magnitude-type, with acceleration maintained at a constant level). Each type was tested at 10 acceleration severities (square cylinder) or 8 (5:1 rectangular cylinder), yielding a total of 20 acceleration conditions for the square cylinder and 16 for the 5:1 rectangular cylinder. The reference time interval for acceleration conditions is defined by the dimensionless time $T = tu^*_\infty/D$, with the square cylinder reference interval being $T = 600$ (from $Re = 17200$ to $Re = 65360$) and the 5:1 rectangular cylinder reference interval being $T = 1350$. The extended Reynolds

number ranges are 60430 to 108590 for the square cylinder and 41230 to 89390 for the 5:1 rectangular cylinder (at $0.500\times$ reference acceleration, PDF p. 5).

The Reynolds number is defined as $Re = u_\infty D/\nu$ (PDF p. 2), where u_∞ is the freestream velocity and D is the square-cylinder side length or the transverse width of the 5:1 cylinder. The dimensionless acceleration has two definitions, both used in the text: aD/u_∞^2 (referenced to the instantaneous freestream velocity) and $aD/u_{\infty,0}^2$ (referenced to the freestream velocity at $t = 0$); the two normalizations are not interconvertible. The strongest Gaussian-type acceleration corresponds to $aD/u_\infty^2 = 7.6 \times 10^{-3}$ (PDF p. 5), falling within the range of 2×10^{-3} to 5×10^{-2} proposed by Mason and Yang (2016).

The sampling frequency is 1000 Hz (PDF p. 3). The CWT analysis uses a Morlet wavelet with $\omega_0 = 6\pi$ (high center frequency to enhance frequency resolution), and it was verified that $\omega_0 = 2\pi$ does not produce significant differences (PDF p. 8). The number of repetitions per acceleration condition, the record length, the tubing transfer function correction, and the synchronization method between conditions are not explicitly reported by the source (not reported). Figure 8 shows the overlay of 10 realizations, but the number of sampling points per realization and the total record duration are not reported (not reported).

Figure 21 shows the DICI wind tunnel test section and model configuration (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 22 shows the formation of constant-frequency time cells under WTG acceleration for the square cylinder (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 23 shows the behavior of Strouhal number increasing from 0.12 to 0.16 under acceleration for the 5:1 rectangular cylinder (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 24 shows the relationship between pressure-integrated drag coefficient C_{Dp} and Reynolds number under steady inflow for the square cylinder (direct source figure excerpt, for internal academic review only, copyright permission required before public publication).

3.2.2 Response and mechanisms

For the square cylinder, the vortex shedding frequency under accelerating inflow exhibits a piecewise constant-frequency time cell structure. The strongest acceleration ($0.500\times$ WTG) produces 8 cells, $0.200\times$ WTG produces 6 cells, and $0.125\times$ WTG produces 4 cells (PDF p. 20). The mean Strouhal number within the cells fluctuates in the range 0.10 to 0.14, while the expected value under steady inflow over the same Reynolds number range is $St \approx 0.12$ to 0.13 (PDF p. 11). Tables 1 through 3 provide the mean Reynolds number, Strouhal number, and pressure-integrated drag coefficient $\langle C_{Dp} \rangle$ within each cell (PDF p. 21). Taking the 8 cells of $0.500\times$ WTG as an example, $\langle Re \rangle$ increases from 17875 to 64963, $\langle St \rangle$ fluctuates between 0.117 and 0.125, and $\langle C_{Dp} \rangle$ increases from 1.892 to 2.132 but drops back to 2.081 in the last cell.

Under steady inflow, the square-cylinder C_{Dp} approaches a plateau value of approximately 2.1 for $Re > 30000$ (PDF p. 20). Under stronger acceleration, C_{Dp} remains below the steady-state value for $Re \gtrsim 30000$, with the difference diminishing as acceleration severity decreases (PDF p. 20).

For the 5:1 rectangular cylinder, the behavior is markedly different. In the initial phase of acceleration ($Re \lesssim 20000$), the cell-averaged Strouhal number remains near $St \approx 0.12$; as acceleration continues, the Strouhal number gradually rises to $St \approx 0.16$ (PDF pp. 12 and 18). In conditions without frequency discontinuities, the Strouhal number increases continuously from 0.12 to 0.16 (PDF p. 18). These values are inconsistent with the $St = 0.114$ reported by Mariotti et al. under steady inflow at $Re = 40000$ (PDF p. 18). That $St = 0.114$ value comes from an independently published experiment.

The acceleration threshold for producing frequency discontinuities differs between the two geometries. The threshold is lower for the square cylinder: only $0.091\times$ WTG and $0.091\times$ WTC (corresponding to

$aD/u_{\infty,0}^2 = 2.03 \times 10^{-3}$ and 9.55×10^{-4}) do not produce discontinuities (PDF p. 15). The threshold is higher for the 5:1 rectangular cylinder: $0.111\times$ WTG and $0.125\times$ WTC (corresponding to 2.48×10^{-3} and 1.31×10^{-3}) do not produce discontinuities (PDF p. 18). The authors noted that the occurrence of discontinuities is more strongly associated with the acceleration history and Reynolds number than with the instantaneous acceleration value (PDF p. 16).

Regarding the pressure distribution of the 5:1 rectangular cylinder, the peak $\langle C_p \rangle$ at the front stagnation point is approximately 1.08 (PDF p. 21), exceeding the theoretical maximum of 1; the authors explained that CWT cell-averaged values may produce values slightly above the theoretical limit (PDF p. 22). In the forward half of the side face ($0.5 \leq s/D \leq 2.5$) and on the rear face ($s/D > 5.5$), $\langle C_p \rangle$ does not show significant deviations across different acceleration severities (PDF p. 22). The peak location of pressure fluctuations shifts slightly downstream under transient conditions compared to steady state, indicating downstream migration of the reattachment point (PDF p. 23).

Results from the extended Reynolds number range indicate that the square cylinder behavior does not differ substantially across Reynolds number intervals (PDF p. 26). In contrast, the 5:1 rectangular cylinder shows significant differences in the overlap region of two different Reynolds number ranges under acceleration, with the constant-magnitude acceleration type exhibiting greater differences than the Gaussian type, confirming that this geometry is more sensitive to unsteady dynamic effects (PDF p. 26).

Figure 25 and Figure 26 are controlled overlay figures for Morello et al. (2025a). Figure 25 overlays $\langle C_{Dp} \rangle$ vs. $\langle Re \rangle$ for the square cylinder at three acceleration severities (L4 same-source comparison). Figure 26 overlays the Strouhal number evolution for the 5:1 rectangular cylinder under WTG and WTC, keeping the two acceleration types separated.

3.2.3 Condition table

Table 8 is the high-completeness transient condition table for Morello et al. (2025a). Facility: DICI wind tunnel, circular cross-section with a diameter of 1.1 m. Model: square cylinder $D = 6$ cm, 5:1 rectangular cylinder 20×4 cm, span 80 cm, edge corner radius $r = 0.002$ cm. B/D: 1 (square cylinder) and 5 (rectangular cylinder). Reynolds number definition: $Re = u_\infty D/\nu$. Reynolds number range: 17200 to 65360 (standard), extending to $\sim 10^5$. Angle of attack: 0°. $U(t)$ family: WTG and WTC, each at 10 severities (square cylinder) or 8 (5:1 cylinder). Turbulence intensity: 0.9%. Blockage ratio: 5% (square cylinder), 3.37% (5:1 cylinder). Sampling frequency: 1000 Hz. Record length: not reported (overall). Number of repetitions: not reported (specific counts per condition; Figure 8 shows 10 realizations). Tubing transfer function correction: not reported. Force channel: P-PIF (Eq.4), not balance. Balance force: not reported (not reported). PIV: planned but not implemented (PDF p. 29).

3.2.4 Limitations

The authors of Morello et al. (2025a) stated the following principal limitations. The physical mechanism of constant-frequency time cells is only partially understood (PDF p. 29). A comprehensive theoretical framework linking geometry, acceleration history, and parameter thresholds has not yet been established (PDF p. 29). The phenomenon of the front stagnation $\langle C_p \rangle$ exceeding 1, while explained, indicates an inherent limitation of the cell-averaging method (PDF p. 22). The lower Reynolds number range of the 5:1 rectangular cylinder is constrained by the maximum freestream velocity of the wind tunnel and the smaller D value (PDF p. 26). The approach of presenting only Gaussian-type acceleration results while stating that the constant-magnitude type "does not show significant differences" may obscure subtle response differences (PDF p. 20).

3.2.5 Nearest-neighbor comparison

The square-cylinder results of Morello et al. (2025a) are compared with the physical source Brusco et al. (2022). Matching conditions: $B/D = 1$, similar Reynolds number range (both approximately 1.7×10^4 to 6.4×10^4), accelerating inflow type. Condition differences: different facilities (DICI circular cross-section vs. TKU-MFWT rectangular cross-section), different turbulence intensities (0.9% vs. 2.5%), different acceleration history profiles (WTG/WTC vs. TKU-specific waveform), different sampling frequencies (1000 Hz vs. 300 Hz), and different normalization methods. The comparison ceiling is L2 (geometry matched but normalization, facility, and turbulence intensity differ). Both sources report that the Strouhal number fluctuates in the range 0.10 to 0.14 under accelerating inflow; Brusco et al. (2022) reports a steady-state $S = 0.129$, and the square-cylinder steady-state St of Morello et al. (2025a) is approximately 0.12. Both sources observed constant-frequency time cell structures and frequency discontinuity phenomena. Residual limitations include the two normalizations (aD/u_∞^2 vs. $K_{a,W}$) not being directly interchangeable, and the turbulence intensity difference potentially affecting the discontinuity threshold. Experimental implication: the two facilities separately report piecewise constant-frequency time cells and Strouhal number variations; because the acceleration parameters, facilities, and turbulence conditions differ, apparent similarity is insufficient to establish quantitative consistency.

Morello et al. (2025a) is also compared with the numerical source Lunghi et al. (2024). The LES square-cylinder simulation of Lunghi et al. (2024) covers the same Reynolds number range (1.720×10^4 to 6.536×10^4), but the inflow is a zero-turbulence smooth inflow. The comparison ceiling is L1 (physical and numerical results do not share axes). The Strouhal number ranges reported by both sources are adjacent in order of magnitude, but this does not constitute consistency or validation evidence; the steady-state C_D of Lunghi et al. (2024) (2.19) exceeds that of Morello et al. (2025a) (C_{Dp} plateau value $\sim$ 2.1), with the difference potentially attributable to the combined effects of turbulence intensity, numerical domain, and normalization, and the existing sources are insufficient for single-factor attribution. Experimental implication: the LES and the experiment each exhibit piecewise vortex shedding behavior, but the zero-turbulence assumption and channel differences in the numerical simulation preclude describing the results as consistent or as validation.

3.3 Brusco et al. (2024): Drag and pressure response of a square cylinder in accelerating inflow in the TKU-MFWT

3.3.1 Precise condition statement

The experiment of Brusco et al. (2024) was also conducted in the TKU-MFWT, with facility parameters identical to those of Brusco et al. (2022): 1.32×1.32 m cross-section, 72 fans, and a freestream turbulence intensity of approximately 2.5%. The model is a square-cross-section cylinder with $b = 6$ cm, 94 pressure taps (46 at the mid-span cross-section), tap spacing of 5 mm, and end-tap spacing of 2.5 mm. Steady inflow conditions SF1 through SF12 (U from 4.3 to 15.9 m/s, $Re = 1.72 \times 10^4$ to 6.34×10^4) were each repeated at least 25 times. Transient inflow includes 8 explicitly reported conditions (UF1 through UF4, UF10 through UF13, Table 2); UF5 through UF9 are referenced before p. 26 of the source document but their condition parameters are not fully reported (not reported).

Condition UF1 uses $U_{\text{low}} = 4.30$ m/s corresponding to SF1 and $U_{\text{high}} = 15.86$ m/s corresponding to SF12 as velocity endpoints, with an acceleration ratio $\alpha = 3.69$, $T_b = 60$ s, and 90 repetitions. Table 2 lists the complete parameters for 8 conditions: UF1/UF2/UF3/UF4 have U_{low} values of 4.30/5.77/6.78/8.26 m/s, U_{high} uniformly at 15.86 m/s, α values of 3.69/2.75/2.34/1.92, and T_b uniformly at 60 s. UF10 through UF13 share the velocity endpoints with UF1 but have T_b values of 30/10/5/1 s, respectively, to investigate duration

effects. Table 3 provides the acceleration statistics ($\langle a \rangle_W$ peak and mean values) for each condition. Table 4 provides the dimensionless acceleration parameter $\langle K_a \rangle_W$.

The response quantities unique to this source include the pressure-integrated drag coefficient c_D (Eq.8, integrated using all 46 mid-span taps, PDF p. 15) and the base pressure coefficient $c_{P,b}$. The steady-state baseline values are $c_D = 2.06$ (range 2.01 to 2.09), $c_{P,b} = -1.30$ (range -1.34 to -1.26), $c_D/c_{P,b} = 1.58$, and $\sigma_{c_L} = 1.03$ (PDF p. 7 and Table 1 "Present" row). The steady-state value of the time-varying pressure-difference drag coefficient c'_{PD} is 2.30 (range 2.26 to 2.35, PDF p. 6); this is a two-tap definition, not an integrated drag. The acceleration and deceleration segments are kept strictly separated in this source.

Figure 27 shows Table 2 condition parameters from Brusco et al. (2024) (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 28 shows the normalized variation of $\langle c \rangle_D$ with velocity under condition UF1 (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 29 shows the response of transient pressure coefficient $\langle c \rangle_{P,29}$ in the acceleration and deceleration segments (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 30 shows the acceleration parameter data from Tables 3 and 4 (direct source figure excerpt, for internal academic review only, copyright permission required before public publication).

3.3.2 Response and mechanisms

The core findings of Brusco et al. (2024) concern the effect of accelerating inflow on the drag coefficient. The general trend of drag coefficient reduction in the acceleration segment is: the greater the acceleration (or deceleration), the larger the drag reduction (PDF p. 16). For UF1 at the U_8 reference wind speed, the ramp-up ensemble-mean pressure difference is 187.7 Pa (standard deviation 0.60 Pa), compared with the corresponding steady reference pressure difference of 204.3 Pa. The ramp-down counterpart is 196.8 Pa (standard deviation 0.55 Pa). The difference between 30-repetition and 90-repetition results is small (ramp-up: 189.3 Pa/1.07 Pa vs. 187.7 Pa/0.60 Pa). The standard deviation for steady SF8 is 0.59 Pa, based on 32 repetitions (PDF p. 14).

The normalized drag coefficient reduction in the acceleration segment reaches 0.92 (i.e., approximately 8% reduction), with the maximum reduction occurring near the U_8 crossing (PDF p. 16, Figure 12). The reduction in the deceleration segment is approximately 4%, with an overshooting phenomenon occurring during the deceleration from U_{11} toward U_2. The contribution of acceleration-induced inertial force is estimated at 0.7% of the steady contribution (PDF p. 16): using $U_W = U_8 = 12.55\,\mathrm{m/s}$, $a_W = 6.71\,\mathrm{m/s^2}$, $c_D = 2.06$, added mass coefficient $c_a = 1.51$ (Blevins, 2001), and $c_m = 2.51$, the inertial force is approximately 1.38 Pa, negligible relative to the steady dynamic pressure of 204.3 Pa.

For pressure coefficients, the C29 position (side-face midpoint) shows a maximum increase of 14% in the acceleration segment and 7% in the deceleration segment (PDF p. 18). The C18 position (near the leading edge) shows a maximum increase of 19% in the acceleration segment and 10% in the deceleration segment. The response at the C23 position is smaller (approximately 8%), indicating a non-uniform distribution of pressure response across different locations. The lift fluctuation coefficient σ_{c_L} shows a maximum reduction exceeding 25% in the acceleration segment and approximately 15% in the deceleration segment (PDF p. 19), with the reduction substantially larger than that of drag. An approximately 8% lift overshoot occurs in the deceleration segment near U_{11}.

The effects of varying U_{low} show that the maximum drag reductions for UF1/UF2/UF3/UF4 during ramp-up/ramp-down are approximately 8%/4%, 6%/4%, 3%/4%, and 1%/2%, respectively (PDF p. 20). The effects of varying T_b show that $T_b = 60$ s and 30 s produce no significant difference in drag reduction over the

tested wind speed range (PDF p. 22); however, the maximum reduction in lift fluctuation coefficient differs markedly across T_b values: approximately 11% for UF10, approximately 5% for UF11, approximately 2% for UF12, and near zero for UF13 (PDF p. 22).

The authors reported that the geometric scale ratio required to simulate wind tunnel acceleration effects is approximately 35 (acceleration segment) to 45 (deceleration segment, PDF p. 13).

Figure 31 and Figure 32 are controlled overlay figures for Brusco et al. (2024). Figure 31 overlays the normalized drag reduction for conditions with different U_{low} (L4 same-source comparison). Figure 32 overlays the lift fluctuation coefficient response for conditions with different T_b.

3.3.3 Condition table

Table 9 is the high-completeness transient condition table for Brusco et al. (2024). Facility and model parameters are identical to those of Brusco et al. (2022). Force channel: P-PIF (c_D, 46-tap integration, Eq.8). Independent balance: none. UF5 through UF9 conditions: not reported. Individual wind speed values for the 12 steady conditions: not reported. Blockage correction: not reported (the source states that the blockage ratio is $< 5\%$). Edge condition (sharp-edged): not reported. Details of the tubing transfer function correction method: the source cites Kay et al. (2020) but does not provide verification data for the corrected results. Assessment of model natural frequency effects: not reported. Data availability status: not reported.

3.3.4 Limitations

Brusco et al. (2024) inherits the limitations of the TKU-MFWT facility: freestream turbulence intensity of 2.5% and inflow non-uniformity of 4% to 5%. Although the acceleration-induced inertial force was estimated as negligible (0.7%), the selection of $c_a = 1.51$ is based on a single reference (Blevins, 2001), and no uncertainty propagation analysis was reported. The mechanism of the overshooting phenomenon in the deceleration segment has not been fully explained. The absence of UF5 through UF9 (the unreported portion of 26 conditions) limits complete coverage of the condition space. The non-coincidence of maximum response locations between Strouhal number variation and c_D reduction (the Strouhal number varies most near U_6 to U_7, while the c_D reduction is greatest near U_8) has limited physical explanation.

3.3.5 Nearest-neighbor comparison

The comparison between Brusco et al. (2024) and Brusco et al. (2022) was conducted in the preceding subsection (L4 same-facility, same-model comparison). Here, the $B/D = 1$ results of Brusco et al. (2024) are compared with those of Yang and Mason (2019). Matching conditions: $B/D = 1$, accelerating inflow. Condition differences: different facilities (TKU-MFWT vs. University of Queensland suction wind tunnel), different model dimensions ($b = 60$ mm identical but end plate and span configurations differ), different freestream turbulence intensities (2.5% vs. $< 0.5\%$), different blockage ratios ($< 5\%$ vs. 7.9%, the latter uncorrected), different $U(t)$ waveforms, different sampling frequencies (300 Hz vs. 600 Hz), and different force channels (P-PIF vs. P-PIF but with different normalizations). The comparison ceiling is L2 (geometry matched but multiple condition differences). Yang and Mason (2019) reports a $B/D = 1$ steady-state $\overline{C_D} = 2.14$, differing from the $c_D = 2.06$ of Brusco et al. (2024) by approximately 4%, which may be partly attributable to blockage ratio and turbulence intensity differences. Under accelerating inflow, Yang and Mason (2019) reports reductions of approximately 10%, 30%, 25%, and 5% in $\langle C_D \rangle$, $\langle C'_L \rangle$, $\langle |\hat{C}_L| \rangle$, and $\langle St \rangle$, respectively (Cases 3 and 4, PDF p. 12), while Brusco et al. (2024) reports an approximately 8% reduction in c_D (acceleration segment). Lift fluctuations in both sources show reductions larger than the drag reduction.

Residual limitations: different blockage correction strategies (uncorrected vs. uncorrected, but with a significant blockage ratio difference), and dynamic pressure definition differences. Experimental implication: the two independent facilities both observe systematic reductions in aerodynamic coefficients for the square cylinder in accelerating inflow, with the lift fluctuation reduction exceeding the drag reduction.

Brusco et al. (2024) is also compared with Yu et al. (2013). Yu et al. (2013) is an LES simulation covering $H/D = 0.3$ to 7.0 ($H/D = 1$ corresponds to the square cylinder), at $Re = 10^5$, in purely uniform inflow. The comparison ceiling is L1 (physical and numerical results do not share axes, and the Reynolds number difference is significant: 10^5 vs. $\sim 4 \times 10^4$). Yu et al. (2013) reports C_D of approximately 2 at $H/D = 1$ (PDF p. 4, stating "flat plate and square are both around 2"), which is adjacent in magnitude to the $c_D = 2.06$ of Brusco et al. (2024), but the Reynolds numbers differ by an order of magnitude. Experimental implication: this numerical C_D and the physical source can only serve as faceted adjacency, cannot be described as consistent, and are not permitted as a validation benchmark.

3.4 Wu et al. (2021): Multi-fan wind tunnel experiment on a 5:1 rectangular cylinder in sinusoidal inflow

3.4.1 Precise condition statement

The experiment of Wu et al. (2021) was conducted in the open-circuit wind tunnel at the State Key Laboratory of Disaster Reduction in Civil Engineering, Tongji University, with a test section of $1.5\,\text{m} \times 1.8\,\text{m} \times 10\,\text{m}$ (PDF p. 2), equipped with 120 independently controlled fans arranged in a 10×12 matrix. The model is a 5:1 rectangular cross-section with dimensions of $0.15\,\text{m}$ (chord) $\times 0.03\,\text{m}$ (depth) $\times 1.3\,\text{m}$ (length), fabricated from ABS material; the edge condition was not specifically reported (not reported). The model is supported by two thin-plate end plates and remains stationary. The blockage ratio was not explicitly reported (not reported) and can be estimated from the cross-section dimensions but was not given.

The inflow is a streamwise sinusoidal oscillation: $u(t) = U + \Delta u \sin(2\pi f_u t)$, with mean velocity $U = 8\,\text{m/s}$, $Re_D = 16080$ (defined based on depth $D = 0.03\,\text{m}$, PDF p. 2). The amplitude ratio $\Delta u/U$ ranges from 0 to 0.25, specifically including 0.0625, 0.125, 0.15, 0.1875, 0.20, and 0.25. The frequency f_u ranges from 0.2 to 1.2 Hz (in 0.2 Hz steps), yielding 6 frequencies. The wind tunnel cannot produce harmonic oscillating inflow above 1.2 Hz, a physical limitation of the multi-fan system (PDF p. 3). Pressure was measured through 6 DSM 3400 scanivalves with a full-scale sensor uncertainty of 0.08% (PDF p. 2). The tubing length is 0.2 m, with 15 spanwise spacings from D to $13.33D$. The sampling frequency is 200 Hz (PDF p. 2).

The natural vortex shedding frequency of the model is $f_0 = 31.5$ Hz, corresponding to a Strouhal number of $St_0 = 0.118$ (PDF p. 4). POD decomposition uses the pressure field from 6 cross-sections; the first two modes $C_P^{(R1+2)}$ represent the vortex-shedding-induced component, while the third mode and above $C_P^{(R\geq 3)}$ represent the inflow-induced component (framework of Eq.1 through Eq.7 in the PDF). The force coefficients obtained through POD decomposition include $C_D^{(R1+2)}$, $C_L^{(R1+2)}$, $C_M^{(R1+2)}$ (vortex-shedding-induced) and $C_D^{(R\geq 3)}$, $C_L^{(R\geq 3)}$, $C_M^{(R\geq 3)}$ (inflow-induced).

A pressure-integrated force channel exists in this source but is not an independent balance channel. The force coefficients in Figures 7, 8, 10, and 14 through 20 all derive from pressure integration. Independent balance force: not reported. PIV: not reported. Hot-wire: not reported. Numerical simulation: not reported (the source cites Wu et al. 2020b/2020c but these are not the source's own numerical results).

Figure 33 shows the wind tunnel and model configuration (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 34 shows the vari-

ation of vortex shedding frequency f_s with sinusoidal inflow parameters (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 35 shows the relationship between POD mode energy fraction and $\Delta u/U$ (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 36 shows the comparison of the two-dimensional aerodynamic admittance function with theoretical functions (direct source figure excerpt, for internal academic review only, copyright permission required before public publication).

3.4.2 Response and mechanisms

Under sinusoidal inflow, the vortex shedding frequency exhibits an abrupt drop: f_s/f_0 decreases to 0.75 to 0.82 in certain conditions (PDF p. 6). Under uniform inflow, the time-averaged force coefficients are $\overline{C_D} \approx 0.225$ and $\overline{C_L} \approx 0.031$; under sinusoidal inflow, $\overline{C_D}$ changes little (approximately 0.231, PDF Figure 10a), but the fluctuating values C'_D, C'_L, and C'_M grow approximately linearly with $\Delta u/U$ (PDF p. 7), with C'_D increasing from 0.010 to 0.074, and C'_L increasing from 0.031 to 0.064–0.070.

The core finding from the POD analysis concerns the redistribution of mode energy with $\Delta u/U$. Under uniform inflow, the cumulative energy of the first two modes is approximately 15% (PDF p. 8). Under the SF$(0.6, 0.1875)$ condition, mode 1 contributes over 80%, the first two modes combined reach 93%, and the total energy of higher-order modes is below 7% (PDF p. 8). The contribution of the inflow-induced component can reach approximately 90% (for $\Delta u/U > 0.15$), while the vortex-shedding-induced component ceases to dominate for $\Delta u/U \geq 0.15$ (PDF p. 9). The spanwise correlation coefficients of the vortex-shedding-induced component, $R_{DD}^{(R1+2)}$ and $R_{LL}^{(R1+2)}$, remain at 1 across all spanwise distances, whereas the correlation coefficients of the inflow-induced component begin to decrease beyond approximately $5D$ (one chord length) (PDF p. 10).

Measurements of the two-dimensional aerodynamic admittance function (2D AAF) show that $|\chi_{Du}|^2$, $|\chi_{Lu}|^2$, and $|\chi_{Mu}|^2$ do not vary significantly with $\Delta u/U$ (PDF p. 12). In the low-frequency region ($\omega < 5$ rad/s), $|\chi_{Lu}|^2$ approaches the Horlock (1968) theoretical value, but none of the three components decreases monotonically with frequency, which differs from the behavior of the Davenport, Sears, or Horlock functions (PDF p. 12). The authors also reported that at angles of attack above 4°, the flow fully separates without reattachment (PDF p. 14).

Figure 37 and Figure 38 are controlled overlay figures for Wu et al. (2021). Figure 37 overlays the C'_D growth trend across different $\Delta u/U$ values (L4 same-source comparison). Figure 38 overlays the POD mode energy fraction evolution under different sinusoidal inflow parameters.

3.4.3 Condition table

Table 10 is the high-completeness transient condition table for Wu et al. (2021). Facility: Tongji University multi-fan wind tunnel, 1.5×1.8 m cross-section, 120 fans. Model: $0.15 \times 0.03 \times 1.3$ m, $B/D = 5$. Reynolds number definition: $Re_D = UD/\nu = 16080$ (single Re). Angle of attack: not reported (inferred as 0°). $U(t)$: $u(t) = U + \Delta u \sin(2\pi f_u t)$, $U = 8$ m/s, $\Delta u/U$: 0 to 0.25, f_u: 0.2 to 1.2 Hz. Turbulence intensity: not reported. Blockage ratio: not reported. Sampling frequency: 200 Hz. Record length: not reported (different durations shown in different figures). Number of repetitions: not reported. Tubing frequency response correction: not reported. Pressure sensor accuracy: $\pm 0.08\%$ FS. Force channel: P-PIF (pressure integration, not balance). Balance force: none. PIV: none. Hot-wire: none. Numerical simulation: none. Air density ρ: not reported.

3.4.4 Limitations

The principal limitations of Wu et al. (2021) include: inflow frequency limited to below 1.2 Hz (physical limitation of the multi-fan system); the synchronization method of the Cobra probe was not explicitly reported; record length and number of repetitions per condition were not reported; the tubing transfer function correction was not discussed (the frequency response characteristics of 0.2 m tubing at 200 Hz sampling were not assessed); the truncation criterion for higher-order POD modes ($R \geq 3$) was not given (PDF p. 12 mentions 52 modes but without elaboration); blockage ratio and model end-plate effects were not quantitatively reported; candidate-marked digitization results (candidate-marked digitization points) serve only as candidate evidence and cannot be used as confirmed values.

3.4.5 Nearest-neighbor comparison

Wu et al. (2021) is compared with the physical source Li et al. (2021a). Matching conditions: $B/D = 5$, streamwise sinusoidal inflow, pressure measurement channel. Condition differences: different facilities (Tongji University multi-fan wind tunnel vs. 0.5×0.5 m gust generator wind tunnel), different Reynolds numbers ($Re_D = 16080$ vs. $Re = 1.3 \times 10^4$, noting that the characteristic lengths may differ: Wu et al. (2021) uses D and Li et al. (2021a) also uses D, but the D values differ: 0.03 m vs. 0.03 m, so the Re values are similar), different inflow generation mechanisms (multi-fan active control vs. airfoil gust generator), different $\Delta u/U$ ranges (0 to 0.25 vs. 5.5% to 19.2%), and different frequency ranges (0.2 to 1.2 Hz vs. 0.4 to 1.0 Hz). The comparison ceiling is L2 (geometry matched but inflow generation and normalization differ).

Wu et al. (2021) reports a steady-state $\overline{C_D} \approx 0.225$ (pressure-integrated), while Li et al. (2021a) reports a steady-state $C_D = 1.108$ (Table 3, PDF p. 5); the two values differ significantly. It should be noted that whether the C_D of Wu et al. (2021) uses the chord B or the depth D as the reference area is not fully clear, and the choice of reference area and reference length in the C_D definition of Li et al. (2021a) also requires confirmation. If Wu et al. (2021) uses B (chord 0.15 m) as reference, conversion to a D-referenced value requires multiplication by $B/D = 5$, yielding $C_D \approx 1.125$, which is close to the 1.108 of Li et al. (2021a). However, this conversion requires explicit support in the source text, which is currently NR. Residual limitation: the difference in force coefficient reference area/reference length is the key factor constraining quantitative comparison. Experimental implication: under streamwise sinusoidal inflow, both sources observe that force coefficient fluctuations grow with increasing amplitude ratio, and the modal energy redistribution revealed by POD decomposition should be compared with the separation bubble response from gust generator experiments.

Wu et al. (2021) is also compared with the numerical source Li et al. (2025) (numerical hypothesis line, $B/D = \mathrm{N/A}$). Li et al. (2025) (numerical hypothesis line, $B/D = \mathrm{N/A}$) simulated the response of rectangular cylinders with $R = 3$ to 16 (B/D definition and orientation unclear, $B/D = \mathrm{N/A}$) under accelerating inflow, but with monotonic acceleration rather than sinusoidal inflow. The comparison ceiling is L0 (inflow type mismatch, B/D definition unclear), permitting only methodological mention with no quantitative comparison. Li et al. (2025) (numerical hypothesis line, $B/D = \mathrm{N/A}$) is labeled as a numerical hypothesis channel, not available for physical validation.

3.5 Yang and Mason (2019): Accelerating-inflow experiment on rectangular cylinders of three aspect ratios

3.5.1 Precise condition statement

The experiment of Yang and Mason (2019) was conducted in the open-circuit suction wind tunnel at the University of Queensland. The test section measures 762×762 mm with a length of 2.36 m, and the freestream turbulence intensity is $u'/U < 0.5\%$ (u' is the root-mean-square velocity fluctuation). The freestream velocity range is $2 < U < 40$ m/s with a velocity uniformity of $\pm 1\%$. The $U(t)$ of transient conditions is controlled to within ± 0.2 m/s precision only.

Three two-dimensional rectangular-cross-section cylinders all have depth $D = 60$ mm, with widths B of 20, 60, and 180 mm, corresponding to $B/D = 1/3$, 1, and 3. The blockage ratio is reported by the source as 7.9% (PDF p. 3, text value) or 7.8% (Table 2, PDF p. 8); the table value of 7.8% is adopted. The source explicitly states that no blockage correction was applied: "No correction for tunnel blockage was made in this study given no such procedure exists for accelerating flow conditions." (PDF p. 3). End plates with a diameter of 250 mm are installed at 300 mm on either side of the model centerline, yielding $AR = L/D = 10$.

Pressure measurement uses a system with a 64-channel ZOC 33 scanivalve, connected by 1.5 m PVC tubing (inner diameter 1.5 mm); the tubing transfer function was corrected following the method of Bergh and Tijdeman (1965). Velocity was measured using a Type 55P11 single-wire hot-wire probe positioned 100 mm downstream of the model. The sampling frequency for both pressure and velocity is 600 Hz, but different acquisition systems were used.

The Reynolds number is defined as $Re = UD/\nu$, with a range of $9.3 \times 10^3 \leq Re \leq 1.23 \times 10^5$ (PDF p. 4). Data at $Re < 6000$ were excluded due to accuracy and synchronization issues (PDF p. 8). Each acceleration condition was repeated 30 times for ensemble averaging. Table 1 lists the parameters for 8 acceleration conditions. The pressure-integrated forces are defined using Eq.(2) and Eq.(3). Steady-state durations are at least 180 s, and acceleration conditions last approximately 60 s.

When analyzing $B/D = 1$ and $B/D = 3$, the two aspect ratios must be treated as separate facets and cannot be merged prior to any permissible same-facility comparison.

Figure 39 shows the wind tunnel configuration and the three rectangular cylinder cross-sections (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 40 shows the relationship between $\overline{C_D}$ and Re under steady inflow for the three B/D values along with the Table 2 summary (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 43 shows the ensemble-averaged $\langle C_D \rangle$, $\langle C_L' \rangle$, $\langle |\hat{C}_L| \rangle$, and $\langle St \rangle$ as functions of Re for $B/D = 1$ under accelerating inflow (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 44 shows the accelerating-inflow response for $B/D = 3$ (direct source figure excerpt, for internal academic review only, copyright permission required before public publication).

3.5.2 Response and mechanisms

Steady-inflow baseline values: $B/D = 1/3$ gives $\overline{C_D} = 2.24$, $C_L' = 0.88$, $|\hat{C}_L| = 1.13$, $St = 0.142$; $B/D = 1$ gives $\overline{C_D} = 2.14$, $C_L' = 1.33$, $|\hat{C}_L| = 1.75$, $St = 0.130$; $B/D = 3$ gives $\overline{C_D} = 1.24$, $C_L' = 0.29$, $|\hat{C}_L| = 0.40$, $St = 0.162$ (Table 2, PDF p. 8). The ratio $|\hat{C}_L|/C_L'$ is 1.3 to 1.4 across all tests. The variation of each coefficient with Re does not exceed 7%, but at $Re > 6 \times 10^4$, C_L' and $|\hat{C}_L|$ exhibit approximately 10% oscillations, possibly associated with the first resonance frequency of the tubing system (approximately 40 Hz) being close to the vortex shedding frequency in that Re range (PDF p. 6).

Under accelerating inflow, the collapse to steady-state values occurs before the midpoint of the acceleration period (Case 1 at $Re \approx 5 \times 10^4$, Case 2 at 3.5×10^4, PDF p. 8), requiring approximately 5 to 6 vortex shedding cycles (PDF p. 8). For $B/D = 1/3$, St increases by approximately 7% in the initial acceleration phase but returns to the steady-state value before $Re = 3 \times 10^4$ (PDF p. 10). For $B/D = 1$ (Cases 3 and 4), $\langle C_D \rangle$, $\langle C_L' \rangle$, $\langle |\hat{C}_L| \rangle$, and $\langle St \rangle$ decrease by approximately 10%, 30%, 25%, and 5%, respectively (PDF p. 12). The suction coefficients on the leeward and side faces decrease significantly during acceleration (approximately 30%, PDF p. 12). $C_{p,s}$ remains at approximately 1 for $Re > 2 \times 10^4$, but $C_{p,s} > 1$ in the initial acceleration phase (PDF p. 8).

For $B/D = 3$, C_L' and $|\hat{C}_L|$ immediately drop to approximately two-thirds of the steady-state values at the onset of acceleration (PDF p. 13), while $\langle C_D \rangle$ exhibits an overshoot of up to 10% (PDF p. 13), and $\langle St \rangle$ initially increases by approximately 6% before collapsing to steady state (PDF p. 13). The $\langle C_p \rangle$ on the rear half of the model exceeds the steady-state value by 0.05 to 0.1 in the initial acceleration phase.

The authors emphasized the distinction between coefficients and loads: in all acceleration tests, no overshoot of total loads or pressures beyond the steady-state conditions at the target velocity was observed (PDF p. 15), but the coefficients (normalized by instantaneous velocity) do exhibit reductions or overshoots. Cases 5 and 6 have minimal effects on each variable (PDF p. 13), with differences from steady-state values being negligible.

For all cylinders, periodic vortex shedding does not commence until $Re = 1 \times 10^4$, which is at least one order of magnitude higher than the expected value under steady inflow (PDF p. 10).

Figure 41 and Figure 42 are controlled overlay figures for Yang and Mason (2019). Figure 41 overlays the $\overline{C_D}$ vs. Re relationship for the three B/D values under steady inflow, with each B/D in separate panels (L4 same-source, faceted treatment). Figure 42 overlays the ensemble-averaged $\langle C_D \rangle$ for $B/D = 1$ under Cases 3 and 4 with the steady-state baseline (L4 same-source comparison).

3.5.3 Condition table

Table 11 is the high-completeness transient condition table for Yang and Mason (2019). Facility: University of Queensland suction wind tunnel, 762 × 762 mm cross-section. Model: three rectangular cylinders with $D = 60$ mm, $B = 20$, 60, 180 mm. B/D: 1/3, 1, 3. Reynolds number definition: $Re = UD/\nu$, ν value NR. Reynolds number range: 9.3×10^3 to 1.23×10^5 ($Re < 6000$ excluded). Angle of attack: 0°. $U(t)$ family: steady and 8 acceleration conditions (Table 1). Turbulence intensity: $< 0.5\%$ (u'/U). Blockage ratio: 7.8% (Table 2), uncorrected. Sampling frequency: 600 Hz. Record length: steady ≥ 180 s, acceleration approximately 60 s. Number of repetitions: 30. Pressure system: ZOC 33, 64 channels. Hot-wire: Type 55P11, 100 mm downstream. Force channel: P-PIF (Eq.2/3). Balance: none. PIV: none. Numerical simulation: none. Mid-span cross-section details: not reported. End plate mounting details: not reported. Morlet wavelet parameters: not reported.

3.5.4 Limitations

The blockage ratio of 7.8% to 7.9% is uncorrected, and the authors noted that no established blockage correction procedure exists for accelerating-inflow conditions (PDF p. 3). The hot-wire probe and the pressure system use different acquisition systems, with a synchronization precision of ± 0.02 s (PDF p. 5), which may introduce phase bias under high-acceleration conditions. The cause of the approximately 10% oscillations in C_L' and $|\hat{C}_L|$ at $Re > 6 \times 10^4$ is uncertain; the authors acknowledged that physically driven factors cannot be ruled out but considered further experimentation unwarranted (PDF p. 6). The cause of the collapse differ-

ence between Cases 1 and 2 is unknown (PDF p. 9). The initial flow development hypothesis for $B/D = 3$ requires further confirmation through additional research (PDF p. 9). The C_p distribution for $B/D = 3$ at certain angles of attack was not presented (PDF p. 13).

3.5.5 Nearest-neighbor comparison

The $B/D = 1$ results of Yang and Mason (2019) are compared with the physical source Brusco et al. (2024) (see the Brusco et al. (2024) subsection). The $B/D = 3$ results are compared with the numerical source Morello et al. (2025b). Morello et al. (2025b) uses the LES method in Nek5000 to simulate a 3:1 rectangular cylinder in accelerating inflow (Case A), with a Reynolds number range of 1.720×10^4 to 6.536×10^4 that partially overlaps with that of Yang and Mason (2019). Condition differences: the numerical simulation uses a zero-turbulence smooth inflow, while Yang and Mason (2019) has $< 0.5\%$ turbulence intensity; the numerical domain span is $5D$, while the experimental $AR = 10$; and the acceleration history profiles differ (Gaussian-type vs. experiment-specific waveform). The comparison ceiling is L1. Morello et al. (2025b) Table 1 reports a 3:1 cylinder steady-state $\langle C_D \rangle = 1.079$, while Yang and Mason (2019) reports a $B/D = 3$ steady-state $\overline{C_D} = 1.24$, a difference of approximately 13%. Morello et al. (2025b) Table 3 reports St values in the range 0.164 to 0.168 within each time cell, while Yang and Mason (2019) reports a steady-state $St = 0.162$; the two are qualitatively consistent. Residual limitation: the blockage ratio difference (7.8% vs. numerical domain $< 1\%$) may partly explain the C_D discrepancy. Experimental implication: physical and numerical results for $B/D = 3$ are qualitatively consistent in Strouhal number, but the C_D discrepancy suggests that the effects of blockage or turbulence intensity on the $B/D = 3$ geometry require further clarification.

3.6 M. Li et al. (2023): Square cylinder experiment in streamwise and vertical sinusoidal gusts

3.6.1 Precise condition statement

The experiment of M. Li et al. (2023) was conducted in a closed-circuit wind tunnel with a test section of $0.5\,\text{m} \times 0.5\,\text{m} \times 5.0\,\text{m}$. A gust generator is installed 2.05 m upstream, consisting of oscillating NACA 0012 airfoil blades (chord 0.1 m, span 0.46 m) whose oscillation frequency and amplitude can be independently controlled to produce streamwise gusts (in-phase airfoil oscillation) and vertical gusts (anti-phase airfoil oscillation). The model is a square-cross-section cylinder with $D = 50\,\text{mm}$ and length $L = 300\,\text{mm}$, instrumented with five rows of pressure taps along the span, each row containing 44 taps, for a total of 220 taps. The sampling frequency is 800 Hz, with a record duration of approximately 60 s. Force coefficients are obtained by integrating the strip cross-sectional pressure distribution.

The inflow conditions are divided into three groups (Table 1 in the PDF): Streamwise gust A/B/C, $U = 6.0\,\text{m/s}$, $f_{\text{os}} = 0.2\,\text{Hz}$, airfoil deflection angle $\theta = 14.4°/18.9°/23.2°$, gust frequency $f_g = 0.4\,\text{Hz}$ ($f_g = 2f_{\text{os}}$), $K_g = 2\pi f_g D/U = 0.021$, $\sigma_g/U = 5.3\%/14.1\%/20.8\%$. Vertical gust A/B/C, $U = 6.0\,\text{m/s}$, $f_{\text{os}} = 2.5\,\text{Hz}$, $\theta = 4.9°/9.7°/23.2°$, $f_g = 2.5\,\text{Hz}$, $K_g = 0.131$, $\sigma_g/U = 3.5\%/6.1\%/21.4\%$. The $f_g = 2f_{\text{os}}$ relationship for streamwise gusts arises from the geometric properties of the airfoil oscillation. The background turbulence intensity is $I_u = 0.6\%$. The Reynolds number is $Re = 2.1 \times 10^4$ (based on model thickness D and mean wind speed U, PDF p. 4).

The streamwise and vertical sinusoidal forcing are maintained as independent conditions in this source; synchronous multi-axis forcing was not implemented. Aerodynamic force coefficients are normalized by $\bar{q} = \rho U^2/2$; the value of ρ was not reported (not reported).

Figure 45 shows the gust generator and model layout (direct source figure excerpt, for internal academic

review only, copyright permission required before public publication). Figure 46 shows the velocity fluctuation characteristics under streamwise gusts (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 49 shows the spanwise distribution of pressure fluctuation RMS under vertical gusts compared with smooth inflow (direct source figure excerpt, for internal academic review only, copyright permission required before public publication). Figure 50 shows the POD mode energy distribution under different gust conditions (direct source figure excerpt, for internal academic review only, copyright permission required before public publication).

3.6.2 Response and mechanisms

Smooth-inflow baseline values: $\overline{C_D} = 2.14$, $C'_D = 0.117$, $C'_L = 1.16$, $St = 0.129$ (Table 2, PDF p. 5). The maximum pressure fluctuation RMS is located at $s/D \approx 1.66$ (side-face midpoint).

Under streamwise gusts ($\sigma_u/U = 20.8\%$), $\overline{C_D}$ decreases by approximately 15.2%, and $\tilde{C}_L$ (peak lift fluctuation) decreases by approximately 17.0% to 19.0% (PDF pp. 11 and 12). Under vertical gusts ($\sigma_w/U = 21.4\%$), $\overline{C_D}$ decreases by approximately 23.5%. Vertical gusts shift the location of maximum pressure fluctuation RMS from $s/D \approx 1.66$ to $s/D \approx 1.34$, with the peak value decreasing by approximately 20% (PDF p. 8).

POD analysis shows that under smooth inflow, the first two modes account for more than 90% of the energy (PDF p. 10). Under vertical gusts with $\sigma_w/U = 21.4\%$, the first POD mode energy fraction decreases to 47.0% (PDF p. 10). Low-frequency pressure fluctuations induced by vertical gusts can be three orders of magnitude higher than under smooth inflow (PDF p. 13).

Regarding spanwise correlation coefficients, under vertical gusts, the correlation coefficient at $\delta/D = 1.25$ drops to 0.29 (PDF p. 16); under streamwise gusts, even at σ_u/U of only 5.3%, the correlation coefficient at $\delta/D = 3.5$ remains above 0.85 (PDF p. 16).

Figure 47 and Figure 48 are controlled overlay figures for M. Li et al. (2023). Figure 47 overlays the $\overline{C_D}$ reduction at three streamwise gust intensities against the smooth-inflow baseline (L4 same-source comparison). Figure 48 overlays the first POD mode energy fraction under vertical gusts A/B/C compared with smooth inflow.

3.6.3 Condition table

Table 12 is the high-completeness transient condition table for M. Li et al. (2023). Facility: closed-circuit wind tunnel, 0.5×0.5 m cross-section, 5.0 m length. Gust generator: NACA 0012 airfoil array, 2.05 m upstream. Model: square cylinder $D = 50$ mm, $L = 300$ mm. B/D: 1. Reynolds number: $Re = 2.1 \times 10^4$. Angle of attack: 0°. $U(t)$ family: streamwise sinusoidal ($f_g = 0.4$ Hz, $K_g = 0.021$) and vertical sinusoidal ($f_g = 2.5$ Hz, $K_g = 0.131$). σ_g/U: 5.3% to 20.8% (streamwise), 3.5% to 21.4% (vertical). Background turbulence intensity: 0.6%. Sampling frequency: 800 Hz. Record duration: approximately 60 s. Pressure taps: $5 \times 44 = 220$. Force channel: P-PIF (pressure integration). Balance: none. PIV: none. Hot-wire: not reported (inflow characterization method not explicitly stated). ρ: not reported. Assessment of nonlinear effects of the gust generator on the inflow: not reported.

3.6.4 Limitations

The integral length scale parameter of the gust generator is $\lambda/D = 300$ for streamwise gusts and $\lambda/D = 48$ for vertical gusts (PDF p. 4), differing by one order of magnitude; this means that the streamwise gust scale is much larger than the model scale, while the vertical gust scale is only approximately 50 times the model

dimension. The spatial coherence and volumetric effects of the two gust types may differ fundamentally. The L_u^x/D ranges cited in the literature (Lee 0.73 to 1.14, Laneville < 5, Vickery 1.33, Gartshore 0.2 to 0.5) are not directly comparable with the gust length scales of this source. The window function and smoothing parameters for PSD analysis were not reported (not reported). The type of velocity measurement instrument was not specified (not reported; inferred to be a hot-wire or similar probe but not confirmed).

3.6.5 Nearest-neighbor comparison

The streamwise gust results of M. Li et al. (2023) are compared with the physical source Li et al. (2021a). Matching conditions: streamwise sinusoidal inflow acting on a 5:1 rectangular cylinder (Li et al. (2021a)) or a square cylinder (M. Li et al. (2023)), both using airfoil gust generators. Condition differences: different B/D (1 vs. 5), different Reynolds numbers (2.1×10^4 vs. 1.3×10^4), different model depths (50 mm vs. 30 mm), different facility cross-sections, and different gust frequencies and K_g values. The comparison ceiling is L1 (geometry mismatch, qualitative adjacency only). Both sources observe that streamwise sinusoidal inflow causes migration of the pressure fluctuation peak location and increases in C_D fluctuations. Residual limitation: the B/D difference gives rise to fundamentally different flow-field topologies (full separation for the square cylinder vs. separation-reattachment for the 5:1 cylinder), rendering quantitative comparison meaningless. Experimental implication: streamwise sinusoidal inflow produced by airfoil gust generators causes measurable changes in aerodynamic response for both the square and the 5:1 rectangular cylinder, but the geometric difference requires separate treatment.

M. Li et al. (2023) is also compared with the numerical source Yu et al. (2013). Yu et al. (2013) is an LES simulation of a square cylinder at $Re = 10^5$ under steady uniform inflow. The comparison ceiling is L0 (inflow type mismatch: sinusoidal vs. steady), permitting only methodological mention. Table 1 of Yu et al. (2013) gives C_D of approximately 2 and St in the range 0.09 to 0.18 (depending on H/D) for $H/D = 1$; M. Li et al. (2023) reports $\overline{C_D} = 2.14$ and $St = 0.129$ under smooth inflow. Because the inflow types are entirely different, no quantitative comparison is made.

4 Steady-State and Mechanism Anchor Sources

This section is organized around tables and cross-study numerical figures, presenting the values that directly answer experimental-design questions before proceeding to source-by-source discussion. The updated extraction completed on August 6, 2026 comprises 45 CSV files covering 15 newly deep-extracted sources; the 15,518-point complete dataset of Yang and Mason (2019) is reused as the 16th deep-extraction source. The remaining sources enter the relevant themes through the 55-paper condition matrix, original figures, direct quotations, and source condition tables. The newly extracted data, existing source figures, and source-level condition records do not share the same numerical depth; all values therefore retain their source, aspect ratio, inflow family, measurement channel, and normalization notes.

4.1 Experimental Data for Three Cross-Sections Under Steady Inflow

Table 1: Steady-state physical experiment reference values for three planned cross-sections

Source	B/D	Re range and definition	I_u	Block.	C_D	St	C'_L	Measurement channel
M. Li et al. (2023)	1	2.1×10^4; $Re = UD/\nu$	0.6%	not reported	2.14	0.129	1.16	P-SP; P-PIF; P-VEL
Brusco et al. (2024)	1	1.72–6.34×10^4; $Re = Ub/\nu$	2.5%	<5%	2.06	not reported	1.03	P-SP; P-PIF; P-VEL
Morello et al. (2025a)	1	1.72–6.54×10^4; $Re = u_\infty D/\nu$	0.9%	5%	2.1	not reported	not reported	P-SP; P-PIF
Han et al. (2025)	5	$\sim 2 \times 10^4$; $Re = Uh/\nu$	<0.4%	1.83%	1.03	0.111	∼0.08	P-SP; P-BAL
Matsumoto et al. (2003)	5	2.28–11.22×10^4; $Re = UD/\nu$	0.7%	∼3.75%	1.05	0.115	0.27–0.33	P-SP; P-BAL
Schewe (2013)	5	3.12×10^5–1.01×10^6; $Re = UD/\nu$	<0.2%	not reported	not reported	0.0878–0.0940	not reported	P-SP; P-BAL
Bruno et al. (2014)	5	ensemble statistics	multiple	multiple	1.074	0.109	not reported	ensemble mean
Wu et al. (2021)	5	$Re_D = 16080$; $Re_D = UD/\nu$	not reported	not reported	not reported	0.118	not reported	P-SP; P-PIF
Zhou et al. (2025)	5	$Re_D = 16080$; $D = 0.03\,\mathrm{m}$	3.5%	not reported	1.034	not reported	not reported	P-SP; P-PIF

Steady-state experimental values for the square cylinder ($B/D = 1$) show consistent magnitudes across three independent sources. M. Li et al. (2023) reported $C_D = 2.14$, $C'_L = 1.16$, and $St = 0.129$ at $I_u = 0.6\%$ and $Re = 2.1 \times 10^4$. Brusco et al. (2024) reported $c_D = 2.06$ and $\sigma_{c_L} = 1.03$ at $I_u \approx 2.5\%$ and $Re = 1.72 \times 10^4$ to 6.34×10^4. Morello et al. (2025a) reported a steady-state C_{Dp} approaching a plateau value of approximately 2.1 for $Re > 30000$ at $I_u = 0.9\%$ and $Re = 1.72 \times 10^4$ to 6.54×10^4. The C_D values from the three sources fall within the range 2.06 to 2.14; the differences can be attributed to variations in turbulence intensity (0.6% to 2.5%) and force definition (pressure integration versus two-tap pressure difference).

The steady-state experiments on the $B/D = 5$ rectangular cylinder span $Re \sim 2 \times 10^4$ to 10^6. Han et al. (2025) reported $C_D = 1.03$ and $St = 0.111$ at $I_u < 0.4\%$ with a low blockage ratio of 1.83%, with no Reynolds number effect observed. Matsumoto et al. (2003) reported $C_D = 1.05$ (balance measurement), $St = 0.115$, and $C'_L = 0.27$ to 0.33 at $I_u = 0.7\%$. Schewe (2013) covered $Re = 3.12 \times 10^5$ to 1.01×10^6 and found that St varied in three stages (0.0878 to 0.0940), with an abrupt drop in the range $Re \approx 3.6 \times 10^5$ to 5.0×10^5 indicating a flow-regime transition. The BARC ensemble mean of Bruno et al. (2014) yields $C_D = 1.074$ and $St = 0.109$. The uniform-flow baseline $St = 0.118$ from Wu et al. (2021) and the quiescent-flow $C_D = 1.034$ from Zhou et al. (2025) serve as steady-state references for their respective streamwise sinusoidal and accelerated inflow studies. Matsumoto et al. (2003) reported smooth-flow $St = 0.132$ and turbulent-flow $St = 0.197$; turbulence shifted the reattachment point from $7B/8$ downstream of the leading edge to $3B/8$, indicating that turbulence intensity exerts a first-order effect on both separation bubble topology and vortex shedding frequency.

Steady-state physical experiment values for $B/D = 3$ are extremely scarce among the sources covered in this review. Da Matha Sant'Anna et al. (1988) provides steady-state pressure distributions for the $H/D = 3$ cross-section, and Nakamura et al. (1991) provides hot-wire wake frequency data for a $c/t = 3$ flat plate, but neither reports C_D, St, or C'_L values that can be directly entered into Table 1. The numerical source Morello et al. (2025b) reports an St range of 0.145 to 0.195 for $B/D = 3$ under accelerating inflow, but this value cannot substitute for a physical measurement baseline. Consequently, Table 1 contains no physical

steady-state numerical entry for $B/D = 3$; this gap directly limits the transient comparison capability for the planned 60×180 mm cross-section.

Table 2: Cross-source numerical comparison of physical experiment steady-state baselines

Source	B/D	Re	C_D	C_L^{rms}	St	Channel	Notes
Yang and Mason (2019)	1/3	9.3×10^3–1.23×10^5	2.24	0.88	0.142	P-PIF	Steady-state baseline for three cross-sections in the same facility
Yang and Mason (2019)	1	same as above	2.14	1.33	0.130	P-PIF	Square cylinder steady-state baseline
Brusco et al. (2024)	1	1.72–6.34×10^4	2.06	1.03	not reported	P-PIF	TKU steady-state benchmark
Li et al. (2023)	1	2.1×10^4	2.14	1.16	0.129	P-PIF	Smooth inflow benchmark
Yang and Mason (2019)	3	9.3×10^3–1.23×10^5	1.24	0.29	0.162	P-PIF	3:1 steady-state baseline
Schewe (2013)	5	4×10^3–4×10^5	approx. 1.03	not reported	0.111	P-BAL/P-SP	Reynolds number sweep
Mannini et al. (2017)	5	1.18×10^4–1.13×10^5	approx. 1.05	not reported	0.115	P-BAL/P-PIF	Low-turbulence zero-angle-of-attack anchor
Bruno et al. (2014)	5	2–6×10^4	1.074	not reported	0.109	composite physical benchmark	BARC ensemble mean
Han et al. (2025)	5	3.12×10^5–1.01×10^6	1.11–0.98	not reported	0.094–0.088	P-PIF/P-BAL	High-Re range
Li et al. (2021a)	5	1.3×10^4	1.108	not reported	0.111	P-PIF	Smooth-flow benchmark for the sinusoidal inflow study
Zhou et al. (2025)	5	not reported	1.034	not reported	not reported	P-PIF	Steady-state benchmark for the acceleration and deceleration study

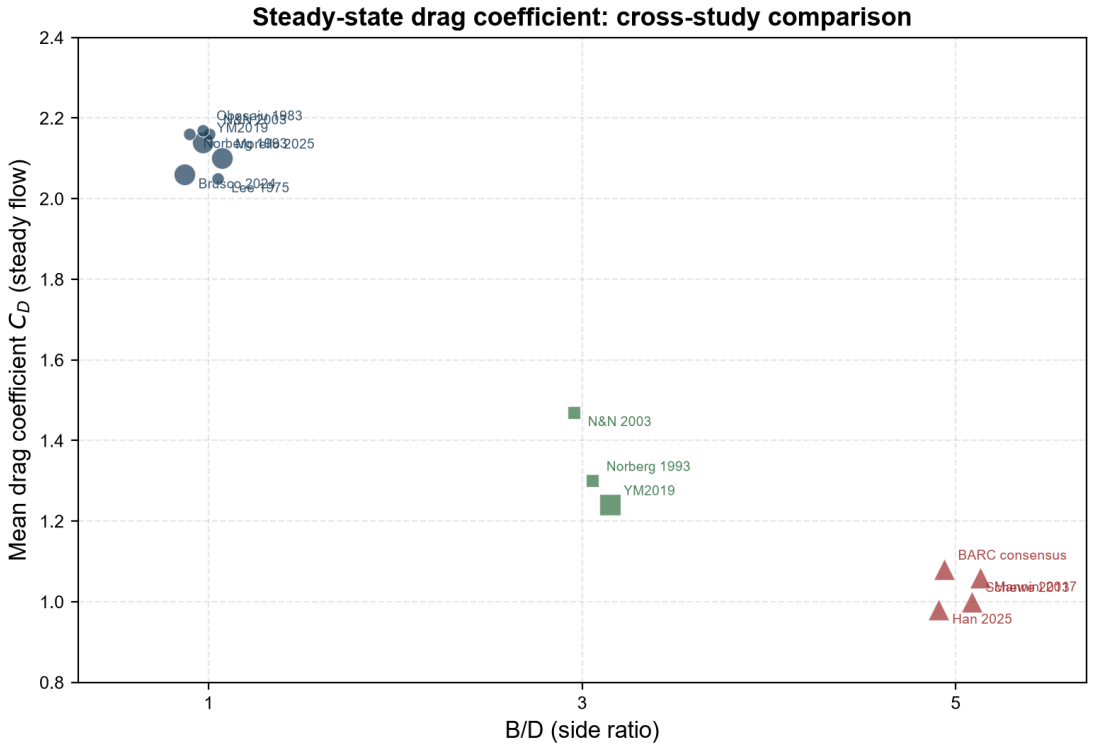


Figure 1: Comparison of steady-state drag coefficients across aspect ratios. Channels for physical and numerical sources are explained separately in the text and data tables.

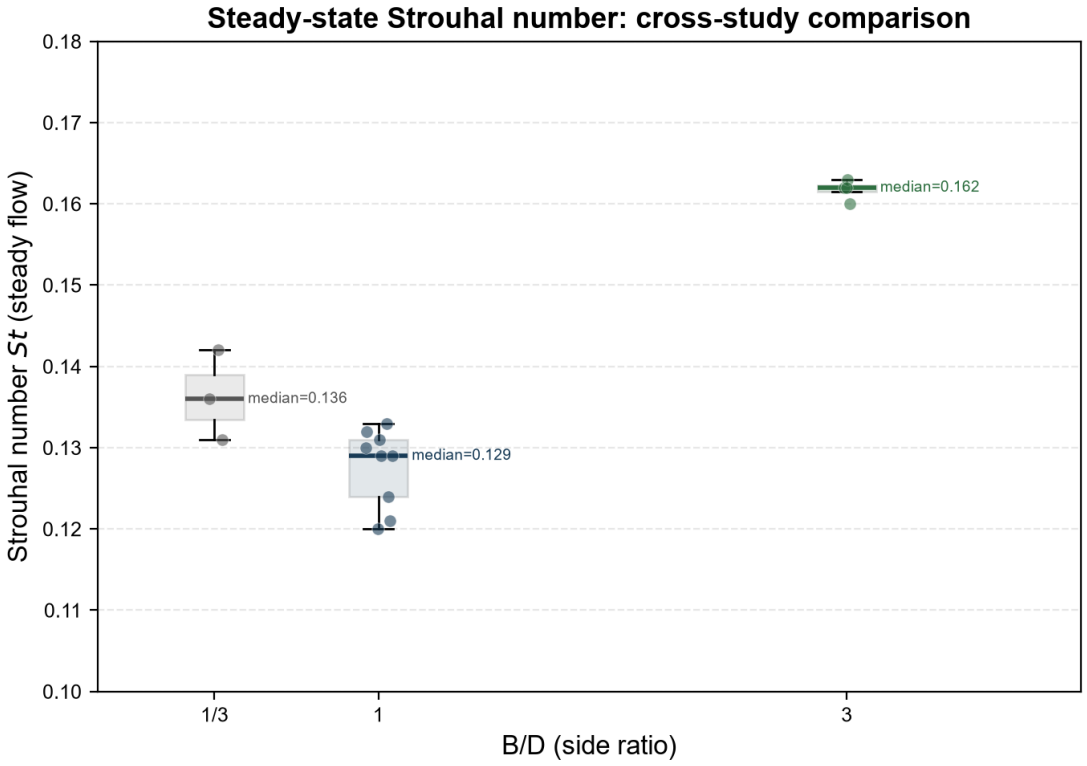


Figure 2: Comparison of steady-state Strouhal numbers across aspect ratios.

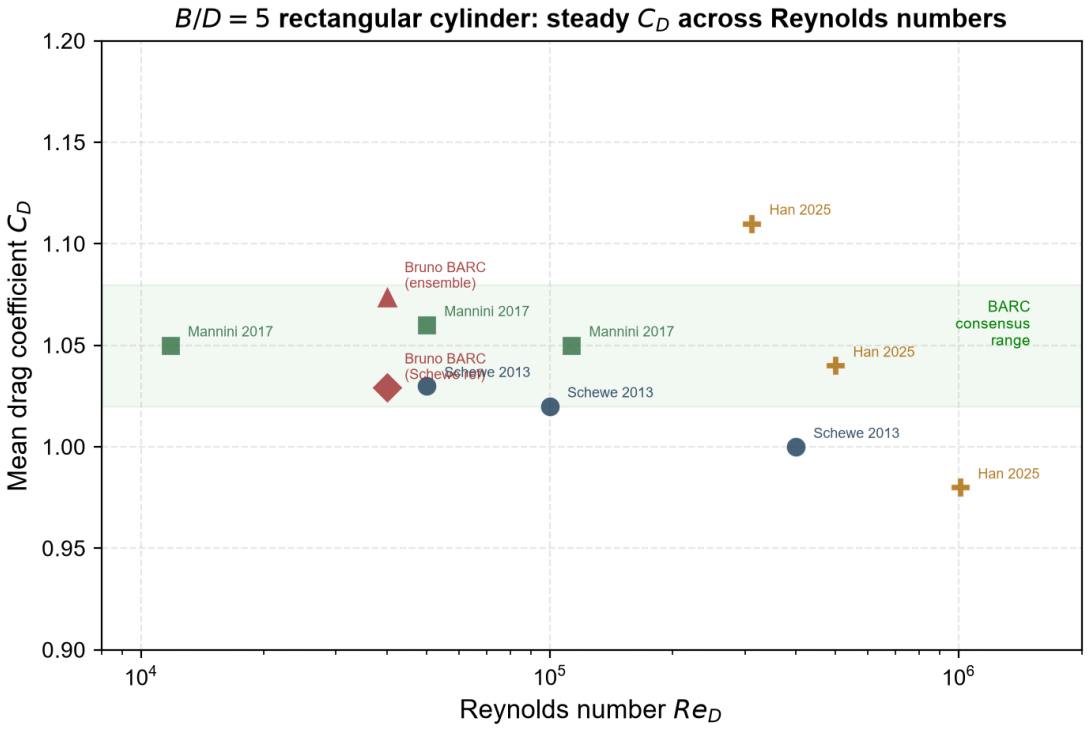


Figure 3: Cross-source comparison of steady-state drag coefficient versus Reynolds number for the 5:1 rectangular cylinder.

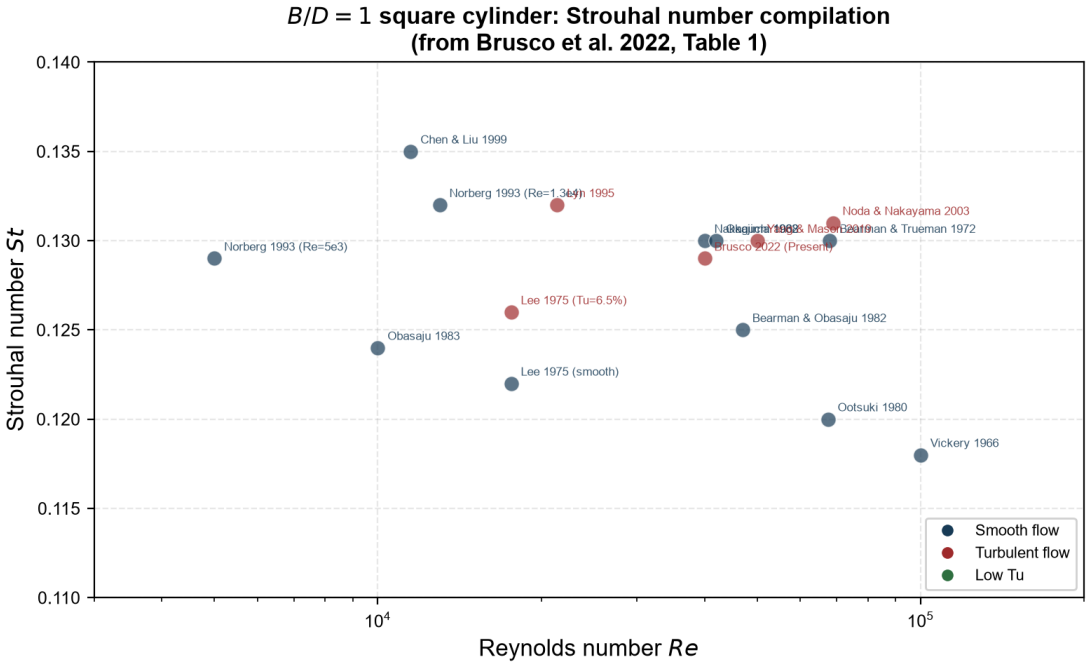


Figure 4: Source compilation of steady-state Strouhal numbers for the square cylinder.

4.2 Geometry-Effect Anchoring Under Steady Uniform Inflow

Da Matha Sant'Anna et al. (1988) provided steady-state pressure distribution data for $H/D = 0.5$ to 3.0 (seven aspect ratios, A through G) at $Re = 10^5$ with turbulence intensity $< 1\%$. The model depth was $D = 120$ mm with an aspect ratio $L/D = 10$, a test section of 1.2×1.2 m, and a blockage ratio $S/C = 9.8\%$. Up to 30 pressure taps per side were concentrated near the corners. The angle of attack ranged from 0° to above 20°. The quantitative anchor points from this source include: C_p at the separation angle (-0.93 for $H/D = 3$ and -1.94 for $H/D = 0.6$), base pressure coefficient (-0.525 for $H/D = 3$ and -2.08 for $H/D = 0.6$), reattachment angle (23° for $H/D = 0.5$ to 0°–1° for $H/D = 3$), and a pressure recovery phenomenon at $X/H \approx 0.7$ (present for $H/D > 0.8$). The source explicitly states that blockage correction was not fully applied (required data unavailable) and suggests a correction magnitude of 15% to 20% of $(1 - \bar{C}_p)$. Sampling parameters are listed as NR in full. Figure 51 shows the variation of C_{pb} with H/D from Da Matha Sant'Anna et al. (1988) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

Robertson et al. (1978) provided steady-state pressure distributions for a $B/D = 1$ square cylinder in the Colorado State University industrial aerodynamics wind tunnel, with a test section of 1.8×1.8 m, $B = 152$ mm, $Re = 1.3$ to 1.4×10^5, under two inflow conditions: low turbulence (0.33%) and high turbulence (10.4%, grid-generated, integral scale approximately 23 mm or $0.15B$). The angle of attack ranged from $-50°$ to $+90°$. Key findings include: minimum reattachment angle of approximately 14° (low turbulence) or approximately 9° (high turbulence, though not confirmed by tuft observation). The pressure distributions were symmetric near $+90°$. The source did not apply blockage correction; a suggested correction magnitude is 15% to 20% of $(1 - \bar{C}_p)$. Sampling parameters were 250 samples/s, 16 s record length, 80 Hz frequency response, and repeatability better than 0.03 (mean) and 0.01 (RMS). Figure 52 shows the variation of S_R/B with angle of attack from Robertson et al. (1978) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

Vickery (1966) provided force and pressure data for a $B/D = 1$ square cylinder (6×6 inches, 7×7 foot wind tunnel) over $Re = 4 \times 10^4$ to 1.6×10^5, in smooth inflow and 10% turbulence intensity inflow (grid scale approximately 8 inches). An independent balance (3-inch active segment) provided direct force measurements. Key quantitative anchor points: smooth-inflow $C_{Lf} = 1.32$ (direct measurement) or 1.27 (pressure-integrated), turbulent-inflow $C_{Lf} = 0.68$ (direct measurement) or 0.63 (pressure-integrated), with a difference of approximately 10%. Spanwise correlation length: $5.6D$ in smooth inflow, $3.3D$ in turbulent inflow. RMS lift fluctuation coefficient approximately 0.17 (identical for both inflow conditions). Strouhal number approximately 0.12 (0° angle of attack), insensitive to Re under steady-state conditions (3% repeatability, St 2% repeatability). Maximum RMS pressure coefficient 1.76 (corresponding to a peak-to-peak value of approximately 6). The source states force coefficient repeatability of approximately 10% and notes that pressure spectra are unreliable for $nD/U > 0.5$. Figure 53 shows the Table 1 force coefficient data from Vickery (1966) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

4.3 Vortex Shedding Branch Anchoring for Flat-Plate Cross-Sections

Nakamura et al. (1991) provided vortex shedding frequency data for flat-plate cross-sections with $c/t = 3$ to 16 ($t = 6$ mm, square leading and trailing edges) at $Re = (1\text{-}3) \times 10^3$, tested in two wind tunnels (frequency measurement: $3 \times 0.7 \times 2$ m, turbulence intensity $< 0.3\%$; flow visualization: $2 \times 4 \times 6$ m, turbulence intensity approximately 0.12%). The core finding is that the chord-based Strouhal number $S(c)$ is approximately constant and equal to 0.6 for $c/t = 3$ to 5, then increases in stepwise increments of approximately integer multiples of 0.6. For

$c/t \geq 16$ the spectrum becomes broadband with no dominant frequency. The vortex shedding mechanism is classified as impinging-shear-layer instability, commencing at $c/t \approx 3$ where the shedding wavelength equals the plate chord length. The lateral convection velocity is $U_c/U \approx 0.63$, and in the wake $U_c/U \approx 0.86$. Figure 54 shows the variation of $S(c)$ with c/t from Nakamura et al. (1991) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

Ohya et al. (1992) used LES to simulate the flow field and force coefficients for flat-plate cross-sections with $d/h = 3$ to 9 at $Re = 10^3$. Table 1 provides complete data: $d/h = 3$ yields $C_D = 1.34$, $S_h = 0.174$, $S_d = 0.52$, $m = 1$ (one lateral vortex); $d/h = 6$ yields $C_D = 1.26$, $S_h = 0.183$, $S_d = 1.10$, $m = 2$ (two lateral vortices); $d/h = 9$ yields $C_D = 0.90$, $S_h = 0.181$, $S_d = 1.63$, $m = 3$. The relation $S_d = S_h \times (d/h)$ can be verified in Table 1. The source is qualitatively consistent with the experiments of Nakamura et al. (1991) in terms of the steady-state Strouhal number (the $1 : 2 : 3$ ratio of S_d), but the Reynolds numbers differ by one order of magnitude (10^3 versus $(1\text{-}3) \times 10^3$). The Kelvin–Helmholtz instability was not observed in the numerical simulation. Figure 55 shows the Table 1 data and Figure 13 experimental comparison from Ohya et al. (1992) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

Zhang and Liu (2017) provided combined PIV and wall-pressure-fluctuation measurements for a blunt-nosed flat plate with $D = 24$ mm in three configurations of $L/D = 3$, 6, and 9, at $Re_D = 1.58 \times 10^4$, $U_0 = 10$ m/s, and freestream fluctuation intensity of approximately 0.6%. The PIV spatial resolution was $0.05D \times 0.05D$ with 6000 images (1 Hz); wall pressure was sampled at 1 kHz with 3×10^6 data points covering more than 150000 large-scale vortex shedding events. At $L/D = 3$ the flow does not reattach and passes directly into the wake, with a shedding frequency $fD/U_0 = 0.16$. At $L/D = 6$ and 9 the reattachment points are located at $x/D = 4.75$ and 4.5, respectively, with a low-frequency shedding of $fD/U_0 = 0.105$. POD analysis shows that the first two modes contain 34.7% and 32.5% of the energy for $L/D = 3$, 9.6% and 8.5% for $L/D = 6$, and 8% and 7.2% for $L/D = 9$. The source did not apply blockage correction (blockage ratio 8%). Figure 56 shows the time-averaged flow field and intermittency factor distribution from Zhang and Liu (2017) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

4.4 High-Reynolds-Number and PIV Anchoring for the 5:1 Rectangular Cylinder Under Steady and Accelerating Inflow

Han et al. (2025) provided high-Reynolds-number data ($Re = 3.12 \times 10^5$ to 1.01×10^6, based on depth D) for a $B/D = 5$ rectangular cylinder in a large Chinese wind tunnel, with angles of attack from 0° to 8°. The test section is an 8×6 m rectangle (filleted corners) with an effective cross-sectional area of 47.40 m^2. The model length was approximately 4 m, instrumented with 442 pressure taps (400 circumferential plus 42 longitudinal). Force measurements were performed simultaneously using a TB1378K external balance and a TH2401A internal balance (P-BAL channel), as well as pressure-integrated forces (P-PIF channel). The sampling frequency was 300 Hz with a 60 s record length. Key findings: at $\alpha = 1°$, C_D decreased from 1.11 at $Re = 3.12 \times 10^5$ to 1.01 at $Re = 1.01 \times 10^6$ (a reduction of 9.01%). The Strouhal number exhibited an abrupt drop in the range $Re = 3.59 \times 10^5$ to 4.99×10^5, indicating a flow-regime transition in this Reynolds number interval. Figure 57 shows the variation of C_D with Re and angle of attack from Han et al. (2025) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

Guissart et al. (2022) provided time-resolved 2D2C PIV data for a $B/D = 5$ rectangular cylinder ($L \times D = 90 \times 15$ mm) in the TU Darmstadt Eiffel wind tunnel, at $Re = 5 \times 10^3$ to 2×10^4 and $\alpha = -6°$ to 6°. The PIV resolution was $D/53$ to $D/116$, sampled at 1 to 2 kHz with 500 to 1500 snapshots per set and 3 independent repetitions. The freestream turbulence intensity was approximately 0.14% (passband [15 Hz, 30 kHz]), with a blockage ratio of 4% to 6% and no correction applied. Key quantitative anchor points: at $\alpha = 0°$ and $Re = 2 \times 10^4$, the reattachment point was located at $x/D = 1.64$, the primary vortex maximum height was $0.57D$, and the vortex center coordinates were $(0.05, 0.80)$. As Re decreased, the reattachment point shifted upstream (extending $> 1D$ from the leading edge at $Re = 5 \times 10^3$). At $\alpha = 2°$ the flow remained fully attached; at $\alpha = 4°$ it no longer did. The source did not report the model width B (not reported). Figure 58 shows the time-averaged streamlines and vorticity field from Guissart et al. (2022) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

Li et al. (2021a) provided experimental data for a $B/D = 5$ rectangular cylinder ($B = 0.15$ m, $D = 0.03$ m, $L = 0.496$ m) in a 0.5×0.5 m wind tunnel using a NACA 0012 airfoil gust generator to produce streamwise sinusoidal inflow. The mean velocity was $U = 6.0$ m/s with $Re = 1.3 \times 10^4$ (based on D) and a blockage ratio of 5.9% (uncorrected). Four SSOF conditions (Table 2) had $f_u = 0.4$ or 1.0 Hz, $\sigma_u/U = 5.5\%$ to 19.2%, and $K_u = 0.012$ to 0.032. Key findings: smooth-inflow $C_D = 1.108$, $St = 0.111$, and reattachment length $L_R^* = 4.0D$ (Table 3). Under SSOF the separation bubble length decreased from $4.0D$ to approximately $2.0D$ ($\sigma_u/U \approx 14\%$), and the pressure recovery onset shifted from $2.0D$ to $0.5D$. The C_p' peak increased from approximately ~ 0.11 in smooth inflow to 0.26–0.45 under SSOF. The Strouhal number increased from 0.111 to 0.144 (5.5%), 0.165 (14%), and 0.179 (19.2%). Figure 59 shows the Table 3 steady-state baseline comparison from Li et al. (2021a) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

Zhou et al. (2025) provided acceleration and deceleration inflow experimental data for a $B/D = 5$ rectangular cylinder (300×60 mm, $L = 1.2$ m) in the Tongji University TJ-5 multi-fan wind tunnel, with $L/B = 4$ and a blockage ratio of 3.33%. Nine test conditions combined three acceleration durations ($dt = 1$, 2, 4 s) with three acceleration and deceleration directions; $U_1 = 6.2$ m/s (low-speed stage), $U_2 = 10.5$ m/s (high-speed stage), and acceleration $a_f = 1.07$ to 4.30 m/s^2. The freestream turbulence intensity was approximately 3.5%, with 5×50 pressure taps. The sampling frequency and record length were not explicitly reported (not reported). HHT (Hilbert–Huang Transform) was used for transient force analysis. Key findings: at $a_f = 4.30$ m/s^2 the acceleration-phase C_D increased to 1.169 (approximately 13.0% above the steady-state value). C_P' peaked at approximately 0.35 during the acceleration phase and approximately 0.29 during the deceleration phase. A linear empirical formula $C_D' = 0.028a_f + 0.258$ ($1.07 \leq a_f \leq 4.30$) describes the acceleration-phase behavior. Figure 60 shows the Table 2 force coefficient summary from Zhou et al. (2025) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

4.5 Numerical Simulation Anchoring

Yu et al. (2013) provided LES results for rectangular cylinders with $H/D = 0.3$ to 7.0 at $Re = 10^5$ under uniform smooth inflow, using a finite-volume method (QUICK scheme, Leith time advancement, SOR solver). The computational domain was $x \in [-8, 16]$, $y \in [-8, 8]$, with grids ranging from 209×81 ($d/h = 3$) to 505×81 ($d/h = 9$). Key findings: no lateral reattachment occurs for $H/D \leq 2$, while reattachment appears for $H/D \geq 3$. C_D reaches a maximum of approximately 3 at $H/D \approx 0.5$ (PDF page 4; Nakaguchi et al. reported approximately 2.8). The Strouhal number exhibits a minimum at $H/D \approx 2$, reaches a maximum ($S \approx 0.187$) at $H/D = 3$, and shows a second discontinuous jump between $H/D = 5$ and 6. The 2D simulation yields $S = 0.18$ at $H/D = 2$, which differs significantly from experimental and 3D simulation results. Figure 61 shows the variation of C_D with H/D from Yu et al. (2013) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

Chiarini and Quadrio (2021) provided DNS data for a $B/D = 5$ rectangular cylinder at $Re = 3000$ (Luchini code, staggered Cartesian grid, $1776 \times 942 \times 150$ grid points, exceeding 250 million), with a computational domain of $x \in [-22.5, 40]$, $y \in [-21, 21]$, $z \in [-2.5, 2.5]$, near-wall spacing $\Delta x = \Delta y \approx 0.0015$, and spanwise spacing $\Delta z/D = 0.558$. The inflow was uniform at $\alpha = 0°$. Statistical samples were collected over 341 to 2345 D/U_∞ time units. Table 1 reports $f_1 = 0.1274$, $\langle C_d \rangle = 0.9437$, and $\sigma(C_l) = 0.2893$. The reattachment length $L = 3.955$ (differing by 8.4% from the value of 3.65 reported by Cimarelli et al. 2018b). The primary vortex center is at $(x_c, y_c) = (-0.143, 0.83)$, and the pressure extremum point is at $(-0.3, 0.99)$. The wall friction c_f distribution reveals a secondary separation region at $-1.87 < x < -0.91$. The source explicitly notes that the Reynolds number is low (PDF page 1). Figure 62 shows the time-averaged streamlines and vorticity field from Chiarini and Quadrio (2021) (direct source figure excerpt, for internal academic review only; copyright clearance required before publication).

Lunghi et al. (2024) used spectral-element LES with Nek5000 to simulate a square cylinder ($B/D = 1$) under accelerating inflow, with Re ranging from 1.720×10^4 to 6.536×10^4, polynomial order $N = 6$, $\Delta t = 0.003$, filter parameter $w = 0.05$, and spanwise domain $\Delta z/D = 0.558$. Three acceleration cases A/B/C had maximum $aD/u^2_{\infty,0}$ values of 2.23×10^{-2}, 2.97×10^{-2}, and 4.46×10^{-2}, respectively. Comparison with the experiments of Brusco et al. (2022) (through the same-facility data of Brusco et al. (2024)) yields $C_{D,\text{sim}} = 2.19$ versus $C_{D,\text{exp}} = 2.06$. During acceleration, St remained in the range 0.10 to 0.14. The source matched the experimental c'_{PL} using a numerical equivalent of a $0.025D$-diameter circle (PDF page 5).

Morello et al. (2025b) used the same numerical framework as Lunghi et al. (2024) (Nek5000) to perform LES of 5:1 and 3:1 rectangular cylinders under accelerating inflow. The steady-state baseline for the 5:1 cylinder was $\langle C_D \rangle = 0.988$ and $\sigma(C_L) = 0.248$; for the 3:1 cylinder, $\langle C_D \rangle = 1.079$ and $\sigma(C_L) = 0.657$ (Table 1). During acceleration, St ranged from 0.066 to 0.178 for the 5:1 cylinder and 0.145 to 0.195 for the 3:1 cylinder. Steady-state simulations (Tables 4 and 5) provide reference values over $Re = 17200$ to 60000. The 5:1 cylinder exhibited a flow-regime transition near $Re = 40000$ to 50000, consistent with the corresponding phenomenon under steady-state conditions but not entirely attributable to the Reynolds number change alone.

Li et al. (2025) (numerical hypothetical line, $B/D = \text{N/A}$) used URANS and LES to simulate rectangular cylinders with $R = 3$ to 16 under accelerating inflow; the precise definition of B/D was not explicitly labeled ($B/D = \text{N/A}$), and the source is therefore tagged as a numerical hypothetical channel. The source reports that R4 ($R = 4$) URANS at $Re = 7.50 \times 10^4$ yields $\overline{C_D} = 1.246$, while LES yields $\overline{C_D} = 1.108$. Morison theory with $C_a = 1.37$ describes the acceleration drag overshoot. The maximum drag overshoot for $R = 3$ to 16 ranges from 22.4% to 109.7% (PDF page 14). When $a_p < 0.01$, the drag coefficient falls below the quasi-steady value (PDF page 11). This source does not provide physical validation or controlled quantitative values.

5 Transient Family Comparison

5.1 Monotonic Acceleration, Acceleration and Deceleration, and Gust-Type Transients

Table 3: Cross-source numerical comparison of physical experiment transient responses

Source	B/D	Inflow family	Parameter	Response quantity	Reported change	Channel	Limitations
Brusco et al. (2022)	1	acceleration/ deceleration	$K_{a,W}$ up to 0.007	St	acceleration -8%, deceleration -5%	P-SP	constant-frequency time cells
Brusco et al. (2024)	1	downburst-like	UF1 peak $a = 7.21$ m/s^2; all cases 4.41–7.78 m/s^2	C_D	acceleration -8%, deceleration -4%	P-PIF	inertia term approximately 0.7% of steady-state drag
Morello et al. (2025a)	1	Gaussian/ constant acceleration	aD/U^2 up to 7.6×10^{-3}	C_{Dp}, St	C_{Dp} approx. -9.9% to $+1.5\%$; $St = 0.10$–0.14	P-PIF	time cell averaging, Re dependence

Source	B/D	Inflow family	Parameter	Response quantity	Reported change	Channel	Limitations
Yang and Mason (2019)	1	monotonic acceleration	eight cases	C_D, C_L^{rms}, St	approx. $-10\%, -30\%, -5\%$	P-PIF	approximately 5–6 cycles from vortex shedding onset to near-steady behavior
Yang and Mason (2019)	1/3	monotonic acceleration	eight cases	St	initially approx. $+7\%$	P-PIF	returns to near-steady values early in the ramp
Yang and Mason (2019)	3	monotonic acceleration	eight cases	C_D, St	C_D up to $+10\%$, St initially approx. $+6\%$	P-PIF	coefficient overshoot does not equal load overshoot
Morello et al. (2025a)	5	Gaussian/ constant acceleration	aD/U^2 up to 7.6×10^{-3}	St	approx. 0.12 increasing to 0.16	P-SP/P-PIF	digitization candidate requiring combination with original figures
Zhou et al. (2025)	5	acceleration/ deceleration	a_f up to 4.30 m/s^2	C_D, C'_D	$C_D + 13.1\%$; C'_D from 0.034 to 0.386	P-PIF	physical overshoot not reported, reattachment point upstream migration

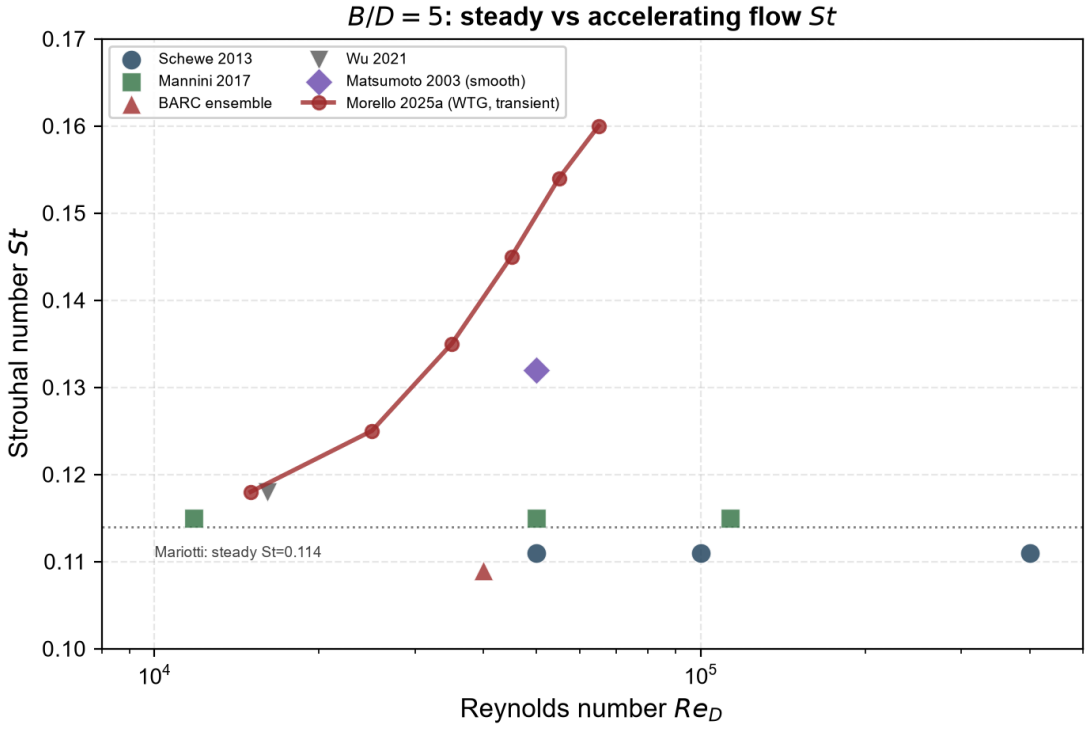


Figure 5: Faceted comparison of steady-state and accelerating-inflow Strouhal numbers for the 5:1 rectangular cylinder.

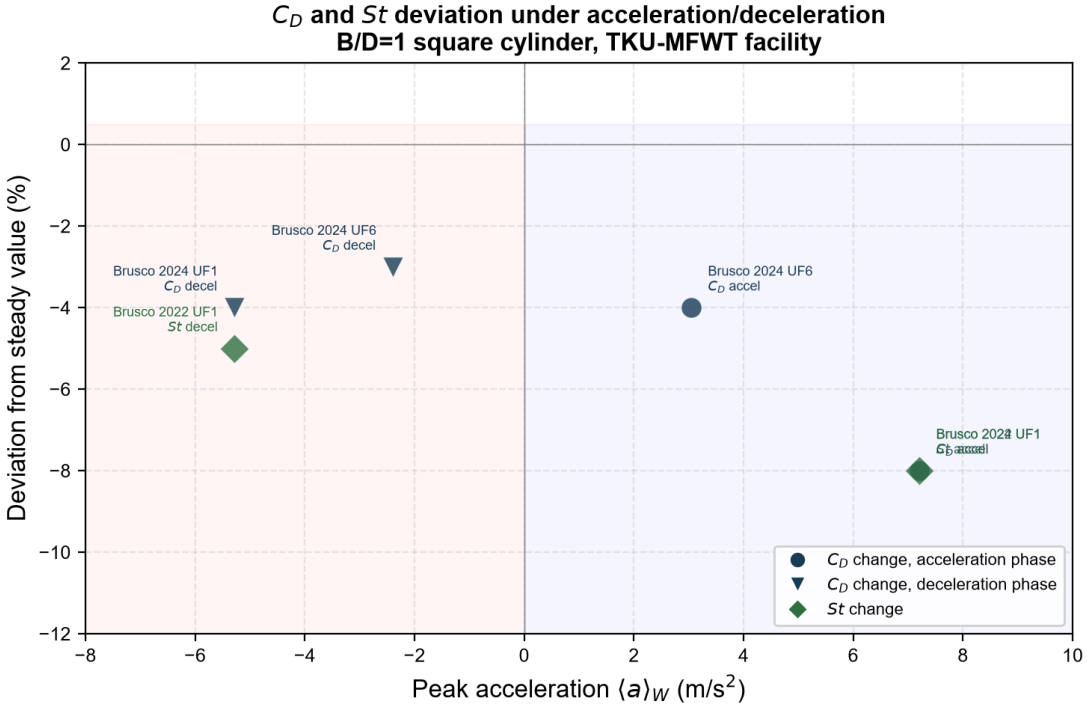


Figure 6: Source-level comparison of transient drag coefficient and Strouhal number changes versus acceleration for the square cylinder.

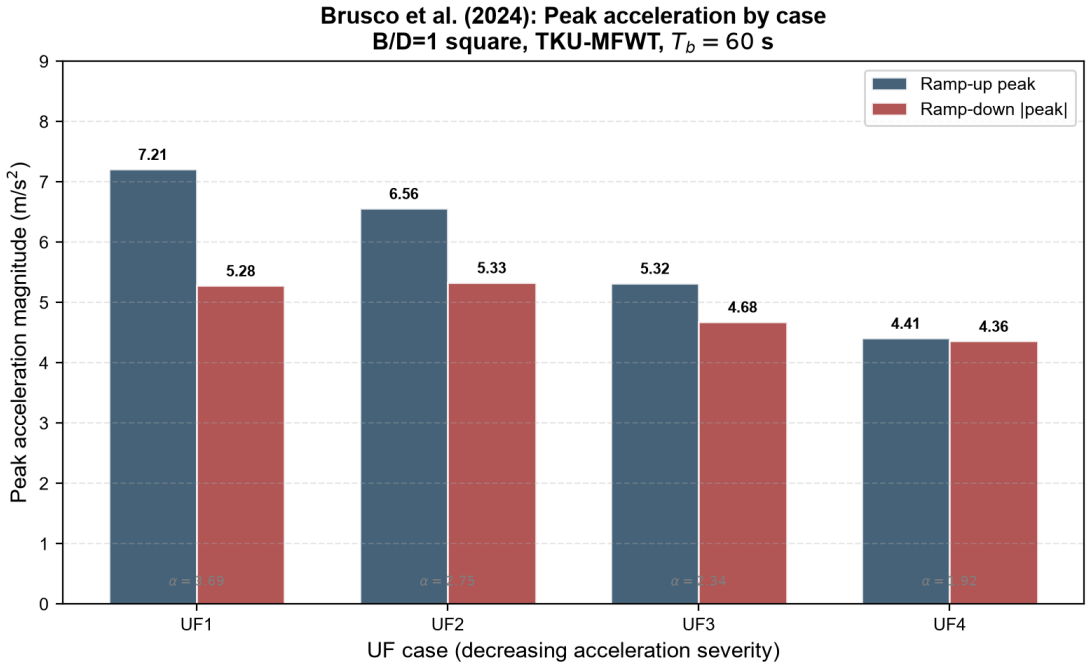


Figure 7: Acceleration statistics by case from Brusco et al. (2024).

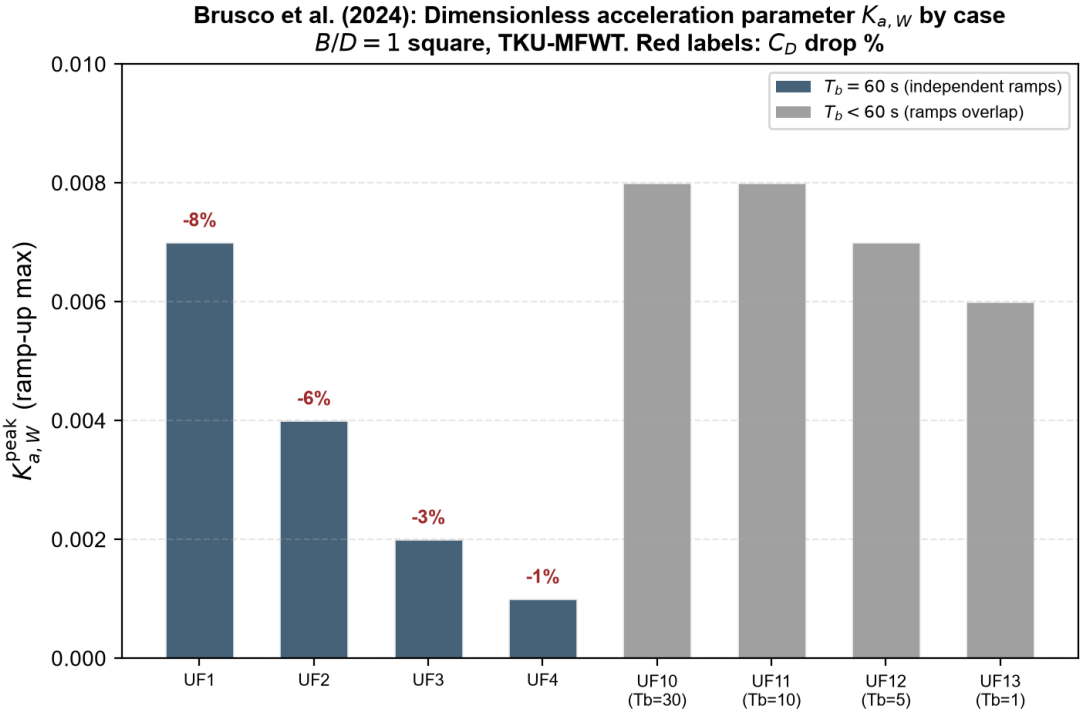


Figure 8: Dimensionless acceleration parameter by case from Brusco et al. (2024).

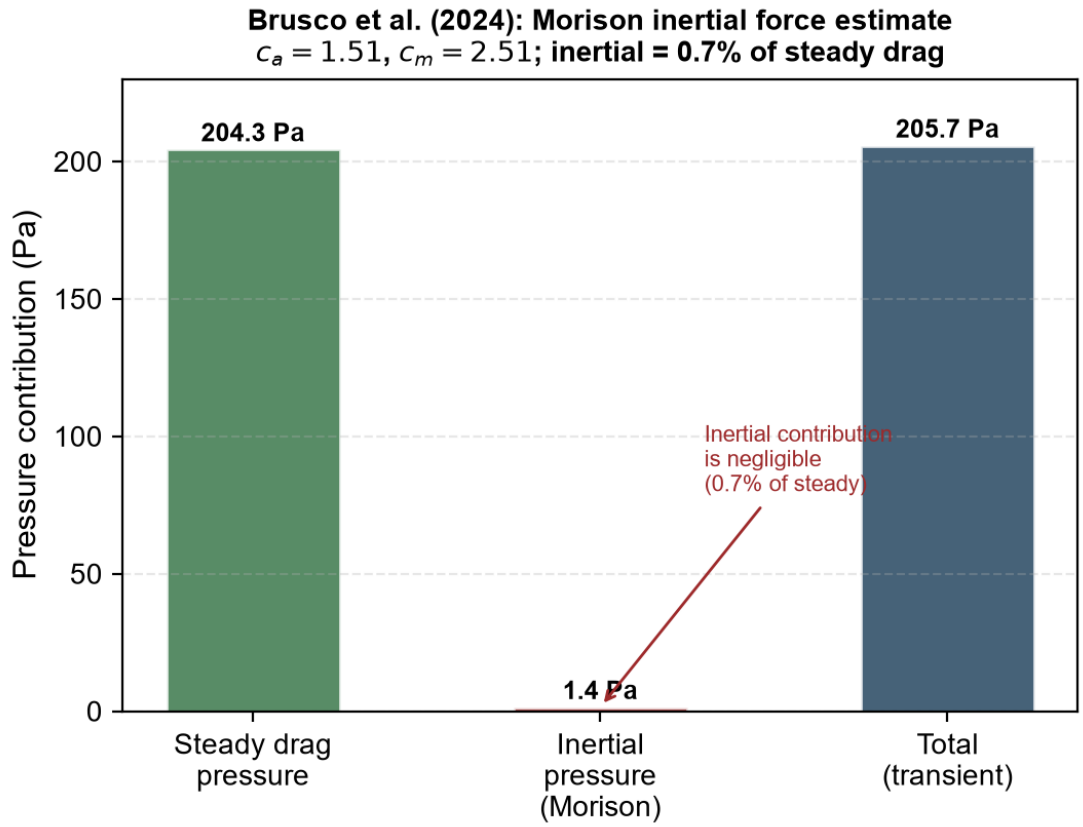


Figure 9: Magnitude comparison of Morison inertia term and steady-state drag.

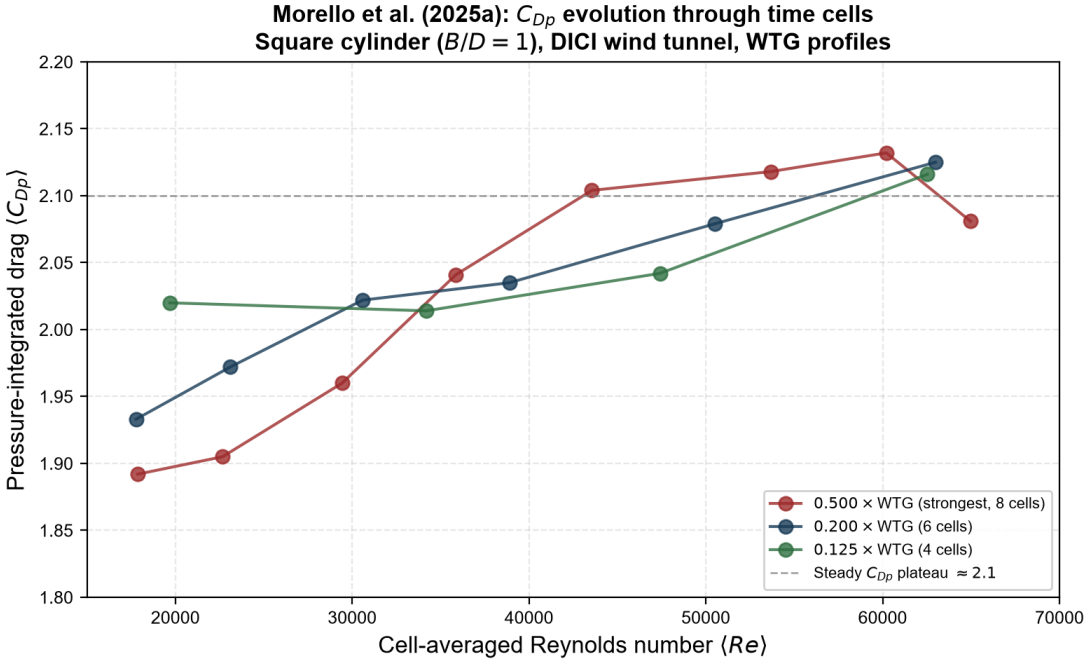


Figure 10: Pressure-integrated drag coefficient versus Reynolds number within time cells for the square cylinder from Morello et al. (2025a).

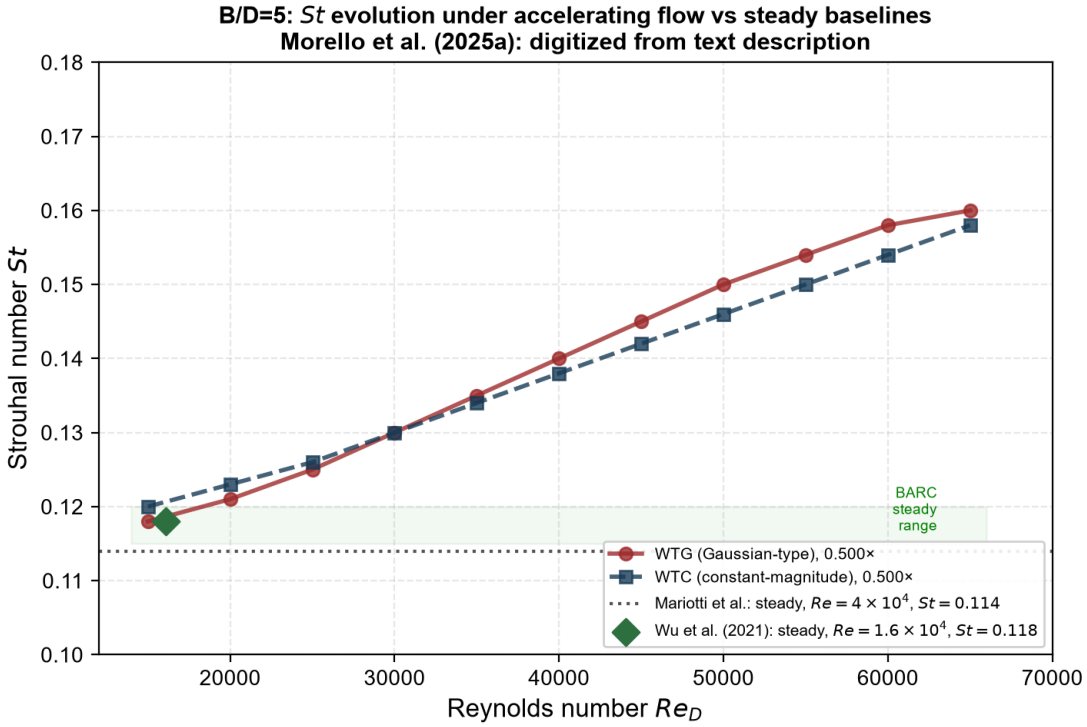


Figure 11: Digitization candidate data for Strouhal number evolution of the 5:1 rectangular cylinder from Morello et al. (2025a).

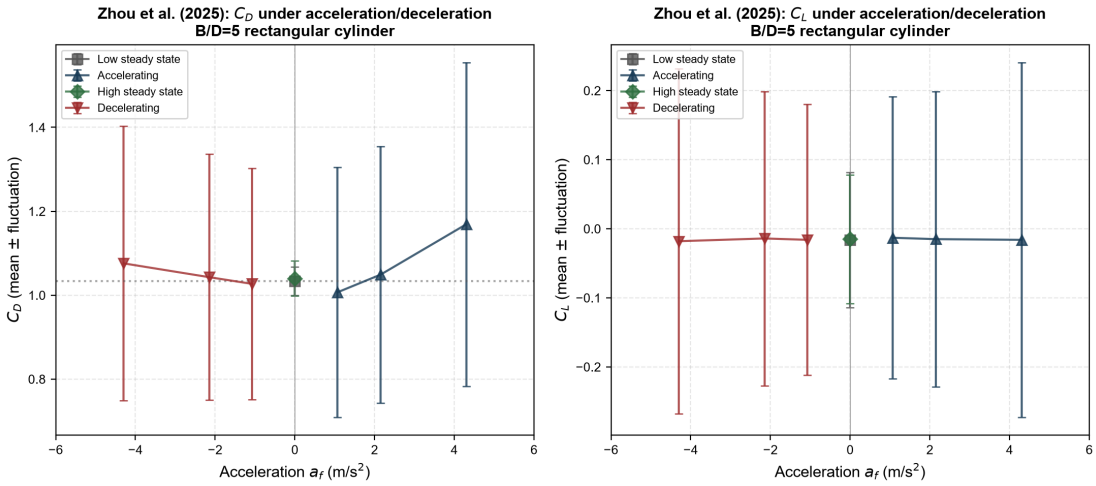


Figure 12: Force coefficient comparison across acceleration and deceleration cases for the 5:1 rectangular cylinder from Zhou et al. (2025).

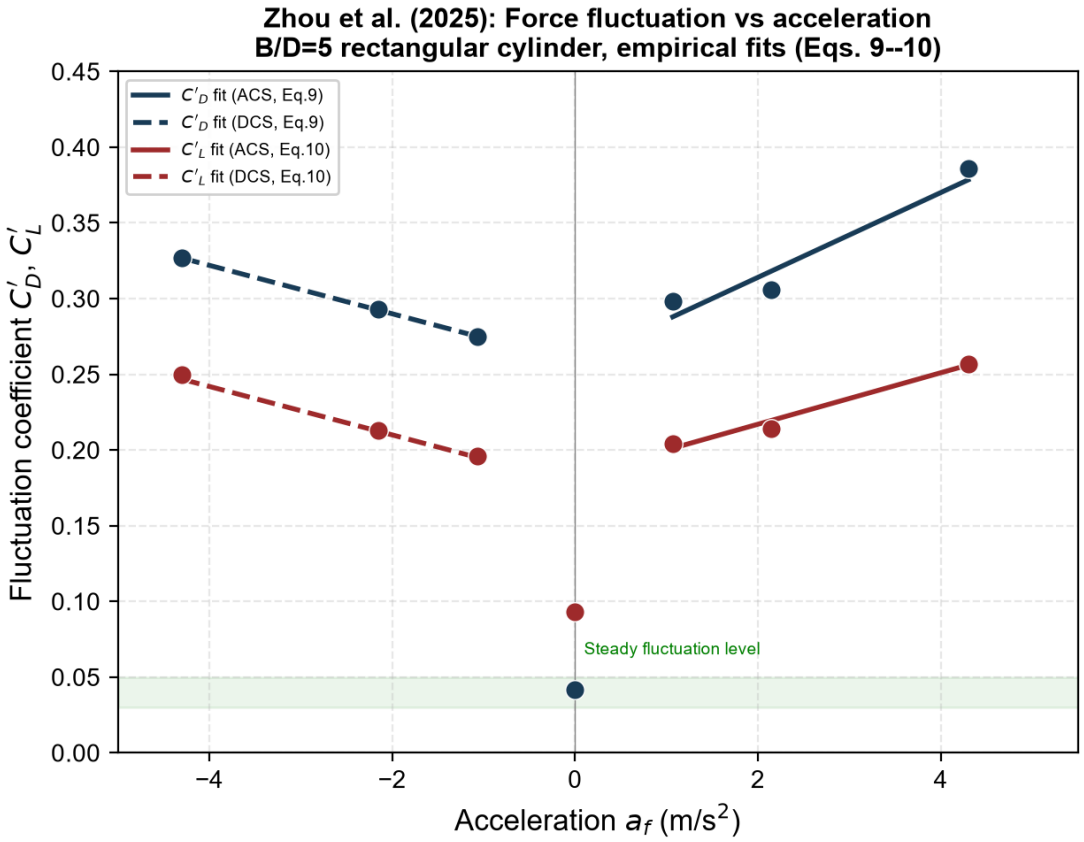


Figure 13: Empirical relationship between force coefficient fluctuations and acceleration from Zhou et al. (2025).

Tables 2 through 6 show that the square cylinder generally exhibits decreased C_D and St during the acceleration phase across multiple independent physical experiments, whereas the 5:1 rectangular cylinder is more sensitive to the inflow family and turbulence conditions. The St increase reported by Morello et al. (2025a), the C_D and fluctuation enhancement reported by Zhou et al. (2025), and the separation bubble shortening reported by Li et al. (2021a) collectively point to the reattachment process as the key controlling mechanism of the transient response for the 5:1 cross-section. Sinusoidal inflow cannot be merged with monotonic acceleration into a single effect size; streamwise sinusoidal inflow and vertical sinusoidal inflow must also be discussed separately.

Table 19 summarizes condition-level information for monotonic acceleration, acceleration and deceleration, and gust-type transients. The physical sources involved include Brusco et al. (2022) (UF1/UF6, square cylinder), Morello et al. (2025a) (WTG/WTC, square cylinder and 5:1 cylinder), Brusco et al. (2024) (UF1 through UF4, UF10 through UF13, square cylinder), Yang and Mason (2019) (Cases 1 through 8, three B/D values), and Zhou et al. (2025) (9 conditions, 5:1 cylinder). Numerical sources include Lunghi et al. (2024) (Cases A/B/C, square cylinder), Morello et al. (2025b) (Cases A/B/C, 5:1 and 3:1 cylinders), and Li et al. (2025) (numerical hypothetical line, $B/D = \mathrm{N/A}$), which is tagged as a hypothetical channel.

For monotonic-acceleration-type transients of the square cylinder ($B/D = 1$), different facilities report different response quantities. Brusco et al. (2022) explicitly identified constant-frequency time cells and reported maximum Strouhal number reductions of approximately 8% and 5% during the acceleration and deceleration phases, respectively. Morello et al. (2025a) reported $St = 0.10$ to 0.14 within time cells but did not provide a unified percentage reduction. Brusco et al. (2024) reported an approximately 8% reduction in pressure-integrated drag during the acceleration phase, without reporting St values usable for the same comparison. The three sources therefore support only a faceted conclusion that accelerating inflow modifies the frequency organization or aerodynamic forces, but do not support a conclusion that the same type of time cell or the same St reduction was observed across all three. Inter-facility differences in turbulence intensity, blockage ratio, $U(t)$ profile, and measurement channel limit the comparison to L2 or below. Yang and Mason (2019) tested the square cylinder at lower turbulence intensity ($< 0.5\%$) and higher blockage ratio (7.8%), reporting approximately 10% reduction in $\langle C_D \rangle$ and approximately 30% reduction in $\langle C'_L \rangle$, reductions larger than those from the TKU facility, but differences in turbulence intensity and blockage preclude quantitative attribution. Figure 19 shows the formalized extracted data from Brusco et al. (2022) FIG09a, retaining the original series and candidate status, permitted only for same-source verification.

For monotonic-acceleration-type transients of the 5:1 rectangular cylinder, Morello et al. (2025a) and Zhou et al. (2025) provide two independent datasets. Morello et al. (2025a) observed at low turbulence intensity (0.9%) that C_{Dp} fell below the steady-state value for $Re > 30000$, with smaller differences at lower acceleration severity. Zhou et al. (2025) observed at higher turbulence intensity (3.5%) that C_D increased to 1.169 (+13%) during the acceleration phase, a direction opposite to the decreasing trend reported by Morello et al. (2025a) at comparable acceleration levels. Condition differences include turbulence intensity (0.9% versus 3.5%), model size ($D = 4$ cm versus $D = 6$ cm), normalization basis (u_∞ reference versus high-speed-stage U_2 reference), and $U(t)$ waveform. The comparison ceiling is L2. Source-definition differences preclude direct merging of trends. Figure 20 shows the formalized extracted data from Brusco et al. (2022) FIG16, permitted only for same-source verification; the 5:1 cross-source statements in this paragraph are not supported by that figure.

Brusco et al. (2024) reported asymmetric drag response between the acceleration and deceleration phases (larger reduction during acceleration than during deceleration, with overshoot during deceleration). Yang and Mason (2019) also observed an initial C_D overshoot (approximately 10%) accompanied by C'_L reduction in the $B/D = 3$ case. The two sources differ in geometry (square cylinder versus $B/D = 3$); the comparison ceiling is L1. Figure 31 shows the formalized extracted data from Brusco et al. (2024) FIG11, examining only the acceleration branch of that source and not representing cross-source asymmetry.

5.2 Streamwise Sinusoidal Inflow

Table 4: Comparison of sinusoidal inflow conditions and aerodynamic responses

Source	B/D	Inflow family	Amplitude	Reduced frequency or frequency	Response	Channel
Li et al. (2021a)	5	streamwise sinusoidal	$\sigma_u/U = 5.5\%$–19.2%	$K_u = 0.012$–0.032	St from 0.111 to 0.179; C_p^{rms} up to 2.6 times; reattachment length from $4D$ to $2D$	P-SP/P-PIF
Wu et al. (2021)	5	streamwise sinusoidal	$\Delta u/U$ up to 0.25	0.2–1.2 Hz	C'_D from 0.010 to 0.074; inflow-induced component approximately 90% when amplitude exceeds 0.15	P-SP/P-PIF
Li et al. (2021b)	5	vertical sinusoidal	$\sigma_w/U = 3.5\%$–21.4%	$K_w = 0.052$–0.078	C_p^{rms} up to 3.8 times that of smooth flow, with strong coupling to angle of attack	P-SP/P-PIF/P-PIV
Li et al. (2023)	1	streamwise sinusoidal	σ_u/U up to 20.8%	$K_g = 0.021$	mean C_D reduced by 15.2%, lift peak reduced by 17%–19%	P-SP/P-PIF
Li et al. (2023)	1	vertical sinusoidal	σ_w/U up to 21.4%	$K_g = 0.131$	mean C_D reduced by 23.5%, POD first mode from above 90% to 47%	P-SP/P-PIF

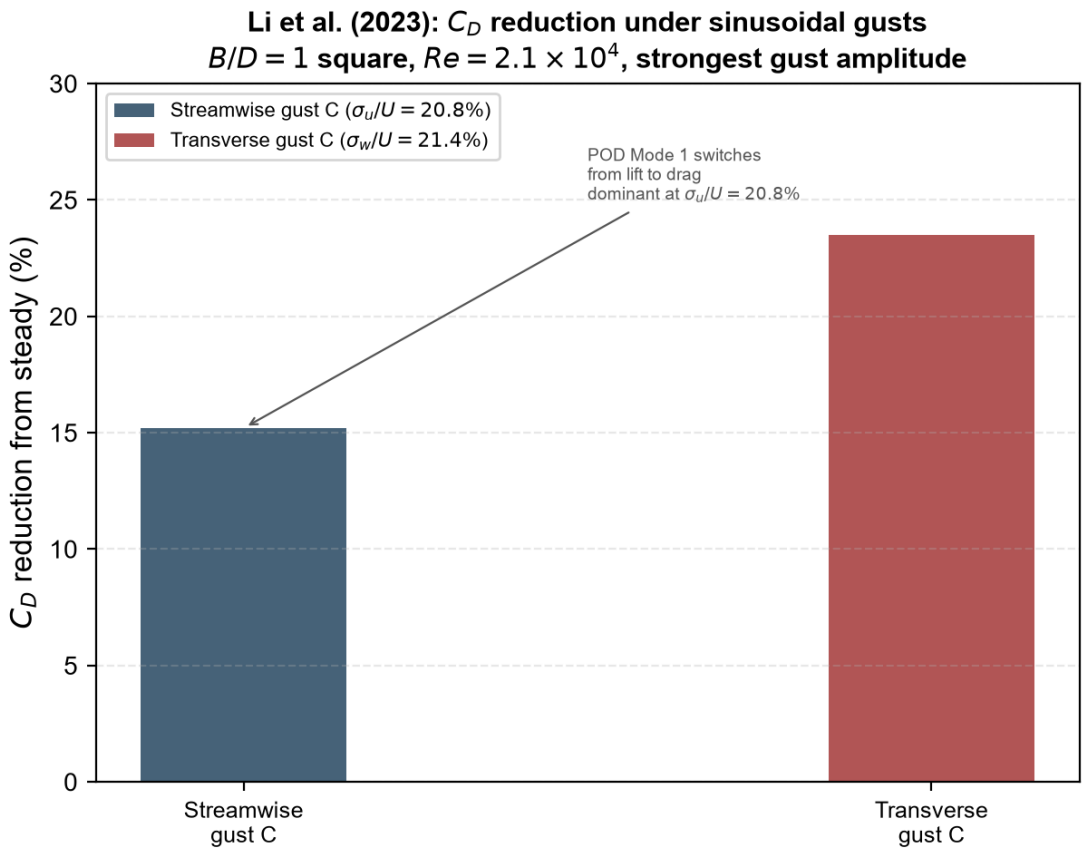


Figure 14: Drag reduction and POD mode variation under streamwise and vertical sinusoidal gusts from Li et al. (2023).

Table 20 summarizes streamwise sinusoidal inflow conditions. The physical sources include Wu et al. (2021) ($B/D = 5$, $f_u = 0.2$ to 1.2 Hz, $\Delta u/U = 0$ to 0.25), Li et al. (2021a) ($B/D = 5$, $f_u = 0.4$ and 1.0 Hz, $\sigma_u/U = 5.5\%$ to 19.2%), and M. Li et al. (2023) ($B/D = 1$, streamwise gust $f_g = 0.4$ Hz, $\sigma_g/U = 5.3\%$ to 20.8%).

Wu et al. (2021) and Li et al. (2021a) both tested the $B/D = 5$ rectangular cylinder under streamwise sinusoidal inflow at similar Reynolds numbers (~ 1.3 to 1.6×10^4), but with different inflow generation mechanisms (multi-fan versus airfoil gust generator), different frequency ranges, and different amplitude parameterizations ($\Delta u/U$ versus σ_u/U). Wu et al. (2021) observed approximately linear growth of C'_D and C'_L with $\Delta u/U$; the POD first two modes became dominated by the inflow-induced component at high amplitudes. Li et al. (2021a) observed a decrease in separation bubble length from $4.0D$ to $2.0D$ and an increase in Strouhal number from 0.111 to 0.179. Both sources indicate that streamwise sinusoidal inflow intensifies pressure fluctuations and modifies the separation bubble topology, but the POD framework of Wu et al. (2021) and the separation bubble length metric of Li et al. (2021a) are not directly interchangeable. The comparison ceiling is L2. Figure 32 shows the formalized extracted data from Brusco et al. (2024) FIG12, used only for same-source deceleration branch verification; the two-source comparison is retained in the text-based faceted discussion.

M. Li et al. (2023) performed streamwise gust experiments on the $B/D = 1$ square cylinder, providing a geometric contrast. The reduced frequency $K_g = 0.021$ (streamwise) is much smaller than the vertical value of $K_g = 0.131$, and the gust length scale $\lambda/D = 300$ is much larger than the model scale. The streamwise gust reduced $\overline{C_D}$ by approximately 15.2% (at $\sigma_u/U = 20.8\%$), a direction opposite to the C_D increase observed by Li et al. (2021a) for the 5:1 cylinder. This difference may originate from the fundamental difference in B/D (fully separated square cylinder versus separated-reattached 5:1 cylinder), or may be influenced by differences in gust length scale and reduced frequency. The comparison ceiling is L1 (geometric mismatch).

5.3 Vertical Sinusoidal Inflow

Table 21 summarizes vertical sinusoidal inflow conditions. The explicit physical sources include the $B/D = 5$ rectangular cylinder from Li et al. (2021b) and the $B/D = 1$ square cylinder from M. Li et al. (2023); the two differ in geometry and response metrics. M. Li et al. (2023) reported at $f_g = 2.5$ Hz, $K_g = 0.131$, and $\sigma_w/U = 3.5\%$ to 21.4% a maximum reduction of approximately 23.5% in $\overline{C_D}$, a decrease in POD first-mode energy from above 90% to 47.0%, and reduced spanwise correlation. Li et al. (2021b) provides pressure and PIV evidence for the 5:1 cylinder under vertical sinusoidal conditions. The two sources cannot be merged into a same-geometry effect size, and each cross-section lacks an independent replication source.

5.4 Multi-Axis Sinusoidal Inflow

Table 22 summarizes multi-axis sinusoidal inflow conditions. Among the sources covered in this review, no test case simultaneously applied both streamwise and vertical sinusoidal forcing. The streamwise and vertical gusts in M. Li et al. (2023) were applied independently in separate test conditions; their combined effect has not been experimentally validated. Table 22 records this gap and notes that the existing evidence is insufficient to support any quantitative statement regarding multi-axis combined effects.

6 Measurement Channel Normalization and Measurement System

6.1 Pressure Channels

Table 16 summarises the surface pressure channel parameters across all sources. Sampling frequencies range from 200 Hz (Wu et al. (2021)) to 1000 Hz (Morello et al. (2025a), Zhang and Liu (2017)), and sensor types include ZOC 33 (Brusco et al. (2022), Brusco et al. (2024), Yang and Mason (2019)), ESP-16HD (Morello et al. (2025a)), DSM 3400 (Wu et al. (2021)), ESP-64HD (Han et al. (2025)), ICP 1/4-inch microphones (Zhang and Liu (2017)), and TFI sensors (M. Li et al. (2023), Li et al. (2021a)). Tubing lengths range from NR (multiple sources) to 1.5 m (Brusco et al. (2022), Brusco et al. (2024), Yang and Mason (2019)) and 0.2 m (Wu et al. (2021), Zhou et al. (2025)). The implementation status of tubing transfer-function corrections is NR in multiple sources. Tap counts range from 30 (Morello et al. (2025a), per face of the square cylinder) to 442 (Han et al. (2025)).

Significant differences exist in the normalisation dynamic pressure definition for C_p. Most sources use $q = 0.5\rho U^2$ (steady reference velocity), but Brusco et al. (2022) and Brusco et al. (2024) use the CWT-extracted instantaneous dynamic pressure $P_{\mathrm{dyn},W}$ in transient analyses, and Yang and Mason (2019) uses an equivalent dynamic pressure based on $(U_s + U_t)/2$ or $(p_s + p_t + 2\sqrt{p_s p_t})/4$. The reference dynamic pressure for C_p in Wu et al. (2021) is $q_{\mathrm{bar}} = 0.5\rho U^2$ (time-averaged), and Li et al. (2021a) likewise uses $q_{\mathrm{bar}} = 0.5\rho U^2$. These normalisation differences must be explicitly noted in cross-source comparisons. The source channel distribution presents the per-source channel matrix for all 55 sources; normalisation differences for pressure channels are governed by the text fields in Table 16.

6.2 Force Channels

Table 17 summarises the force measurement channels. Force source types include: pressure-integrated forces (P-PIF): Brusco et al. (2024) (46-tap integration), Wu et al. (2021) (6-section pressure integration), Yang and Mason (2019) (Eq.2/3), Han et al. (2025) (Type IV, 442-tap integration), M. Li et al. (2023) (strip integration), Li et al. (2021a) (56-tap integration), Zhou et al. (2025) (50-tap integration). Independent balance forces (P-BAL): Vickery (1966) (3-inch active-segment balance), Han et al. (2025) (TB1378K external balance and TH2401A internal balance). Sources without an

independent balance: Brusco et al. (2022) (only two-tap c'_{PL}), Morello et al. (2025a), Wu et al. (2021), Yang and Mason (2019), M. Li et al. (2023), Li et al. (2021a). The absolute difference between P-PIF and P-BAL force coefficients is approximately 10% in Vickery (1966) (PDF p. 7); in Han et al. (2025) both balance types are used simultaneously but the systematic bias between channels is not quantitatively reported (not reported). P-PIF and P-BAL are not interchangeable. The master source table presents the channel counts by evidence tier, and the master source table presents the transient source channel matrix; per-source force source type information is governed by Table 17.

6.3 Numerical Channels

Table 18 summarises the numerical channel parameters. The numerical sources involved include Yu et al. (2013) (LES, $Re = 10^5$, finite volume method), Ohya et al. (1992) (LES, $Re = 10^3$, finite difference/finite volume method), Chiarini and Quadrio (2021) (DNS, $Re = 3000$, finite difference method), Lunghi et al. (2024) (LES, Nek5000 spectral element method, $Re \sim 2 \times 10^4$ to 6.5×10^4), Morello et al. (2025b) (LES, Nek5000, same Reynolds number range), and Li et al. (2025) (numerical hypothesis line, $B/D = \mathrm{N/A}$) (URANS and LES, $Re = 7.5 \times 10^4$, OpenFOAM). The Reynolds numbers across sources span from 10^3 to 10^5, spatial resolutions from $\Delta x \approx 0.0015D$ (Chiarini and Quadrio (2021), near-wall) to $\Delta x/D = 0.125$ (Lunghi et al. (2024) and Morello et al. (2025b), wall), and time steps from $\Delta t \approx 0.0013$ (Chiarini and Quadrio (2021)) to $\Delta t = 0.003$ (Lunghi et al. (2024), Morello et al. (2025b)) or $(1\text{-}2) \times 10^{-3}$ (Ohya et al. (1992)). Subgrid-scale model differences are substantial: Yu et al. (2013) uses the Smagorinsky model (constant NR), Lunghi et al. (2024) and Morello et al. (2025b) use explicit filtering ($w = 0.05$), and Chiarini and Quadrio (2021) is DNS with no subgrid-scale model. The upper bound of cross-numerical-source comparison is limited by differences in Reynolds number, subgrid-scale model, domain size, and boundary conditions. The master source table presents the source type and evidence tier distributions, and the master source table presents the comparison tier distributions; numerical channel parameters are governed by Table 18.

7 Evidence and Gaps in PIV Measurements

Table 5 summarises the PIV channels. Seven sources provide direct PIV data: Moore et al. (2019), Li et al. (2021b), van Oudheusden et al. (2005), van Oudheusden et al. (2008), Zhang and Liu (2017), Lander and Letchford (2016), and Guissart et al. (2022). Among these, Zhang and Liu (2017) reports 6000 snapshots, 1 Hz, $0.05D \times 0.05D$ resolution, and a $12D \times 10D$ field of view, while Guissart et al. (2022) reports 500 to 1500 snapshots, 1 to 2 kHz, $D/53$ to $D/116$ resolution, and three FOVs. The geometry, sampling, and PIV usage of the remaining sources are recorded per-source in Table 5 and are not merged on the basis of shared channel names. Morello et al. (2025a) proposed PIV only in a subsequent plan and explicitly did not implement it; this source is therefore not counted as a PIV source. Cross-source comparison remains limited by differences in geometry, temporal resolution, sample size, and uncertainty definitions.

Table 5: PIV-channel condition matrix. Comparison upper bound: L0–L3, no channel pooling. Vacancies uniformly recorded as not reported.

source id	tier	B over D	flow family	channel identity	normalization or reference	limitation
Moore et al. (2019)	D	1,3,5	steady uniform	P-SP;P-PIV;P-HW	as defined in source; not reported items remain not reported	included
Li et al. (2021b)	D	5	vertical sinusoidal	P-SP;P-PIF;P-PIV;P-VEL	as defined in source; not reported items remain not reported	included
van Oudheusden et al. (2005)	B	1	steady uniform	P-PIV	as defined in source; not reported items remain not reported	included
van Oudheusden et al. (2008)	B	1	steady uniform	P-PIV	as defined in source; not reported items remain not reported	included
Zhang and Liu (2017)	D	3	steady uniform	P-SP;P-PIV	as defined in source; not reported items remain not reported	included
Lander and Letchford (2016)	D	1	steady turbulent	P-SP;P-PIF;P-PIV	as defined in source; not reported items remain not reported	included
Guissart et al. (2022)	D	5	steady uniform	P-PIV	as defined in source; not reported items remain not reported	included

7.1 Steady PIV Evidence for Planned Cross-Sections

Guissart et al. (2022) employed 1 to 2 kHz high-speed PIV on a 5:1 rectangular cylinder ($Re = 2 \times 10^4$) under steady smooth inflow, with spatial resolution of $D/53$ to $D/116$, covering three fields of view (FOVs) and uncertainty $< 4\%$. This source identifies the reattachment point at $x/D = 1.64$. van Oudheusden et al. (2005) and van Oudheusden et al. (2008) each provide PIV flow-field data for a square cylinder under steady smooth inflow. Lander and Letchford (2016) acquired PIV data for a square cylinder under steady turbulent inflow, simultaneously with pressure and pressure-integrated force channels. The PIV from all four sources above is limited to steady inflow and cannot be directly extended to flow-field evolution under accelerating or sinusoidal conditions.

7.2 PIV Evidence under Unsteady Inflow

Li et al. (2021b) simultaneously provides pressure and PIV evidence for a 5:1 rectangular cylinder under vertical sinusoidal inflow and is the only physical source in this review that possesses PIV data under unsteady inflow. The PIV coverage area, temporal resolution, and uncertainty for this source are governed by Table 5. Under monotonic acceleration inflow, the PIV of Morello et al. (2025a) remained at the planning stage and was not implemented. No published PIV data exist for $B/D = 1$, $B/D = 3$, or $B/D = 5$ under accelerating inflow.

7.3 Adjacent-Geometry and Multi-Section PIV Evidence

Zhang and Liu (2017) provides PIV data for blunt flat plates with $L/D = 3$, 6, and 9 under steady smooth inflow, reporting reattachment points at $x/D = 4.75$ and 4.5 for $L/D = 6$ and $L/D = 9$ respectively, with spatial resolution $0.05D \times 0.05D$ and 6000 snapshots. Moore et al. (2019) simultaneously provides pressure, PIV, and hot-wire data for three cross-sections with $B/D = 1$, 3, and 5 under steady smooth inflow and is the only single PIV source covering all three planned cross-section geometries.

7.4 Accelerating-Flow PIV Gaps and Planned Measurements

PIV data under accelerating inflow are entirely absent for all three planned cross-sections of 60×60, 60×180, and 60×300 mm. The existing steady PIV parameters (1 to 2 kHz sampling rate, $D/53$ to $D/116$ resolution, $< 4\%$ uncertainty from Guissart et al. (2022)) and unsteady PIV parameters (vertical sinusoidal pressure plus PIV from Li et al. (2021b)) together form the parametric basis for the planned 300 Hz synchronised PIV and pressure measurements. The planned experiments require synchronous acquisition of surface pressure and PIV under measured $U(t)$, covering all three cross-sections, using consistent coordinate systems, dynamic pressure definitions, and time bases, in order to fill this systematic gap.

8 Separation of Physical and Numerical Evidence

Table 6: Principal quantitative anchor points and evidence roles of numerical studies

Source	B/D	Method	Re	C_D	St	Permitted use
Chiarini and Quadrio (2021)	5	DNS	3000	0.9437	0.1274	low-Re mechanism
Lunghi et al. (2024)	1	LES	1.72–6.54×10^4	2.19 steady	0.10–0.14 transient	mechanism comparison, not a substitute for physical validation
Morello et al. (2025b)	3	LES	1.72–6.54×10^4	1.079 steady	0.145–0.195 transient	mechanism comparison
Morello et al. (2025b)	5	LES	same as above	0.988 steady	0.066–0.178 transient	mechanism comparison
Li et al. (2025)	not reported	URANS/LES	approx. 7.5×10^4	overshoot hypothesis	not reported	numerical hypothesis only, not primary evidence

This review strictly enforces the principle that physical experimental data and numerical simulation data do not share coordinate axes. The source channel distribution uses all 55 sources as rows and annotates direct channels; physical and numerical sources are kept separate in text, tables, and direct data plots. Across the 18 detailed comparison tables and 20 project-generated figures, physical and numerical sources consistently occupy separate panels or rows.

For the $B/D = 1$ condition, the physical sources Brusco et al. (2022), Morello et al. (2025a) (square-cylinder portion), Brusco et al. (2024), Yang and Mason (2019) ($B/D = 1$ portion), and M. Li et al. (2023) provide pressure or pressure-integrated force data, Vickery (1966) simultaneously provides balance force and pressure data, and Robertson et al. (1978) provides pressure data. The numerical sources Lunghi et al. (2024), Yu et al. (2013) ($H/D = 1$ portion), and Ohya et al. (1992) ($d/h \geq 3$ portion) provide numerical force and flow-field data. Cross-facility comparisons among physical sources are limited by differences in turbulence intensity, blockage ratio, sampling frequency, and normalisation, reaching at most L2 or L3 (L4 for same-facility sources). Comparisons between physical and numerical sources are invariably L1 or L0.

For the $B/D = 5$ condition, the physical sources Morello et al. (2025a) (5:1 portion), Wu et al. (2021), Han et al. (2025), Guissart et al. (2022), Li et al. (2021a), Li et al. (2021b), and Zhou et al. (2025) cover a range from $Re \sim 10^4$ to 10^6, but the $U(t)$ type, normalisation method, and response quantity definitions differ substantially across sources. The numerical sources Chiarini and Quadrio (2021) ($Re = 3000$), Morello et al. (2025b) (5:1 portion), and the hypothesis channel Li et al. (2025) (numerical hypothesis line, $B/D = \mathrm{N/A}$) provide numerical reference, but the Reynolds number disparity limits physical/numerical comparison to L1-level qualitative adjacency only.

Figure 37 and Figure 38 present the formalised extracted data from Wu et al. (2021) FIG09 and FIG10a respectively, used solely for within-source verification of the streamwise sinusoidal inflow condition, and do not constitute a physical versus numerical baseline comparison.

9 Bounded Experimental Implications and Planned Cross-Section Mapping

9.1 Bounded Implications of Current Evidence

Based on the condition-level reconstruction and comparison upper-bound constraints described above, this review distils the following bounded implications within declared boundaries; each implication is accompanied by its supporting sources, comparison upper bound, residual limitations, and unverified assumptions.

First, multiple physical experiments on the square cylinder ($B/D = 1$) indicate that accelerating inflow alters vortex shedding frequency organisation or aerodynamic forces, though the evidence carriers differ. Brusco et al. (2022) explicitly reports constant-frequency time cells and St reduction, Morello et al. (2025a) reports St ranges within time cells and Reynolds number dependence, Brusco et al. (2024) reports a reduction in pressure-integrated drag without reporting the corresponding St, and Yang and Mason (2019) reports reductions of varying magnitude in C_D, lift fluctuation, and St. Cross-facility comparison can only compare separately reported response quantities; it is not possible to generalise all sources as having jointly observed constant-frequency time cells. The LES source Lunghi et al. (2024) provides only numerical mechanism adjacency.

Second, for the 5:1 rectangular cylinder ($B/D = 5$), the effects of accelerating and sinusoidal inflow on separation bubble topology, pressure fluctuations, and force coefficients depend on the combination of inflow type, acceleration magnitude, and turbulence intensity. Morello et al. (2025a) observed C_{Dp} below the steady-state value at low turbulence intensity, while Zhou et al. (2025) observed C_D above the steady-state value at high turbulence intensity, in the opposite direction. Li et al. (2021a) found that streamwise sinusoidal inflow shortens the separation bubble length and increases the Strouhal number. These seemingly contradictory results may stem from non-linear modulation of the separation-reattachment mechanism by turbulence intensity, but the existing evidence is insufficient to establish a unified parametric relationship.

Third, vertical sinusoidal inflow (only M. Li et al. (2023), $B/D = 1$) disrupts vortex shedding structures more strongly than streamwise sinusoidal inflow, and the sharp drop in spanwise correlation (decreasing to 0.29 at $\delta/D = 1.25$) indicates that vertical perturbations introduce significant three-dimensional effects. This conclusion is based solely on data from a single source at a single Reynolds number (2.1×10^4).

Fourth, the reduction or overshoot of coefficients normalised by instantaneous velocity under accelerating inflow does not equate to exceedance of overall loads or pressures. Yang and Mason (2019) explicitly states that no overshoot of overall loads beyond the target-velocity steady-state condition was observed in any accelerating test (PDF p. 15),

while also noting that generalisation of this conclusion requires more extensive investigation (PDF p. 15).

Fifth, the 5:1 rectangular cylinder may undergo a flow-regime transition in the $Re \sim 3.5 \times 10^5$ to 5×10^5 range (Han et al. (2025) observed an abrupt drop in Strouhal number), qualitatively similar to the St jump observed in the numerical simulation of Morello et al. (2025b) at $Re = 40000$ to 50000, but the Reynolds numbers differ by an order of magnitude.

9.2 Planned Cross-Section Mapping

Table 23 maps the three planned experimental cross-sections of 60×60 mm, 60×180 mm, and 60×300 mm to existing measured evidence, gaps, and comparison upper bounds. The 60×60 mm cross-section ($B/D = 1$) possesses the richest measured evidence, including accelerating and sinusoidal inflow data from Brusco et al. (2022), Morello et al. (2025a) (square-cylinder portion), Brusco et al. (2024), Yang and Mason (2019) ($B/D = 1$), and M. Li et al. (2023), as well as steady baselines from Vickery (1966) and Robertson et al. (1978). The principal gap is that vertical sinusoidal inflow has only a single source (M. Li et al. (2023)), and no multi-axis combined inflow data exist.

The 60×180 mm cross-section ($B/D = 3$) has less measured evidence; only the physical source Yang and Mason (2019) ($B/D = 3$) provides data under accelerating inflow (blockage ratio 7.8%, uncorrected), and Da Matha Sant'Anna et al. (1988) provides pressure distributions under steady $H/D = 3$ conditions. The numerical source Morello et al. (2025b) provides LES data for the 3:1 cylinder. The principal gaps are the absence of sinusoidal inflow data, PIV data, and high-Reynolds-number data.

The 60×300 mm cross-section ($B/D = 5$) possesses relatively rich measured evidence, encompassing steady conditions (Han et al. (2025), Guissart et al. (2022), Li et al. (2021a) smooth-inflow baselines), monotonic acceleration (Morello et al. (2025a) 5:1 portion, Zhou et al. (2025)), streamwise sinusoidal (Wu et al. (2021), Li et al. (2021a)), and vertical sinusoidal (Li et al. (2021b)). Numerical sources include Chiarini and Quadrio (2021) (DNS, $Re = 3000$) and Morello et al. (2025b) (LES). The principal gaps are the lack of independent replicate sources with the same geometry for each sinusoidal direction, the absence of synchronous multi-axis inflow data, and the absence of accelerating inflow data at $Re > 10^6$.

The master source table and master source table present the aspect ratio, evidence tier, and inflow family coverage. Figure 47 and Figure 48 present the formalised extracted data from M. Li et al. (2023) F06-A and F07-A respectively, used solely for within-source verification, and are not planned cross-section coverage maps.

10 Synthesis and Discussion

This review has conducted a condition-level reconstruction audit of 55 steady and unsteady rectangular-cylinder aerodynamic sources. Six Tier-A sources passed both post-hoc condition verification (6/6) and condition-level independent re-examination (6/6), achieving complete condition reconstruction; 12 anchor sources provide critical comparison anchor points for the steady, mechanism, and transient families. Across all 55 sources, cross-source numerical values are used only for internal technical comparison under explicitly matched conditions and channels.

The condition-level audit reveals several core findings. Systematic differences exist across sources in Reynolds number definition (choice of characteristic length and reference velocity), dynamic pressure normalisation, force coefficient source channel (P-SP, P-PIF, P-BAL, N-F), and non-dimensionalisation of acceleration parameters; these differences restrict most cross-source comparisons to L2 or below. For the square cylinder ($B/D = 1$), different facilities separately report constant-frequency time cells, St variations, or pressure-integrated force variations, and these response quantities cannot serve as proxies for one another; inter-facility differences in turbulence intensity, blockage ratio, $U(t)$ waveform, and measurement channel also preclude quantitative merging. For the 5:1 rectangular cylinder ($B/D = 5$), the evidence indicates that the effect direction of accelerating inflow on drag coefficient may depend on turbulence intensity: C_D decreases at low turbulence intensity and increases at high turbulence intensity, a contradiction that remains unresolved in the existing evidence. Vertical sinusoidal inflow has only a single source for $B/D = 1$ and $B/D = 5$ respectively (M. Li et al. (2023) and Li et al. (2021b)), and independent replicate evidence at the same geometry is absent; experimental or numerical evidence for synchronous multi-axis combined inflow is also entirely absent. Some physical and numerical sources separately report Strouhal number trends in approximately the same di-

rection, but differences in condition definitions and channel types prevent these from being interpreted as quantitative agreement or mutual validation; C_D baseline discrepancies may be jointly attributable to turbulence intensity, blockage ratio, numerical domain size, and normalisation method.

All analyses and inferences in this review are restricted to the scope of an internal academic review draft and do not constitute facility safety determinations, hardware engineering application authorisations, engineering design conclusions, or public dissemination permissions. The source annotations, comparison upper-bound assignments, and interpretation boundaries applied to direct source figures, formalised extraction overlays, newly generated cross-paper numerical figures, and detailed comparison tables collectively form the evidence boundary framework of this review.

Physical experiment evidence. Pressure and force experiments have revealed multiple facets of reattachment, vortex shedding frequency, and pressure recovery mechanisms. Brusco et al. (2022) and Brusco et al. (2024), at the same facility, separately report constant-frequency time cells and a reduction in pressure-integrated drag for the square cylinder under accelerating inflow; Yang and Mason (2019) observes cross-section-dependent coefficient reduction and overshoot at $B/D = 1$ and $B/D = 3$; and Morello et al. (2025a) simultaneously tracks time cells and the migration of pressure fluctuation peaks in both the square and 5:1 cylinders. For the 5:1 cross-section, Zhou et al. (2025) reports a 13.1% increase in C_D at high turbulence intensity, opposite in direction to the below-steady-state C_D observed by Morello et al. (2025a) at low turbulence intensity, indicating that the modulation of the separation-reattachment mechanism by turbulence intensity is a currently unresolved core contradiction. The streamwise sinusoidal experiments of Li et al. (2021a) and Wu et al. (2021) respectively reveal separation bubble shortening and non-monotonic frequency response of aerodynamic admittance, while the vertical sinusoidal experiment of M. Li et al. (2023) confirms a sharp drop in spanwise correlation. These physical experiments collectively support the qualitative conclusion that accelerating inflow alters vortex shedding organisation and force coefficients, but they do not support a unified transient correction relationship across cross-sections and inflow families.

PIV evidence. Guissart et al. (2022) identifies the reattachment point at $x/D = 1.64$ ($Re = 2 \times 10^4$) for the 5:1 rectangular cylinder under steady inflow; van Oudheusden et al. (2005) and van Oudheusden et al. (2008) provide square-cylinder steady PIV flow fields; and Lander and Letchford (2016) acquires square-cylinder PIV data under turbulent inflow. Li et al. (2021b) is the only source that simultaneously provides pressure and PIV under unsteady inflow (vertical sinusoidal), and its 5:1 rectangular cylinder data support the correlation between pressure fluctuation peak location and flow-field reattachment point. Zhang and Liu (2017) reports a reattachment point at $x/D = 4.75$ on a blunt flat plate with $L/D = 6$. The existing PIV evidence does not support a quantitative description of reattachment point migration rate under accelerating inflow, nor does it support flow-field topology claims for $B/D = 1$ or $B/D = 3$ under any transient inflow.

CFD evidence. The LES of Lunghi et al. (2024) reproduces the time cell structure in the square cylinder under accelerating inflow, with St varying in the 0.10 to 0.14 range and a steady-state $C_{D,\text{sim}} = 2.19$ that exceeds the experimental value of 2.06, a discrepancy attributed to differences in inflow turbulence intensity. The LES of Morello et al. (2025b) covers $B/D = 3$ and $B/D = 5$ under accelerating inflow, reporting St from 0.066 to 0.178 for the 5:1 cylinder and St from 0.145 to 0.195 for the 3:1 cylinder. The DNS of Chiarini and Quadrio (2021) at $Re = 3000$ provides a low-Re mechanism reference for the 5:1 cylinder, with the authors stating that the low Re and finite integration time limit the conclusions. The overshoot numerics of Li et al. (2025) remain designated as a mechanism hypothesis. Numerical studies support the qualitative mechanism hypotheses of segmented frequency behaviour and aspect-ratio-dependent differences in transient response, but cannot substitute for physical validation, cannot share coordinate axes with physical sources, and cannot fill the gap in $B/D = 3$ physical evidence.

11 Conditions and Quantitative Evidence of Key Transient Experiments

11.1 Brusco et al. (2022), Time Cells and Frequency Jumps during Monotonic Acceleration and Deceleration

This source organises transient vortex shedding from a square cylinder into constant-frequency time cells and discontinuous jumps. A quantitative interpretation must simultaneously retain the sign of the acceleration, the repeat index, the instantaneous-frequency extraction method, and the steady-state reference. The pressure-difference quantity c_{DPL} is derived from two pressure taps and is not equivalent to the sectional lift coefficient. The original text reports that a single repeat exhibits an abrupt jump between approximately 20 Hz and approximately 25.7 Hz, and reports a finite-amplitude variation in the ensemble-averaged Strouhal number. The source-constrained time–frequency curves can be used for within-source kernel pairing but do not constitute a universal frequency correction across cross-sections.

> *"the black dots in Fig. 9a and b estimate the instantaneous frequency to be locked on ~ 20 Hz (τ < 13.8 s), before abruptly swapping to ~ 25.7 Hz (τ > 13.8 s)"*
>
> Source Brusco et al. (2022), PDF p. 10, print p. 10, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: UF1 exhibits an abrupt lock-on frequency transition from approximately 20 Hz to approximately 25.7 Hz within a single repeat.

> *"its maximum drop, around 8%, is observed in correspondence of the sixth reference wind speed U6, when up-crossed, for an acceleration < a>W = 7.11 m/s^2"*
>
> Source Brusco et al. (2022), PDF p. 14, print p. 14, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: the ensemble-averaged Strouhal number exhibits a maximum drop of approximately 8% at the up-crossing of U6.

Table 7: Brusco et al. (2022) high-completeness transient condition and evidence table. Comparison ceiling: within-source L0-L4. Absent entries are uniformly recorded as not reported.

source id	condition field	source recorded content	status
Brusco et al. (2022)	methods and configuration	CWT uses a complex Morlet wavelet, Eq.(1) : ψ(t) = e^{i ω0 t} e^{-t^2/2}.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	methods and configuration	HT instantaneous frequency Eq.(2) : n_{x,H}(t) = (1/2π) dφ_x(t)/dt.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	methods and configuration	CWT calibration band for cΔPL spans 5.5 Hz to 55.1 Hz, centre frequency ω0 = 6π.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	methods and configuration	After a first estimate of the steady-state Strouhal number, the wavelet analysis is repeated within a 2.5 Hz narrow band around the original peak for refinement.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	methods and configuration	ω0 sensitivity test covers 2π, 4π, 6π; 6π is selected.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	inflow and Reynolds-number definitions	Steady-state reference wind speed U range: 4.3 m/s to 15.9 m/s.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	inflow and Reynolds-number definitions	Reynolds-number definition (p. 3): Re = U b / ν, where ν is taken as 1.5 x 10^-5 m^2/s.	recorded within source; not reported items remain not reported

Table 7: Brusco et al. (2022) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Brusco et al. (2022)	inflow and Reynolds-number definitions	Corresponding Re range: 1.72 x 10^4 to 6.34 x 10^4.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	inflow and Reynolds-number definitions	Reference dynamic-pressure definition: ”it was defined as the difference between the pressure recorded by the C06 pressure tap and the average mean static pressure recorded by P2 and P3”.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	inflow and Reynolds-number definitions	Pitot-static tube arrangement: P2 and P3 are located 20 cm above the model axis, on opposite sides of the centreline, with tube openings strictly aligned with the model windward face; P1 measures total pressure 2 b upstream of the model, closer to the inlet section.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	inflow and Reynolds-number definitions	The time-varying mean wind speed U_W for unsteady conditions is back-calculated from the dynamically filtered P_dyn,W; acceleration a_W is obtained by simple numerical differentiation of U_W.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	inflow and Reynolds-number definitions	No instantaneous Reynolds-number range is separately reported for unsteady conditions (not reported).	recorded within source; not reported items remain not reported
Brusco et al. (2022)	inflow and Reynolds-number definitions	Air density, temperature, and atmospheric pressure are all not reported; ν is an ”assumed value” rather than a measured quantity.	recorded within source; not reported items remain not reported
Brusco et al. (2022)	measurement channels inventory	cΔPL (steady, Eq.3) = [P_ C18(t) - P_C40(t)] / $\bar{P}$_dyn	recorded within source; not reported items remain not reported
Brusco et al. (2022)	measurement channels inventory	cΔPL (unsteady, Eq.4) = [P_ C18(t) - P_C40(t)] / P_ dyn,W(t)	recorded within source; not reported items remain not reported
Brusco et al. (2022)	measurement channels inventory	cΔPL,W: narrow-band asymptotic signal obtained by CWT dynamic filtering of cΔPL	recorded within source; not reported items remain not reported
Brusco et al. (2022)	measurement channels inventory	S_W (Eq.5) = n_{cΔPL}(t) b / U_W(t)	recorded within source; not reported items remain not reported

Table 7: Brusco et al. (2022) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Brusco et al. (2022)	sampling, synchronization and filtering	Pressure scanner ZOC 33 with built-in 45 kHz high-speed multiplexer.	recorded within source; not reported items remain not reported

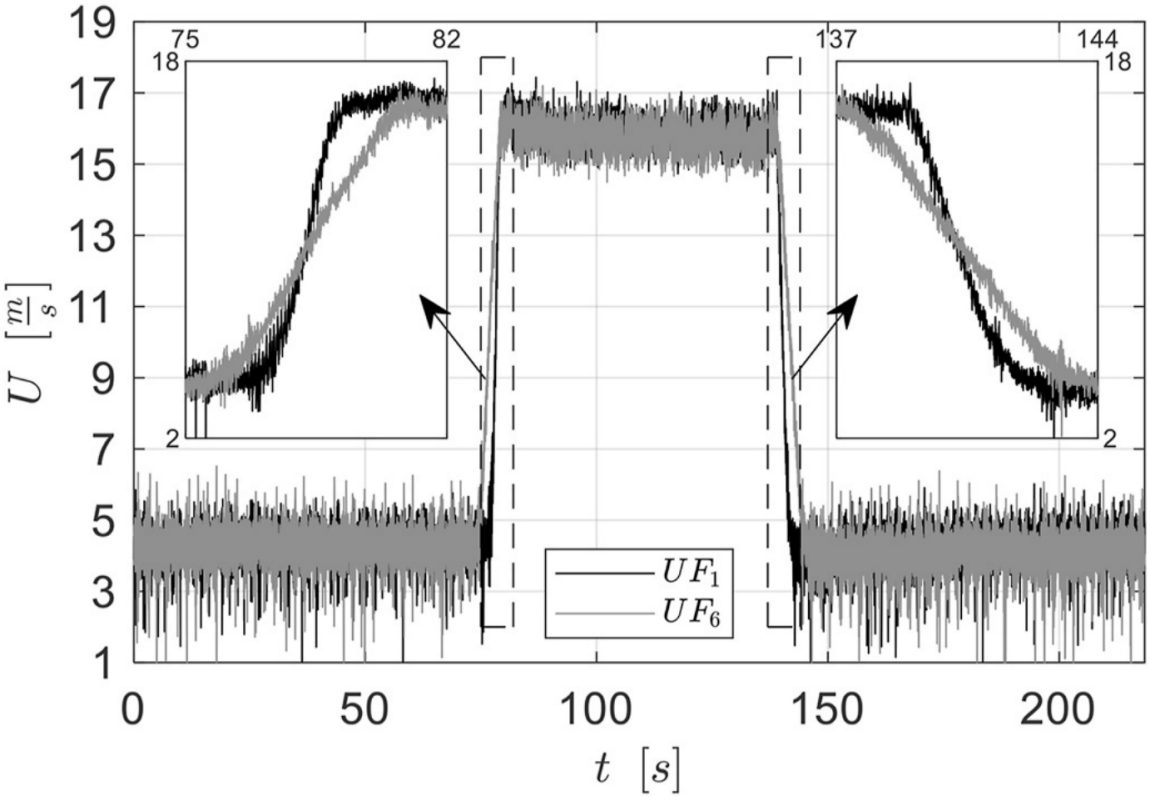


Figure 15: Source figure excerpt. Source Brusco et al. (2022), original figure/table Fig. 4, panel single, PDF p. 4, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

The closest physical neighbour source is Zhou et al. (2025); both discuss acceleration and deceleration, but differ in cross-sectional ratio, pressure tap layout, and normalisation, limiting the comparison ceiling to L2. The numerical neighbour source Lunghi et al. (2024) reuses similar acceleration conditions to examine time cells but can only provide numerical-mechanism adjacency and cannot validate the physical experiment.

11.2 Morello et al. (2025a), Time Cells, Pressure-Fluctuation Peak Location, and Reattachment under Monotonic Acceleration of Square and 5:1 Rectangular Cylinders

This source separates, within the same study, the square cylinder and the 5:1 rectangular cylinder, and further separates the WTG and WTC acceleration profiles. The duration of time cells shortens as the rate of velocity variation increases; for the 5:1 cross-section, the peak location of pressure fluctuations shifts downstream in most unsteady conditions,

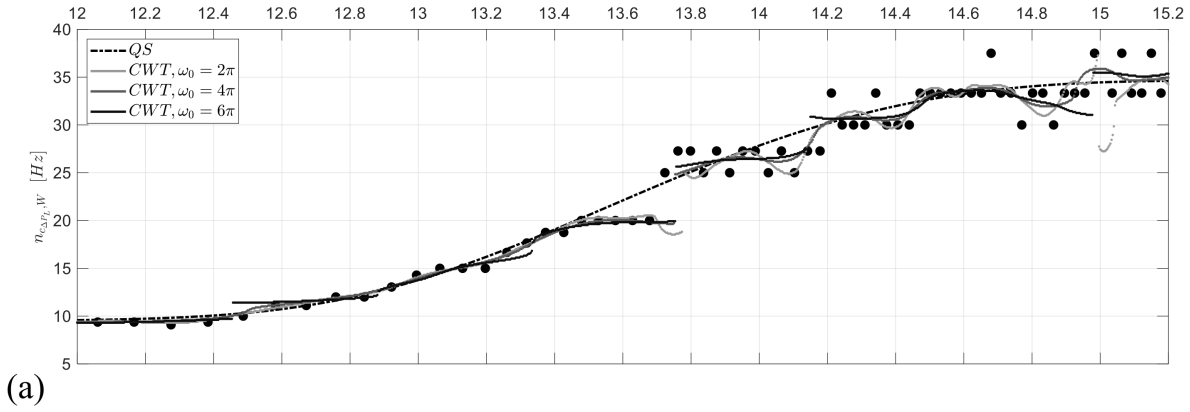


Figure 16: Source figure excerpt. Source Brusco et al. (2022), original figure/table Fig. 9, panel a, PDF p. 10, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

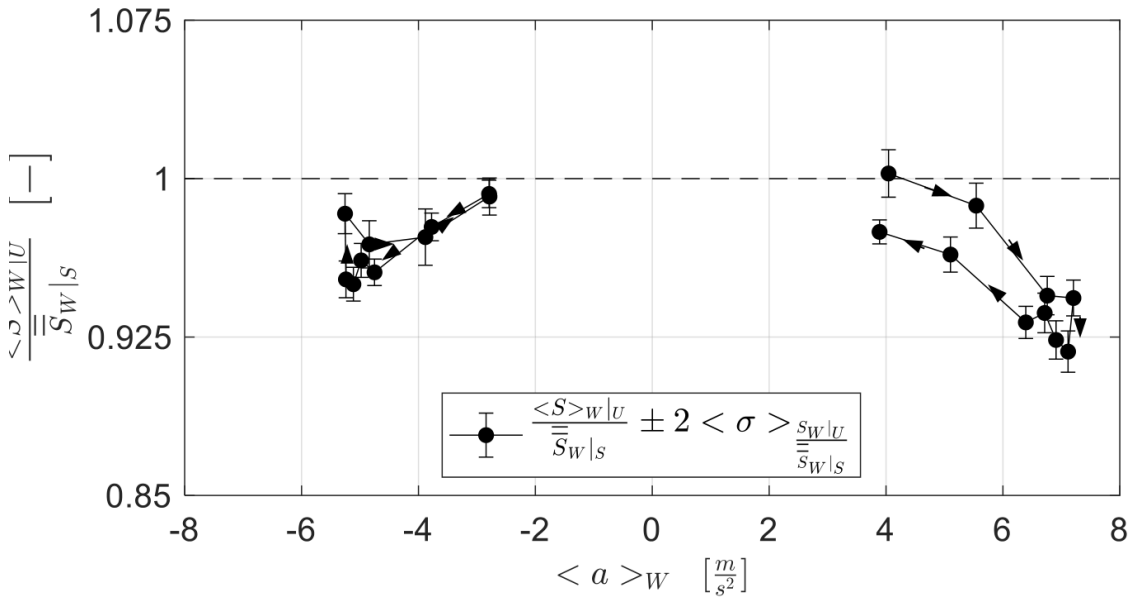


Figure 17: Source figure excerpt. Source Brusco et al. (2022), original figure/table Fig. 16, panel single, PDF p. 14, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

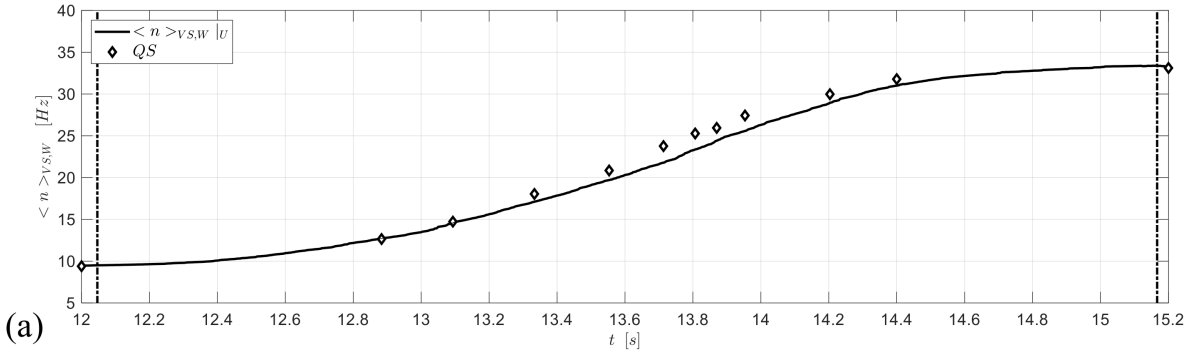
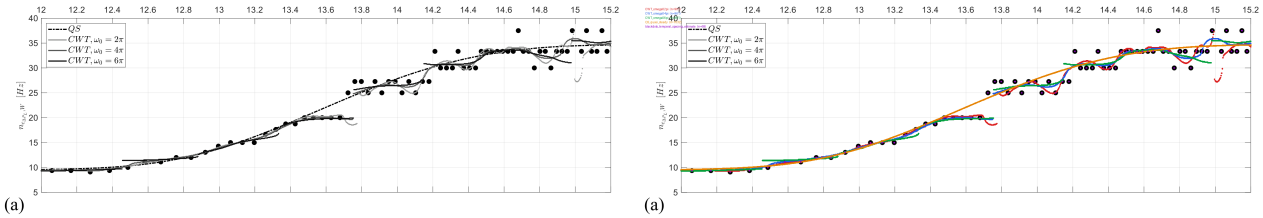
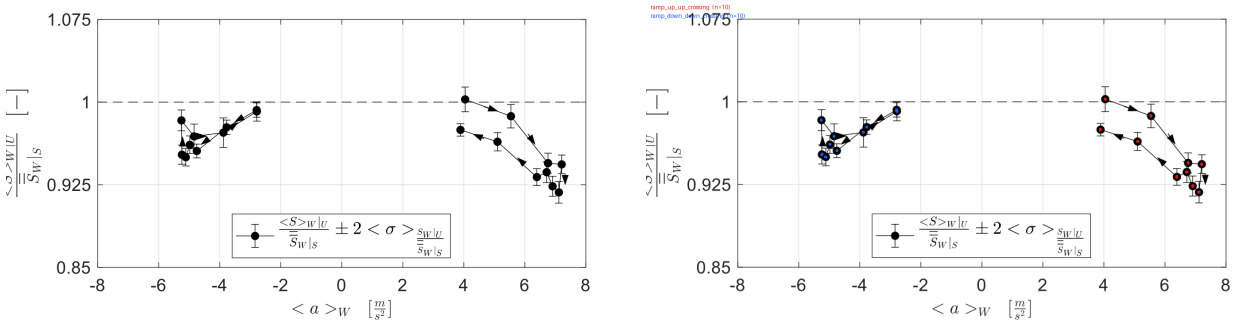


Figure 18: Source figure excerpt. Source Brusco et al. (2022), original figure/table Fig. 18, panel a, PDF p. 15, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.



Source figure Source-constrained extraction overlay

Figure 19: Source figure and source-constrained extraction overlay shown side by side. Source Brusco et al. (2022), PDF p. 10. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.



Source figure Source-constrained extraction overlay

Figure 20: Source figure and source-constrained extraction overlay shown side by side. Source Brusco et al. (2022), PDF p. 14. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

which the authors interpret as a change in the reattachment position. Any threshold statement is constrained by the profile definition, the cross-sectional ratio, and the acceleration normalisation specific to this source.

> *"The cells do not have uniform duration in time; their temporal length tends to decrease as the rate of velocity variation increases."*
>
> Source Morello et al. (2025a), PDF p. 10, print p. not reported, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: constant-frequency time cells have non-uniform durations and tend to shorten as the rate of velocity variation increases.

> *"Almost all unsteady cases exhibit a slight downstream shift in the position where pressure fluctuations reach their maximum, and this result indicates a downstream movement of the flow reattachment point along the lateral surface."*
>
> Source Morello et al. (2025a), PDF p. 23, print p. not reported, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: for the 5:1 rectangular cylinder, the peak location of pressure fluctuations shifts slightly downstream in most unsteady conditions, interpreted by the authors as a downstream migration of the reattachment point.

Table 8: Morello et al. (2025a) high-completeness transient condition and evidence table. Comparison ceiling: within-source L0-L4. Absent entries are uniformly recorded as not reported.

source id	condition field	source recorded content	status
Morello et al. (2025a)	WTG(Gaussian-type)	Velocity rises monotonically following a standard Gaussian cumulative distribution function shape; the acceleration is bell-shaped.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	WTC(constant-magnitude)	Velocity rises linearly; the acceleration forms a rectangular plateau.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	Re = u∞·D/ν (p. 2 verbatim	The Reynolds number, defined as Re = u ∞ D/ν). Here u∞ is the instantaneous free-stream velocity, D is the cylinder depth, and ν is the kinematic viscosity of air.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	main Re range	17200 ≤ Re ≤ 65360, shared by both cross-sections. p. 6 explicitly states that, unless otherwise noted, results in each section pertain to this range.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	extended Re range (0.500x intensity only)	Square cylinder Re = 60430 to 108590; 5:1 rectangular cylinder Re = 41230 to 89390.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	inflow and Reynolds-number definitions	Free-stream velocity measured by a Pitot tube (p. 9: calculated from the free-stream velocity measured by the Pitot tube). The legend in Figs. 17a to 22b directly labels the black line as Pitot.	recorded within source; not reported items remain not reported

Table 8: Morello et al. (2025a) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Morello et al. (2025a)	two normalisations exist for the dimensionless acceleration; this review uses both interchangeably and they must be strictly distinguished	not reported	recorded within source; not reported items remain not reported
Morello et al. (2025a)	inflow and Reynolds-number definitions	Free-stream turbulence intensity 0.9%.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	inflow and Reynolds-number definitions	The paper states that the WTG profile was designed following the apparatus and inflow methodology of Brusco et al. (2022) and Brusco et al. (2024) to approximate acceleration conditions observed in the atmosphere, and cites Brusco et al. (2022) to indicate that the WTG flow has qualitative adjacency with full-scale downburst measurements in terms of slowly varying characteristics.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	square cylinder	7 on the front face, 7 on the rear face, 8 on each lateral face, totalling 30.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	5:1 rectangular cylinder	9 on the front face, 9 on the rear face, 27 on each lateral face, totalling 72.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	sampling frequency	1000 Hz (p. 3 verbatim: The sampling frequency is set at 1000 Hz, which easily resolves vortex shedding from the studied configurations).	recorded within source; not reported items remain not reported

Table 8: Morello et al. (2025a) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Morello et al. (2025a)	reference-pressure synchronisation	The pressure sensors inside the model share the same static pressure as the Pitot tube used to determine the free-stream velocity; this is achieved by connecting the Pitot-tube static-pressure port to the reference-pressure port of the pressure sensors via tubing. This is the only synchronisation mechanism explicitly described in the paper, and it pertains to aerodynamic referencing rather than time-base synchronisation.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	not reported	The time-synchronisation scheme, trigger method, and time-delay calibration between the pressure-scanner sampling clock and the Pitot-tube velocity-signal acquisition are not reported.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	not reported	No digital or analogue filters are reported (no cut-off frequency, no filter order, no anti-aliasing filter description).	recorded within source; not reported items remain not reported
Morello et al. (2025a)	not reported	The tubing transfer function correction, tube length, and tube diameter are not reported. For a configuration with 1000 Hz sampling and embedded pressure scanners, this is a clearly missing methodological element.	recorded within source; not reported items remain not reported
Morello et al. (2025a)	not reported	The total sampling duration for each case is not reported (it can only be indirectly estimated from the T-axis extent of each figure and the definition of T; this card does not perform such estimation).	recorded within source; not reported items remain not reported
Morello et al. (2025a)	number of repeats	Only Fig. 8 explicitly uses 10 realizations for ensemble averaging. The number of repeats for all other results is not reported.	recorded within source; not reported items remain not reported

The closest physical neighbour source is Yang and Mason (2019), but the cross-section combinations, inflow definitions, and acceleration parameters of the two sources cannot be directly converted, so only facet-wise comparisons are possible. The numerical neighbour source Morello et al. (2025b) covers numerical acceleration for 3:1 and 5:1 cross-sections but cannot be co-axially aligned with physical pressure channels, nor can it supplement the physical measurements.

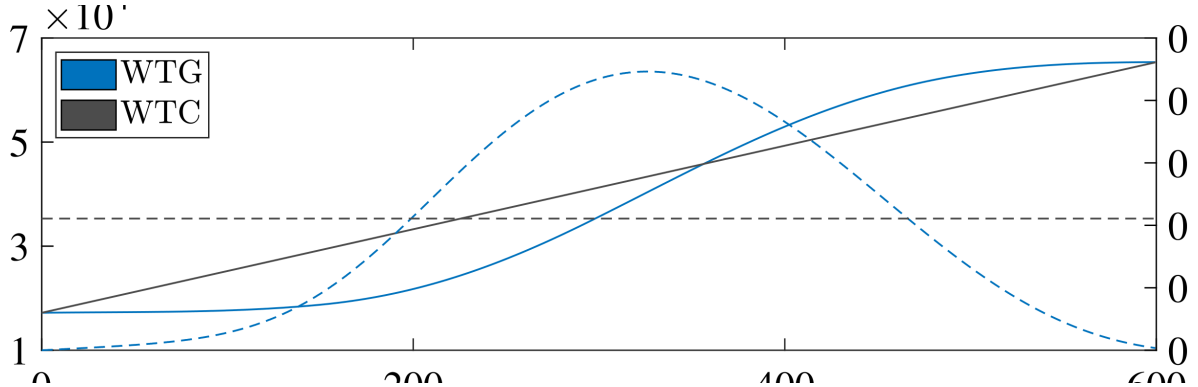


Figure 21: Source figure excerpt. Source Morello et al. (2025a), original figure/table Fig. 3, panel single, PDF p. 5, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

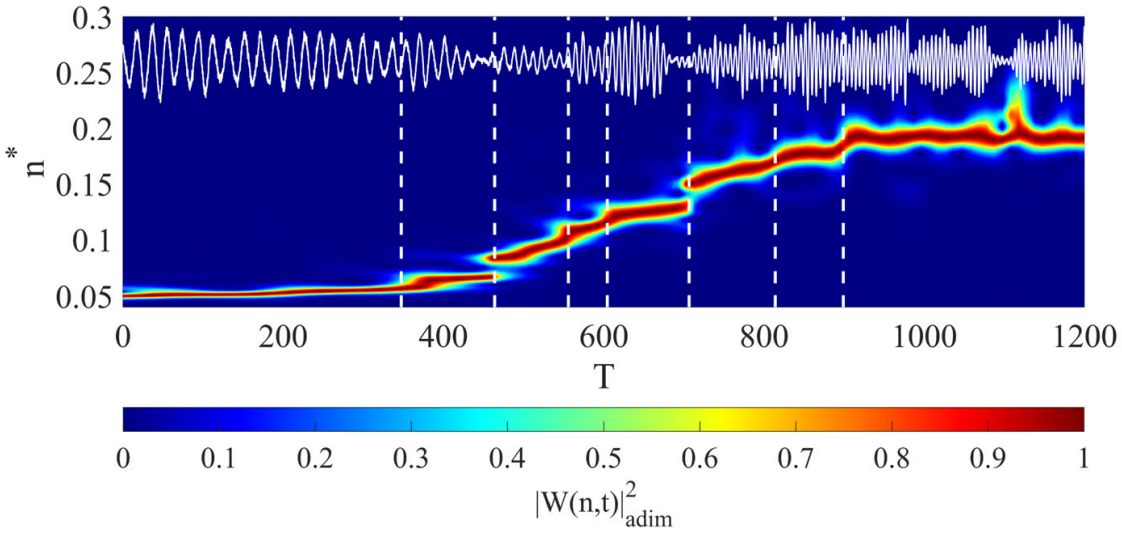


Figure 22: Source figure excerpt. Source Morello et al. (2025a), original figure/table Fig. 6, panel a, PDF p. 8, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

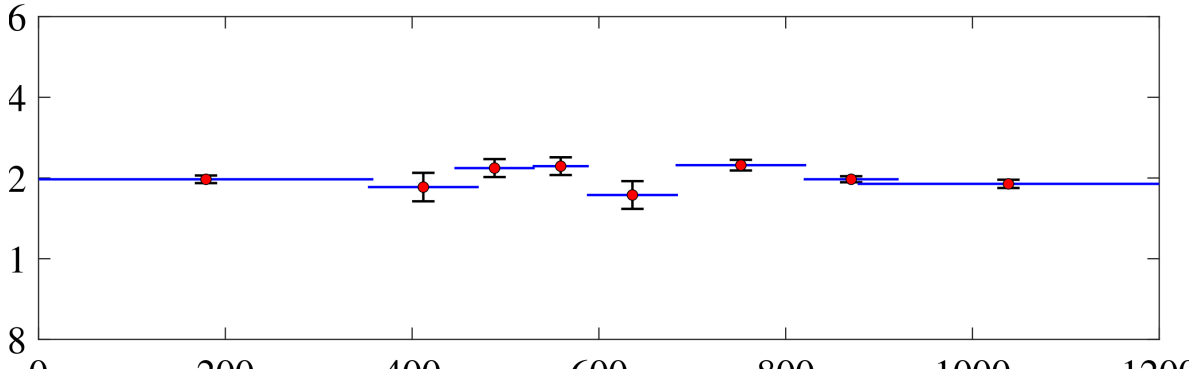


Figure 23: Source figure excerpt. Source Morello et al. (2025a), original figure/table Fig. 8, panel a, PDF p. 10, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

11.3 Brusco et al. (2024), Local Pressure and Pressure-Integrated Drag in the Acceleration and Deceleration Branches of a Downburst-Like Transient

This source establishes ensemble statistics for the UF condition from 90 repeats and strictly separates the ramp-up and ramp-down branches. The original text reports, at U8, an ensemble-mean pressure difference of 187.7 Pa with a standard deviation of 0.60 Pa and a steady-state reference of 204.3 Pa during the acceleration segment, and simultaneously notes that the increase in local pressure coefficients and the decrease in overall drag are not the same quantity. Figures and extracted points can only be used for within-source branch verification and must not be distilled into universal correction factors.

> *"the ensemble mean of the values gathered from the 90 repeats under the ramp-up condition is 187.7 Pa, with a standard deviation of 0.60 Pa, while the corresponding steady reference pressure difference is 204.3 Pa"*
>
> Source Brusco et al. (2024), PDF p. 14, print p. 14, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: the ensemble-mean pressure difference of UF1 at U8 during the acceleration segment is lower than the corresponding steady-state reference pressure difference.

> *"the maximum increase of the time-varying mean pressure coefficient <c>P29,W (14% in the ramp-up, and 7% in the ramp-down) is almost double than the corresponding reductions of the aerodynamic drag"*
>
> Source Brusco et al. (2024), PDF p. 18, print p. 18, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: the maximum increase in the C29 time-varying mean pressure coefficient is 14% and 7% in the acceleration and deceleration segments, respectively, approximately double the corresponding drag reduction.

Table 9: Brusco et al. (2024) high-completeness transient condition and evidence table. Comparison ceiling: within-source L0-L4. Absent entries are uniformly recorded as not reported.

source id	condition field	source recorded content	status
Brusco et al. (2024)	methods and configuration	Steady: reference wind speed U from 4.3 m/s to 15.9 m/s, comprising 12 settings SF1 to SF12, each with at least 25 repeats, each repeat containing 2^16 samples, i.e. 218.4 s.	recorded within source; not reported items remain not reported

Table 9: Brusco et al. (2024) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Brusco et al. (2024)	methods and configuration	Unsteady: 13 conditions in total, each with no fewer than 30 repeats; each test likewise contains 2^16 samples with the same duration as steady tests. Conditions are described by Ulow, Uhigh, the velocity-change time Δτ, and the high-speed plateau duration Tb (Fig. 2).	recorded within source; not reported items remain not reported
Brusco et al. (2024)	inflow and Reynolds-number definitions	Reynolds-number definition (p. 5, inline formula): Re = U b / ν, where the characteristic length is the square-cylinder side length b = 6 cm and the kinematic viscosity is assumed as ν = 1.5e-5 m2/s. Original text: ”where ν is assumed as 1.5 · 10^-5 m2/s”. Air temperature, atmospheric pressure, and density are all not printed, recorded as not reported.	recorded within source; not reported items remain not reported
Brusco et al. (2024)	inflow and Reynolds-number definitions	Steady Re range: 17,200 to 63,440 (corresponding to U from 4.3 to 15.9 m/s). The ”Present” row in Table 1 writes the same range as 1.72 to 6.34 e4; the two entries are consistent.	recorded within source; not reported items remain not reported
Brusco et al. (2024)	inflow and Reynolds-number definitions	Unsteady Re: the paper does not separately report unsteady Re. It can be inferred from Ulow and Uhigh in Table 2 together with the above definition, but this review does not print that inference, and the entry is therefore recorded as not reported within the source.	recorded within source; not reported items remain not reported

Table 9: Brusco et al. (2024) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Brusco et al. (2024)	inflow and Reynolds-number definitions	Reference dynamic pressure Pdyn definition (p. 5): taken as the difference between the C06 tap pressure and the average static pressure of P2 and P3. The authors state that this choice is motivated by good repeatability, shared acquisition with load data to ensure simultaneity, and all three signals lying in the same vertical plane. This means the reference velocity is not an independent anemometer channel but is partly defined by a model-surface tap, a methodological characteristic recorded here.	recorded within source; not reported items remain not reported
Brusco et al. (2024)	inflow and Reynolds-number definitions	Turbulence characteristics: only the mean turbulence intensity of approximately 2.5% is given (Table 1 ”Present” row writes 2). Turbulence integral scale, power spectrum, and anisotropy are not reported, recorded as not reported.	recorded within source; not reported items remain not reported
Brusco et al. (2024)	inflow and Reynolds-number definitions	Full-scale benchmarking (p. 13): the accelerations reproduced in the wind tunnel are generally higher than full-scale values, but the dimensionless acceleration parameter Ka,W matches well at the maximum time-varying wind speed. The geometric scaling required for similarity is approximately 35 for the Livorno ramp-up and approximately 45 for the La Spezia ramp-down.	recorded within source; not reported items remain not reported
Brusco et al. (2024)	sampling, synchronization and filtering	Sensors: two 64-channel pressure sensors, each with a built-in 45 kHz high-speed multiplexer. The choice to use the same acquisition system is explicitly stated as ensuring simultaneity between dynamic-pressure and load data.	recorded within source; not reported items remain not reported
Brusco et al. (2024)	sampling, synchronization and filtering	Accuracy: stated as 0.15% of full scale (2500 Pa), i.e. accuracy ±3.75 Pa.	recorded within source; not reported items remain not reported

Table 9: Brusco et al. (2024) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Brusco et al. (2024)	sampling, synchronization and filtering	Sampling rate: 300 Hz. Original text: ”The sampling frequency is set at 300 Hz, which easily resolves vortex-shedding from the studied configuration.”	recorded within source; not reported items remain not reported
Brusco et al. (2024)	sampling, synchronization and filtering	Tubing: each tap is connected to the sensor via a 1.5 m PVC tube of 1.5 mm inner diameter; raw pressure signals are corrected for both amplitude and phase frequency response according to tube length (citing Kay et al., 2020).	recorded within source; not reported items remain not reported
Brusco et al. (2024)	sampling, synchronization and filtering	Record length: 2^16 points = 218.4 s per steady repeat; transient sub-signal 2^13 points = 27.3 s.	recorded within source; not reported items remain not reported
Brusco et al. (2024)	sampling, synchronization and filtering	Filter A (vortex-shedding extraction, p. 8): complex Morlet mother wavelet $\psi(t) = e^{i\omega 0 t} e^{-t2/2}$, centre frequency $\omega 0 = 6\pi$ (rather than the commonly used 2π, to improve frequency resolution because the shedding frequency changes abruptly during transients). Frequency band 5.5 Hz to 55.1 Hz, discretised into 501 scales. The shedding frequencies corresponding to the measured wind-speed range are approximately 9 Hz to 33 Hz. Energy threshold set at 3 times the mean of the energy map. Result: $\sigma cL,VS,W$ accounts for approximately 96% of σcL at low wind speeds, decreasing to 87% at high wind speeds.	recorded within source; not reported items remain not reported
Brusco et al. (2024)	sampling, synchronization and filtering	Filter B (time-varying mean extraction, p. 9): also a continuous wavelet transform, but with an upper frequency limit of 2.51 Hz, below the shedding spectral peak at the lowest wind speed; energy threshold set at 5% of the mean of the energy map. Used to obtain $Pdyn,W$, $\Delta PD,W$, Pi,W, Ps,W, and UW.	recorded within source; not reported items remain not reported

Table 9: Brusco et al. (2024) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Brusco et al. (2024)	sampling, synchronization and filtering	Secondary smoothing: on p. 18, Filter B is applied again to <σ>cL,VS,W to obtain <σ>cL,VS,W,W, on the grounds that the original ensemble standard deviation ”remain strongly intermittent and oscillating”.	recorded within source; not reported items remain not reported
Brusco et al. (2024)	sampling, synchronization and filtering	Transient boundary criterion: ensemble-averaged acceleration <a>W = ±0.25 m/s2 (p. 9).	recorded within source; not reported items remain not reported
Brusco et al. (2024)	quantitative anchors (including verbatim short excerpts and physical PDF pages)	p. 6: cΔPD = 2.30, range 2.26 to 2.35. Verbatim: ”cΔPD is found to be 2.30, ranging from 2.26 to 2.35.”	recorded within source; not reported items remain not reported

The closest physical neighbour sources are Brusco et al. (2022) and Zhou et al. (2025), but the inflow shape, cross-sectional ratio, and force construction differ, so only the directional asymmetry can be compared. Li et al. (2025) is only a numerical hypothesis line with B/D marked as not reported and must not serve as validation of this physical source.

11.4 Wu et al. (2021), Pressure-Integrated Forces, Component Decomposition, and Two-Dimensional Aerodynamic Admittance of a 5:1 Rectangular Cylinder under Streamwise Sinusoidal Inflow

The overall forces in this source are obtained from surface pressure integration, not from an independent balance channel. Within-source comparisons cover sinusoidal amplitude and frequency, and the source reports the proportion of the inflow-induced drag component as well as the non-monotonic frequency response of the two-dimensional aerodynamic admittance. The source-constrained point set contains candidate marker identities whose candidate status must accompany each figure and must not be promoted to verified series through formatting.

> *“The ratio can reach about 90% when Delta u/U is higher than 0.15.”*
>
> Source Wu et al. (2021), PDF p. 9, print p. 9, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: when the streamwise sinusoidal fluctuation amplitude exceeds 0.15, the inflow-induced component of drag can reach approximately 90%.

> *“the three components of 2D AAF do not decrease monotonously with the frequency, which is different from Davenport’s, Sears’ or Horlock’s functions”*
>
> Source Wu et al. (2021), PDF p. 12, print p. 12, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: the three two-dimensional aerodynamic admittance components do not decrease monotonically with frequency and differ from the three classical functions.

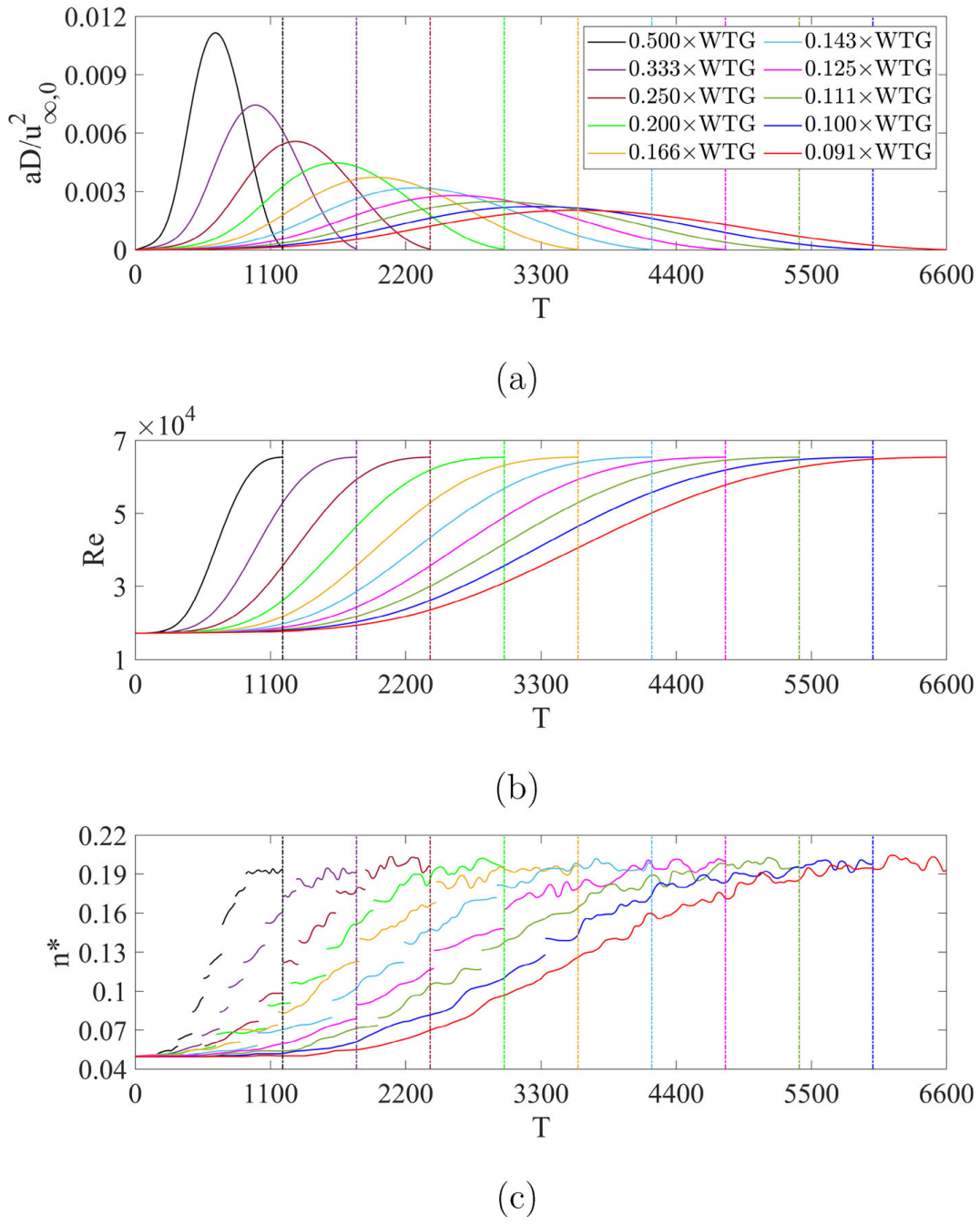
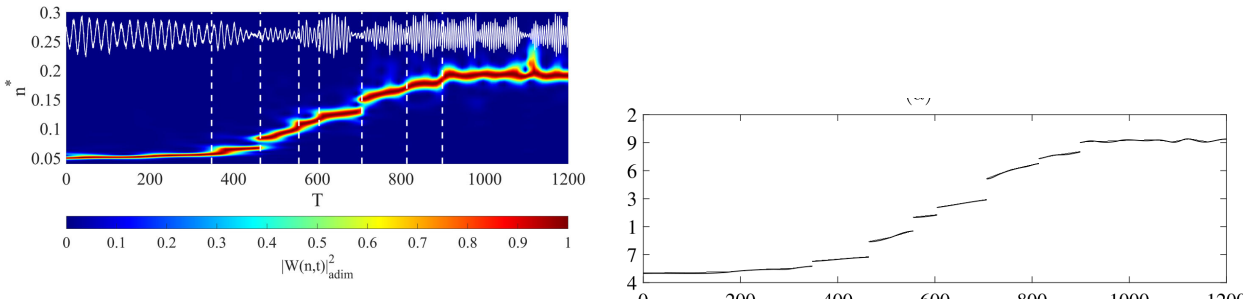
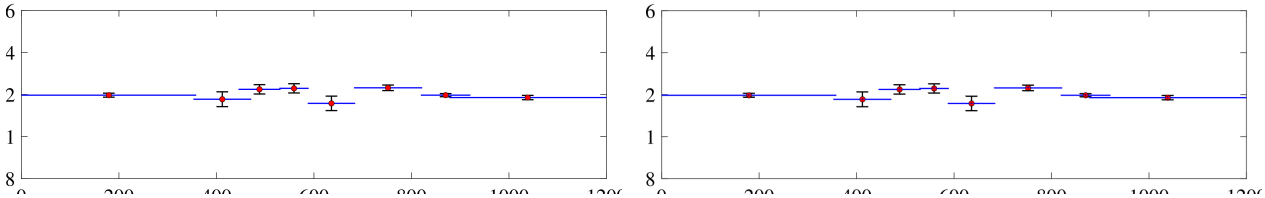


Figure 24: Source figure excerpt. Source Morello et al. (2025a), original figure/table Fig. 9, panel a, PDF p. 11, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.



Source figure Source-constrained extraction overlay

Figure 25: Source figure and source-constrained extraction overlay shown side by side. Source Morello et al. (2025a), PDF p. 8. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.



Source figure Source-constrained extraction overlay

Figure 26: Source figure and source-constrained extraction overlay shown side by side. Source Morello et al. (2025a), PDF p. 10. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

Table 1
List of estimated force coefficients from previous studies (steady flows).

Author(s)	$\frac{\sqrt{U^2}}{\overline{U}}$ [%]	*Re*	$\bar{\bar{c}}_D$	$\bar{\bar{c}}_{P_b}$	$\bar{\sigma}_{c_L}$
Vickery, 1966	Smooth	10^5	–	−1.31	1.31
–	10	–	–	−0.71	0.68
Nakaguchi et al., 1968	Smooth	2–6 10^4	2.10	−1.50	–
Bearman and Trueman, 1972	Smooth	6.8 10^4	2.19	−1.40	–
Lee, 1975	<0.5	1.76 10^4	2.05	−1.30	1.23
–	6.5	1.76 10^4	1.93	−1.18	0.95
Ootsuki et al., 1980	0.2	6.5 –7 10^4	2.08	−1.35	0.82
Bearman and Obasaju, 1982	Smooth	4.7 10^4	–	−1.65	–
Mizota and Okajima, 1982	Smooth	7.14 10^4	–	−1.64	–
Okajima, 1982	<0.5	4.2 10^4	–	−1.47	–
Obasaju, 1983	<0.04	10^4	2.17	–	–
Norberg, 1993	<0.06	5 10^3	2.21	−1.45	–
–	<0.06	13 10^3	2.16	−1.38	–
Tamura and Miyagi, 1999	0.4	3 10^4	2.09	–	1.05
–	6.5	–	1.79	–	0.74
–	14	–	1.49	–	0.34
Noda and Nakayama, 2003	0.2	6.89 10^4	2.16	−1.48	1.18
Dutta et al., 2003	<0.05	1340 – 9980	2.21–2.27	–	–
Yang and Mason, 2019	<0.5	9300 – 12.3 10^4	2.14	–	1.33
Sohankar, 2006 (CFD)	smooth	10^3 – 5 10^6	2.20	−1.50	1.25
Present	2	1.72 – 6.34 10^4	2.06	−1.30	1.03

Figure 27: Source figure excerpt. Source Brusco et al. (2024), original figure/table Table 1, panel not reported, PDF p. 7, print p. not reported. Data origin: values compiled from existing studies by the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

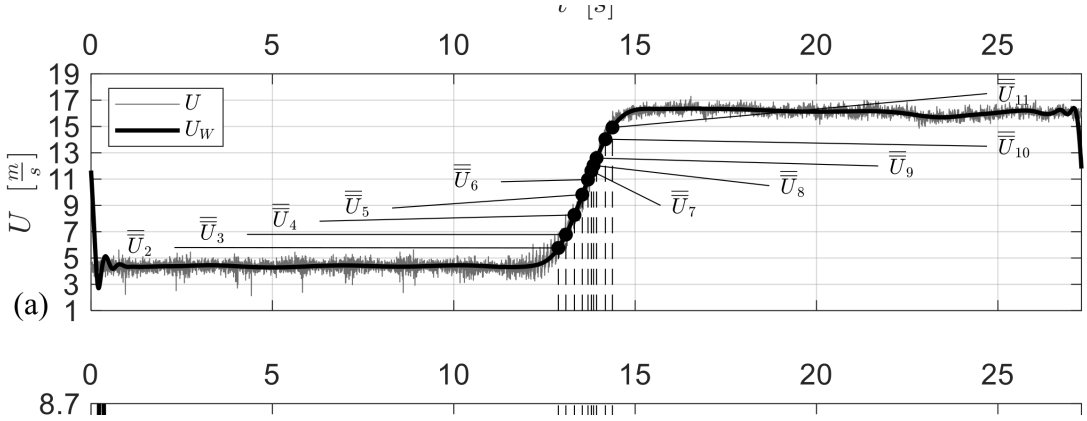


Figure 28: Source figure excerpt. Source Brusco et al. (2024), original figure/table Fig. 8, panel a, PDF p. 12, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

speed (U_8). Hence, the equation is numerically solved by easily replacing the terms linked with t

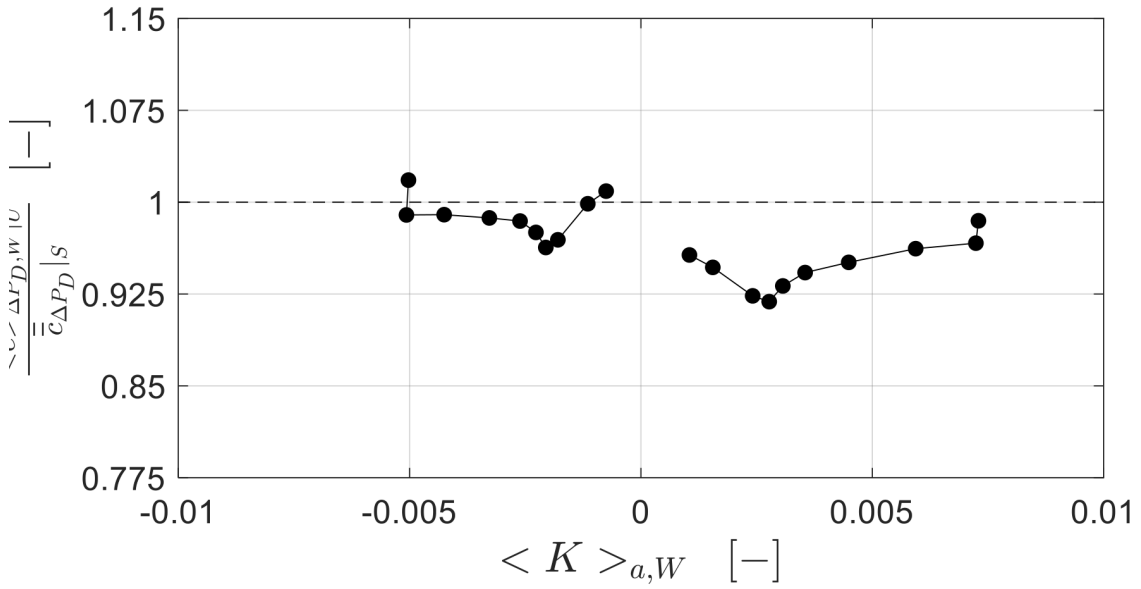


Figure 29: Source figure excerpt. Source Brusco et al. (2024), original figure/table Fig. 11, panel not reported, PDF p. 15, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

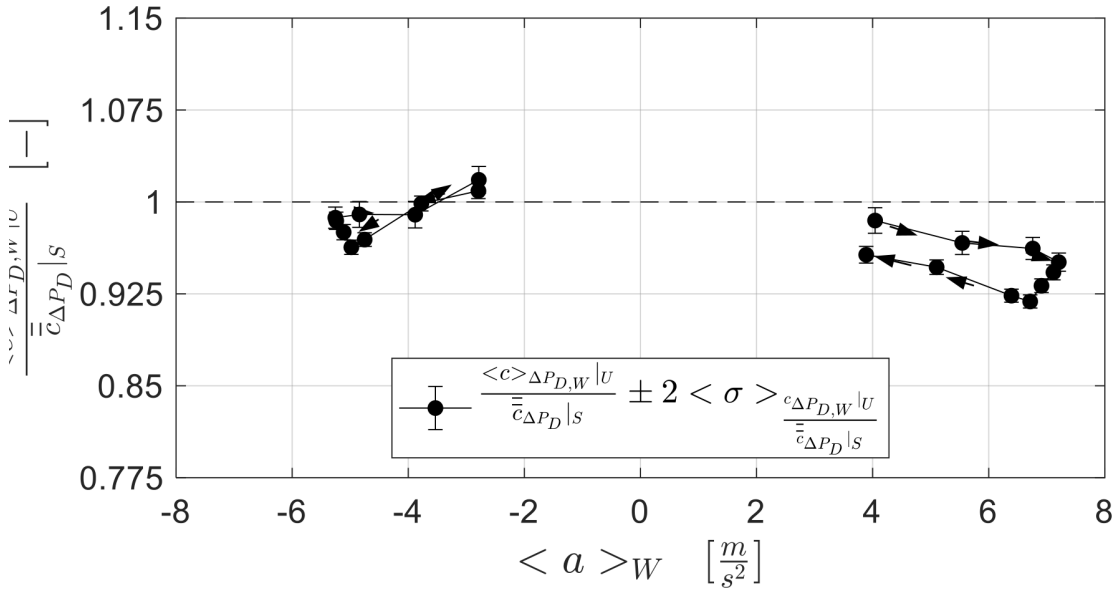
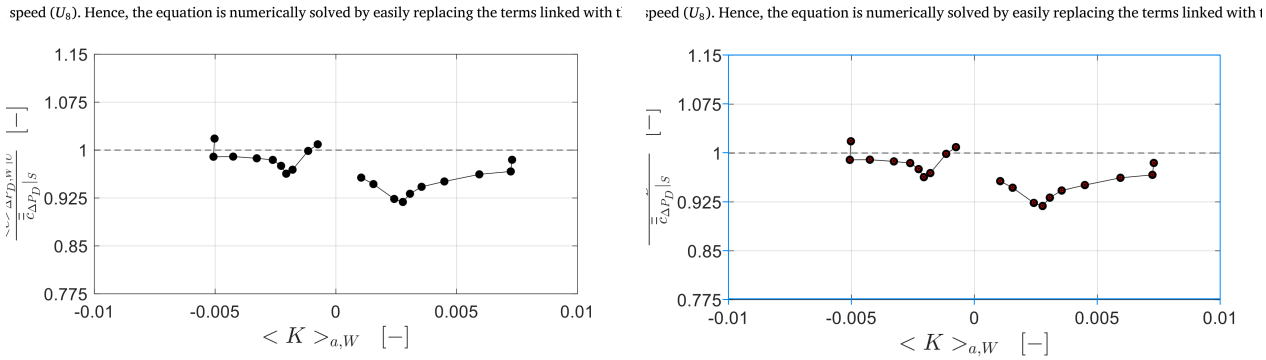


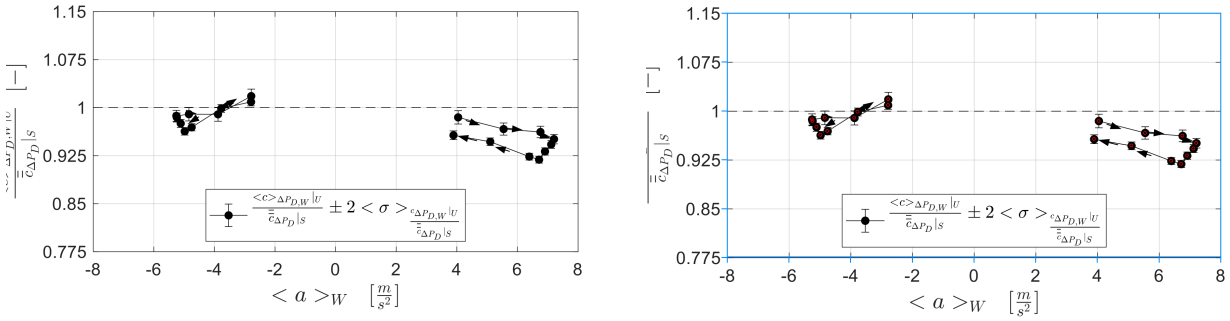
Figure 30: Source figure excerpt. Source Brusco et al. (2024), original figure/table Fig. 12, panel not reported, PDF p. 16, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.



Source figure

Source-constrained extraction overlay

Figure 31: Source figure and source-constrained extraction overlay shown side by side. Source Brusco et al. (2024), PDF p. 15. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.



Source figure

Source-constrained extraction overlay

Figure 32: Source figure and source-constrained extraction overlay shown side by side. Source Brusco et al. (2024), PDF p. 16. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

Table 10: Wu et al. (2021) high-completeness transient condition and evidence table. Comparison ceiling: within-source L0-L4. Absent entries are uniformly recorded as not reported.

source id	condition field	source recorded content	status
Wu et al. (2021)	inflow and Reynolds-number definitions	Inflow definition (p. 2, unnumbered equation): u(t) = U + Delta_u . sin(2 pi f_u t).	recorded within source; not reported items remain not reported
Wu et al. (2021)	inflow and Reynolds-number definitions	Mean velocity: U = 8 m/s.	recorded within source; not reported items remain not reported
Wu et al. (2021)	inflow and Reynolds-number definitions	Reynolds number: the original text writes the Reynolds number Re_D = 16080, with subscript D, i.e. defined using the cross-section depth D = 0.03 m. The original text provides no definition formula for the Reynolds number, nor does it report the air density, temperature, or kinematic viscosity; all three are not reported. A Reynolds number defined using the chord length B does not appear anywhere in the paper and is not reported.	recorded within source; not reported items remain not reported
Wu et al. (2021)	inflow and Reynolds-number definitions	Amplitude range: 0 <= Delta_u/U <= 0.25. The discrete amplitudes actually appearing in figures are 0.0625, 0.125, 0.15, 0.1875, 0.20, 0.25 (six for force-coefficient figures), whereas the legend in Fig. 19 lists only 0.0625, 0.125, 0.15, 0.1875, 0.25 (five, missing 0.20).	recorded within source; not reported items remain not reported

Table 10: Wu et al. (2021) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Wu et al. (2021)	inflow and Reynolds-number definitions	Frequency range: f_u from 0.2 to 1.2 Hz; the discrete frequencies actually appearing in figures are 0.2, 0.4, 0.6, 0.8, 1.0, 1.2 Hz, corresponding to circular frequencies omega of approximately 1.26, 2.51, 3.77, 5.03, 6.28, 7.54 rad/s, consistent with the six data-point positions on the horizontal axes of Figs. 19 and 20.	recorded within source; not reported items remain not reported
Wu et al. (2021)	inflow and Reynolds-number definitions	Origin of the frequency upper limit: the original text explicitly states it was not possible to oscillate the generated flow harmonically at frequencies higher than 1.2Hz. This is due to the limitations of the multiple-fan system.	recorded within source; not reported items remain not reported
Wu et al. (2021)	inflow and Reynolds-number definitions	Condition naming: SF(f_u, Delta_u/U).	recorded within source; not reported items remain not reported
Wu et al. (2021)	inflow and Reynolds-number definitions	Baseline condition: uniform flow corresponds to Delta_u/U = 0, appearing as the data point at the horizontal-axis origin in Figs. 9, 10, 13, 14, 15, and as an independent Uniform flow curve in Figs. 16 and 18. The free-stream turbulence intensity of the uniform flow is not reported.	recorded within source; not reported items remain not reported
Wu et al. (2021)	inflow and Reynolds-number definitions	Spanwise coherence: the inflow is explicitly stated to be one-dimensional, containing only streamwise fluctuations with no lateral fluctuation component; this statement is the prerequisite for Eq. (12) to degenerate into Eq. (13).	recorded within source; not reported items remain not reported
Wu et al. (2021)	sampling, synchronization and filtering	Sampling frequency: the original text states The sampling frequencies of the Cobra Probe and DMS3400 pressure scanners are both set to 200Hz, i.e. 200 Hz for both instrument types.	recorded within source; not reported items remain not reported

Table 10: Wu et al. (2021) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Wu et al. (2021)	sampling, synchronization and filtering	Synchronisation: pressure is explicitly described as measured simultaneously by DSM 3400 scanners; the empty-tunnel inflow is explicitly described as measured simultaneously using five Turbulent Flow Instrument (TFI) Cobra Probes. Whether the pressure channels and the Cobra channels are mutually synchronised is not stated and is not reported.	recorded within source; not reported items remain not reported
Wu et al. (2021)	sampling, synchronization and filtering	Sampling duration and sample count: the total record length is not stated anywhere in the paper. The only visible information is the figure window lengths: 30 s in Fig. 4, 60 s in Fig. 5, 20 s in Fig. 7, and 25 s in Fig. 11. These are plotting windows and do not equal the record length. Total duration, number of segments, and number of repeats are all not reported.	recorded within source; not reported items remain not reported
Wu et al. (2021)	sampling, synchronization and filtering	Pre-processing: the only explicitly stated pre-processing step is mean removal prior to POD (the random pressure field is pre-processed by removing the mean components). Beyond this, no low-pass, high-pass, or band-pass filtering, no detrending, and no outlier removal are mentioned anywhere in the paper; all are not reported.	recorded within source; not reported items remain not reported
Wu et al. (2021)	sampling, synchronization and filtering	Spectral estimation parameters: the window function, segment length, overlap ratio, number of averages, and frequency resolution for the spectra in Figs. 6, 8, and 11 are all not reported.	recorded within source; not reported items remain not reported

Table 10: Wu et al. (2021) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Wu et al. (2021)	sampling, synchronization and filtering	Tubing transfer-function correction: the tubing length of 0.2 m is explicitly given, but whether any correction for the amplitude attenuation and phase lag of this tube length was applied is not mentioned anywhere in the paper and is not reported. This is particularly relevant for signals containing vortex-shedding components near 30 Hz at 200 Hz sampling.	recorded within source; not reported items remain not reported
Wu et al. (2021)	sampling, synchronization and filtering	Nyquist margin: 200 Hz sampling corresponds to a Nyquist frequency of 100 Hz, while the uniform-flow vortex-shedding frequency f0 = 31.5 Hz; the horizontal-axis upper limit in Fig. 8 is approximately 5 x 10^1 Hz. The anti-aliasing filter settings are not reported.	recorded within source; not reported items remain not reported
Wu et al. (2021)	explicit not reported fields	Dimensions, shape, thickness, and mounting position of the end plates (two thin plates): not reported.	recorded within source; not reported items remain not reported
Wu et al. (2021)	explicit not reported fields	Gap between the model ends and the tunnel walls, and whether the model spans the full width of the test section: not reported. The plan-view drawing in Fig. 3 shows a printed total length of 1500 mm, which is inconsistent with the model length of 1.3 m stated on p. 2; no explanation is given in the text.	recorded within source; not reported items remain not reported

The closest physical neighbour source is Li et al. (2021a); both belong to the streamwise sinusoidal family with a cross-sectional ratio of 5, but the reduced frequencies and reference-length definitions still require item-by-item alignment, limiting the comparison ceiling to L2. Numerical neighbour sources can only provide mechanistic context and cannot be co-axially aligned with pressure-integrated forces.

11.5 Yang and Mason (2019), Monotonic Acceleration Comparison of B/D=1 and B/D=3 in the Same Facility

This source retains separate B/D=1 and B/D=3 cross-section facets within the same facility and records the wind-tunnel setup, velocity reference, and pressure-integrated forces. The original text reports that the coefficient variations for

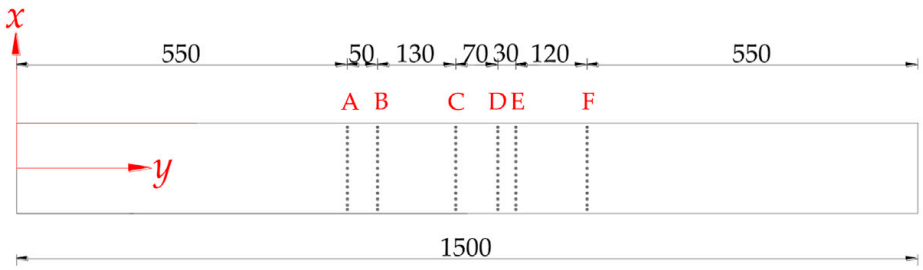


Figure 33: Source figure excerpt. Source Wu et al. (2021), original figure/table Fig. 3, panel plan view (bottom), PDF p. 3, print p. not reported. Data origin: not reported in the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

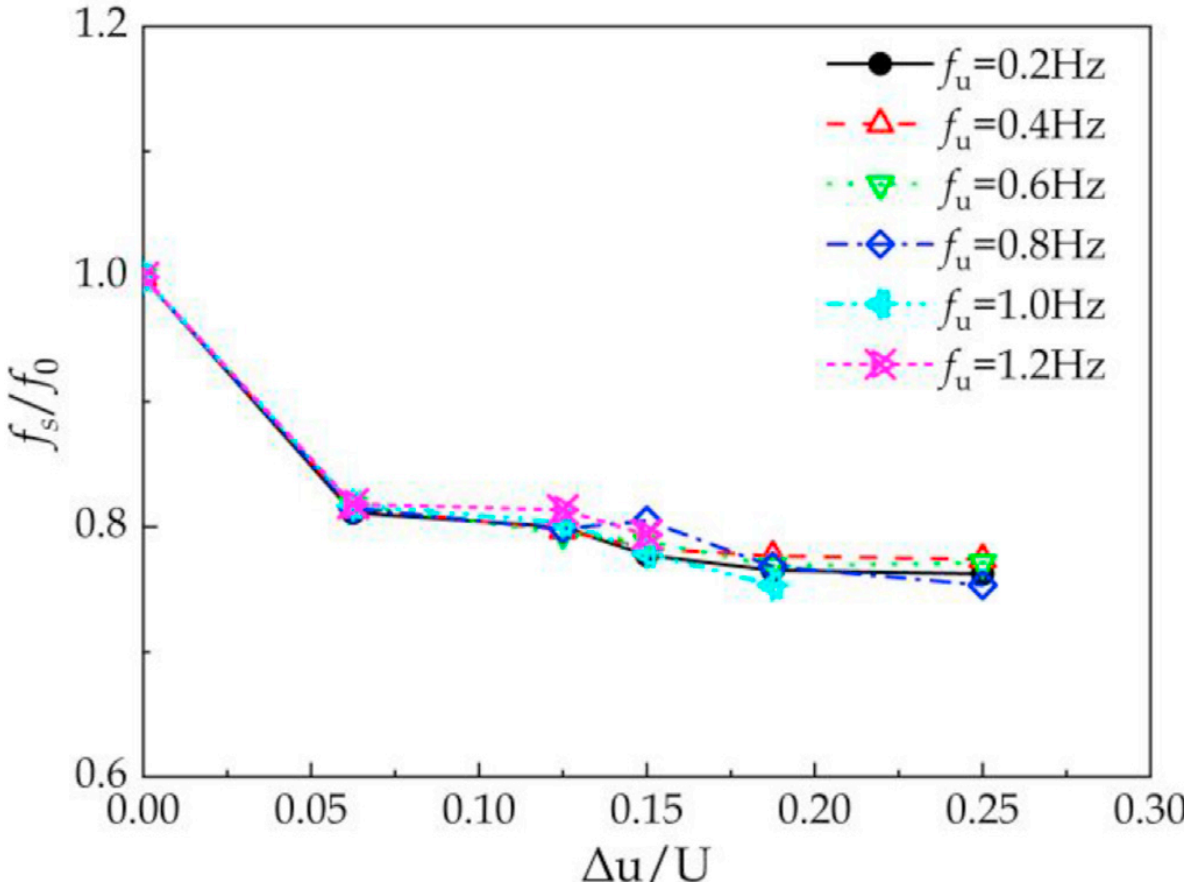


Figure 34: Source figure excerpt. Source Wu et al. (2021), original figure/table Fig. 9, panel single, PDF p. 6, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

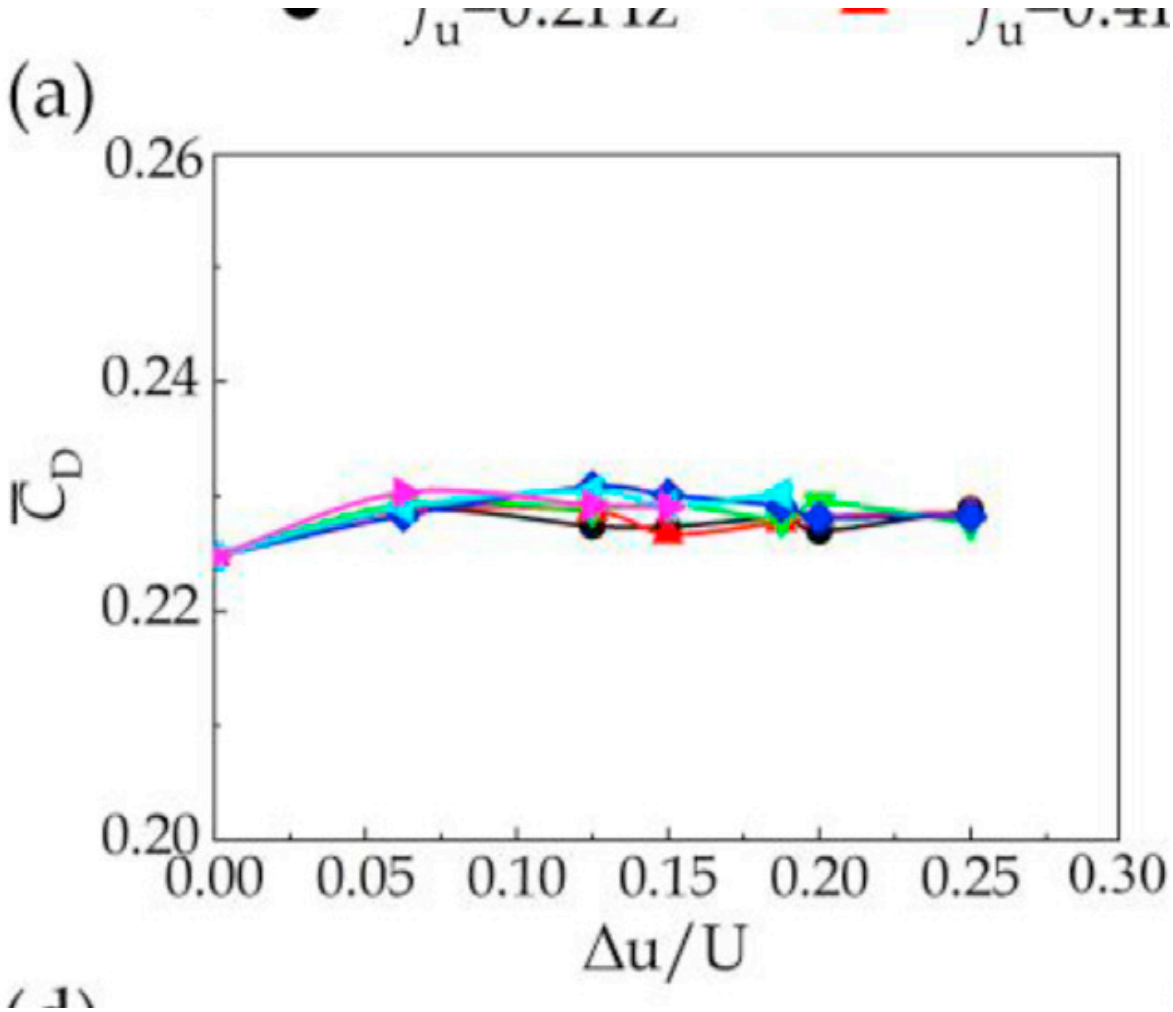


Figure 35: Source figure excerpt. Source Wu et al. (2021), original figure/table Fig. 10, panel a, PDF p. 7, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

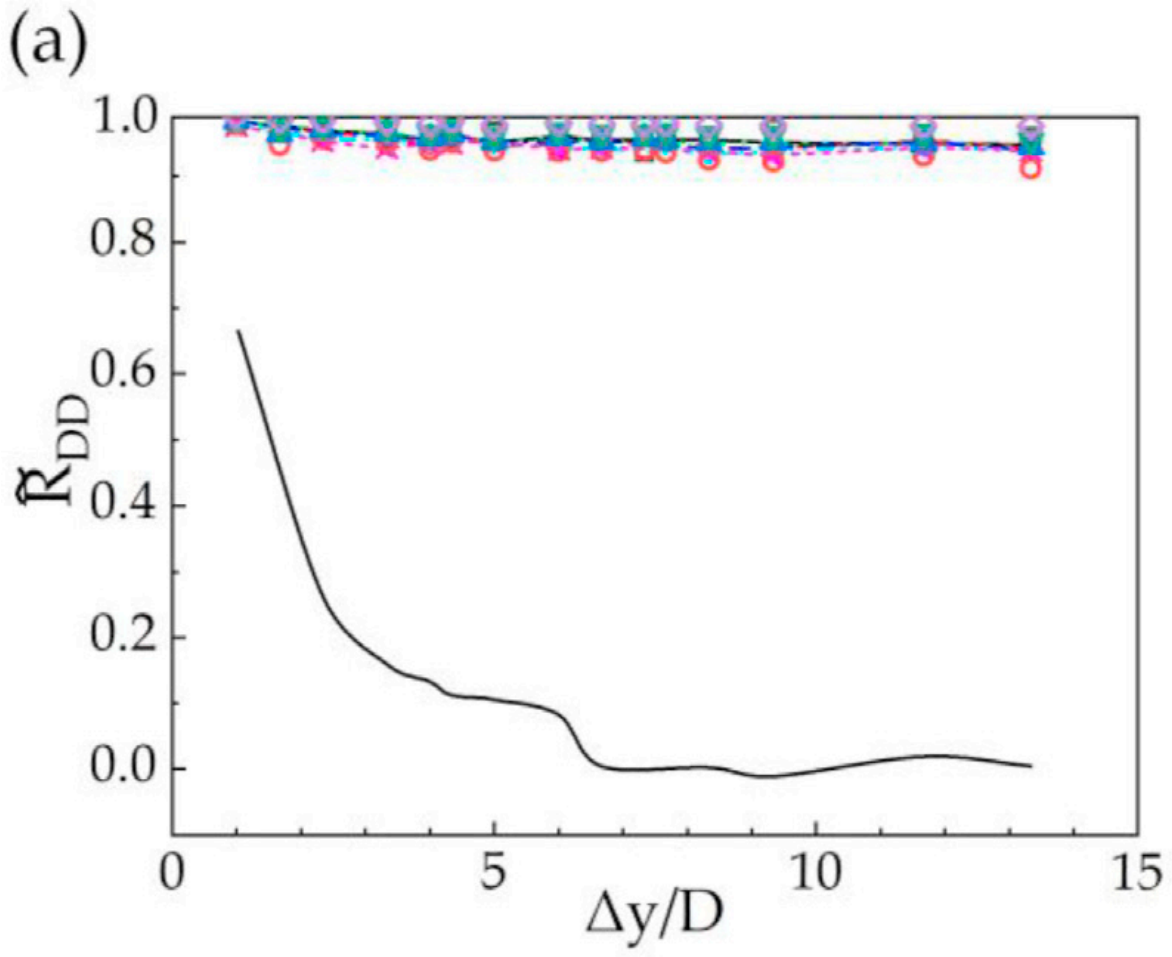
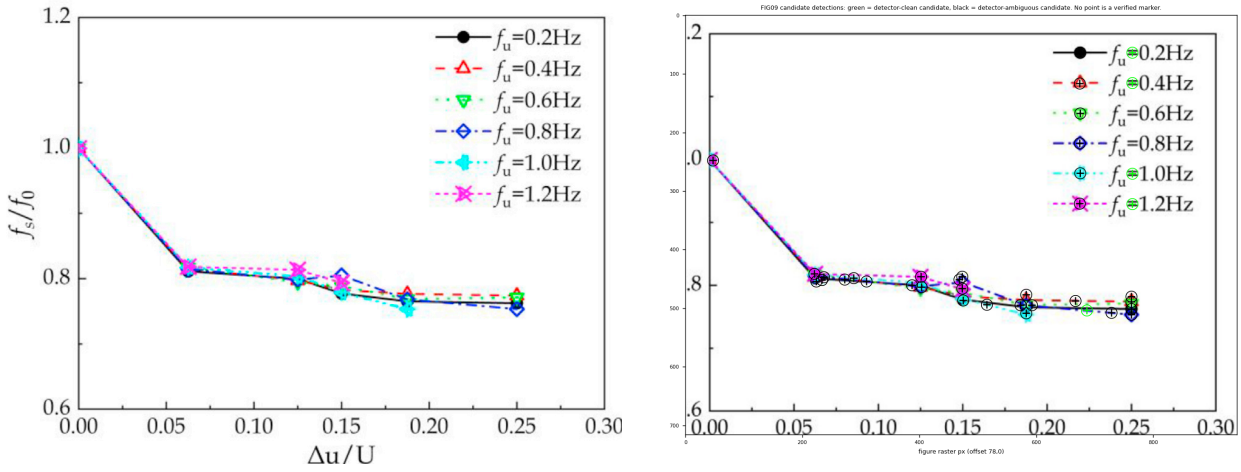


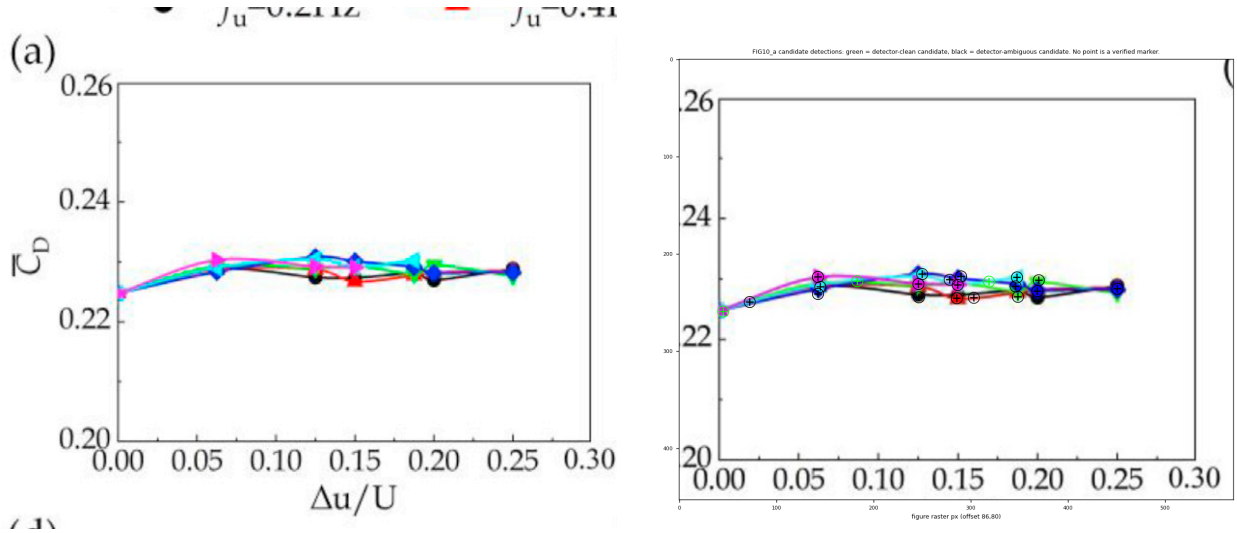
Figure 36: Source figure excerpt. Source Wu et al. (2021), original figure/table Fig. 16, panel a, PDF p. 10, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.



Source figure

Source-constrained extraction overlay

Figure 37: Source figure and source-constrained extraction overlay shown side by side. Source Wu et al. (2021), PDF p. 6. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.



Source figure Source-constrained extraction overlay

Figure 38: Source figure and source-constrained extraction overlay shown side by side. Source Wu et al. (2021), PDF p. 7. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

the two cross-section types are not consistent: B/D=3 exhibits a combination of drag overshoot and lift decrease at the initiation of acceleration. The source-constrained package contains only two finally approved original setup figures; response evidence is supplemented through source-constrained masks and source records, and the report does not treat masks as independent measurements.

> *"Coefficient values, in both flow cases, are reduced by up to approximately 10%, 30%, 25% and 5% for <CD>, <CL'>, <absCLhat> and <St>, respectively."*
>
> Source Yang and Mason (2019), PDF p. 12, print p. 257, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: in both acceleration conditions for B/D=1, drag, fluctuating lift, peak lift, and Strouhal number decrease by different amounts relative to steady-state values.

> *"Coupled with the drop in lift coefficients at the initiation of flow acceleration is an overshoot of up to 10% in <CD>."*
>
> Source Yang and Mason (2019), PDF p. 13, print p. 258, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: at the initiation of acceleration, B/D=3 exhibits a lift-coefficient decrease coupled with a drag-coefficient overshoot of up to 10%.

Table 11: Yang and Mason (2019) high-completeness transient condition and evidence table. Comparison ceiling: within-source L0-L4. Absent entries are uniformly recorded as not reported.

source id	condition field	source recorded content	status
Yang and Mason (2019)	wind tunnel	The University of Queensland open-circuit suction wind tunnel.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	test section	762 x 762 mm cross-section, 2.36 m long.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	free-stream turbulence intensity	u'/U < 0.5%, applicable range 2 < U < 40 m/s; u' is the measured wind-speed standard deviation.	recorded within source; not reported items remain not reported

Table 11: Yang and Mason (2019) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Yang and Mason (2019)	uniformity	In steady state, u' and U are uniform to +/- 1% over the spanwise width of the test body; during acceleration, instantaneous velocity U(t) is uniform to +/- 0.2 m/s over the same width.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	model	Three two-dimensional rectangular cylinders, all with D = 60 mm; B = 20, 60, 180 mm, corresponding to B/D = 1/3, 1, 3.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	blockage ratio	The text on PDF p. 3 records 7.9%; Table 2 (PDF p. 8) Present row records BR 7.8%. The two values are inconsistent within the source; this card lists both without making a selection.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	blockage correction	Explicitly not applied. Original text:	recorded within source; not reported items remain not reported
Yang and Mason (2019)	mounting	Mounted horizontally at mid-height of the tunnel, with the geometric centre located 580 mm (9.67D) downstream of the test-section inlet. Fig. 1 also labels a 1,000 mm streamwise dimension measured from the inlet, which is inconsistent with the 580 mm stated in the text; this is recorded as a within-source discrepancy and WP1a does not resolve it.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	end plates	Two circular end plates of 250 mm diameter, located 300 mm on each side of the model centreline (the inset in Fig. 1 labels R=125mm, consistent with the 250 mm diameter).	recorded within source; not reported items remain not reported
Yang and Mason (2019)	aspect ratio	AR = L/D = 10, where L is the spanwise distance between the two end plates.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	pressure measurement	Circumferential pressure taps on the model, synchronously acquired by a 64-channel ZOC 33 pressure scanner (Scanivalve Corporation).	recorded within source; not reported items remain not reported

Table 11: Yang and Mason (2019) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
Yang and Mason (2019)	tubing	1.5 m PVC tube, 1.5 mm inner diameter; the Bergh and Tijdeman (1965) linearised tubing-response correction function is used to simultaneously correct both amplitude and phase distortion; the tunnel-floor static-pressure taps use the same tubing system and the same correction.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	velocity measurement	Hot-wire anemometer (Type 55P11, Dantec Dynamics), positioned 100 mm from the tunnel ceiling, with the probe tip aligned with the windward face of each model.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	reference static pressure	Tunnel-floor pressure tap, also aligned with the windward face of each model.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	sampling	Both pressure and velocity at 600 Hz, but acquired by two separate acquisition systems requiring post-hoc synchronisation.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	angle of attack	All three models are tested with the windward face perpendicular to the incoming flow (normal incidence); no other angles of attack are tested in the source.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	inflow and Reynolds-number definitions	Re = UD/nu, where U is the reference velocity, D is the cross-flow depth (= 60 mm), and nu is the kinematic viscosity of air (PDF p. 4). The value of nu, air temperature, and air density rho are all not reported.	recorded within source; not reported items remain not reported
Yang and Mason (2019)	inflow and Reynolds-number definitions	The reference velocity is taken from the hot-wire probe, positioned 100 mm from the ceiling and streamwise-aligned with the windward face of each model. Re is therefore based on the free-stream reference velocity rather than the local velocity near the model.	recorded within source; not reported items remain not reported

The closest physical neighbour source is Da Matha Sant'Anna et al. (1988), whose steady-state multi-section pressure

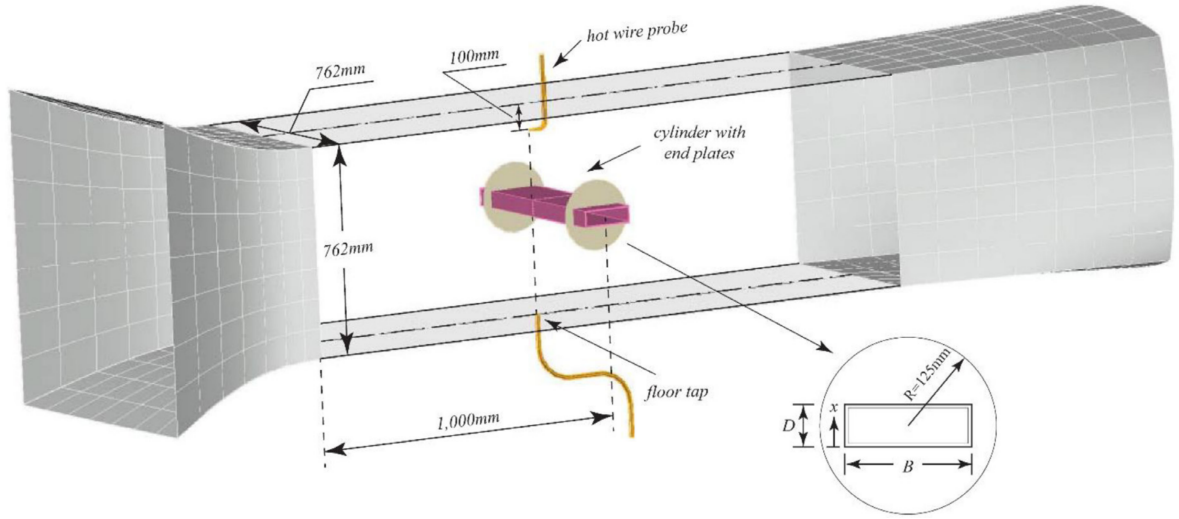


Figure 39: Source figure excerpt. Source Yang and Mason (2019), original figure/table Fig. 1, panel main perspective view, PDF p. 3, print p. 248. Data origin: not reported in the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

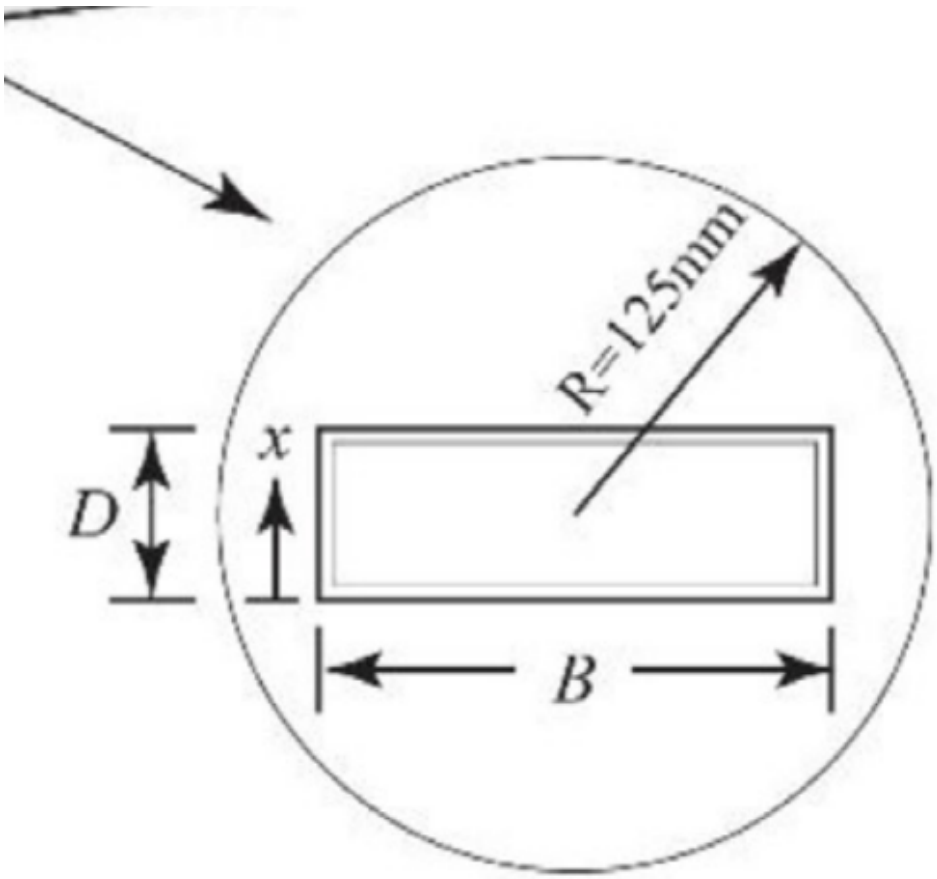
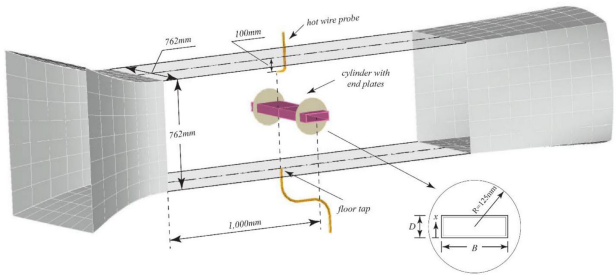
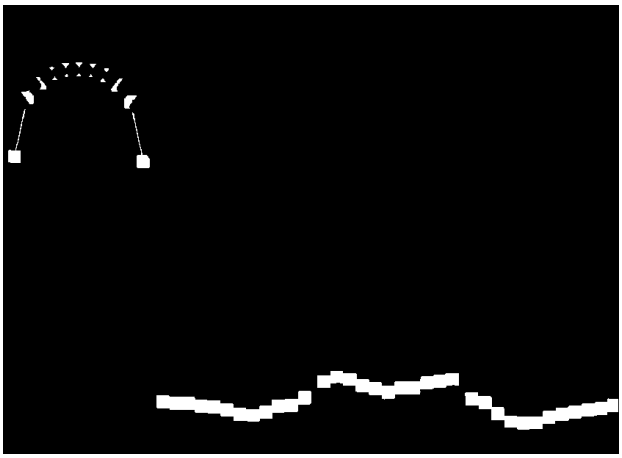


Figure 40: Source figure excerpt. Source Yang and Mason (2019), original figure/table Fig. 1, panel cross-section detail inset, PDF p. 3, print p. 248. Data origin: not reported in the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

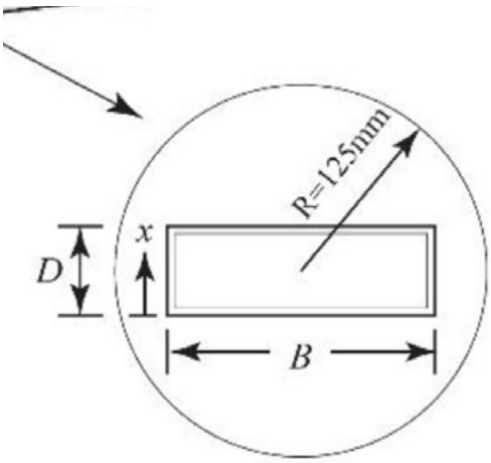


Source figure

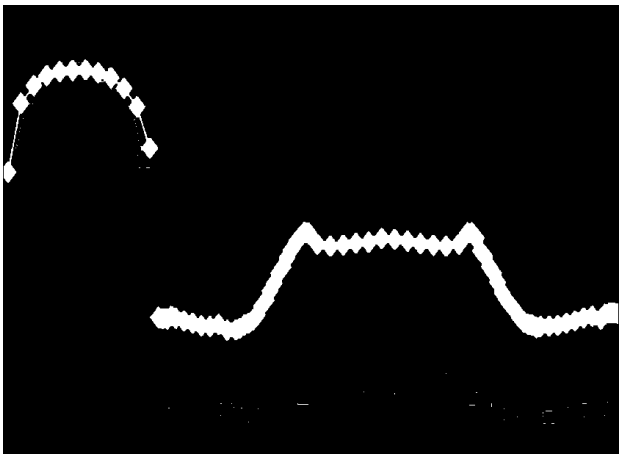


Source-constrained extraction overlay

Figure 41: Source figure and source-constrained extraction overlay shown side by side. Source Yang and Mason (2019), PDF p. not reported. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.



Source figure



Source-constrained extraction overlay

Figure 42: Source figure and source-constrained extraction overlay shown side by side. Source Yang and Mason (2019), PDF p. not reported. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

measurements can provide geometric adjacency, but the inflow conditions differ. The numerical neighbour source Yu et al. (2013) discusses the effect of cross-sectional ratio on reattachment flow regimes and can only provide steady-state numerical mechanism context.

11.6 M. Li et al. (2023), Mean and Fluctuating Force Response of a Square Cylinder in Independent Streamwise and Vertical Sinusoidal Gusts

This source applies streamwise and vertical sinusoidal gusts separately in different test conditions, with no simultaneous multi-axis forcing; the two sets of amplitude and frequency parameters cannot be collapsed into a single gust intensity. The original text reports within-source variations of mean drag and fluctuating lift under high-amplitude sinusoidal components. Where figures contain imported literature series, those imported series must be identified separately from the present source's physical measurements.

> *"This is reduced by approximately 15.2% and 23.5% when sigma_u/U = 20.8% and sigma_w/U = 21.4%, respectively."*
>
> Source M. Li et al. (2023), PDF p. 11, print p. 11, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: the mean drag is reduced by approximately 15.2% and 23.5% under high-amplitude streamwise and lateral sinusoidal gusts, respectively.

> *"The value of C_L tilde is reduced from approximately 17.0% to 19.0% when sigma_u/U increases from 5.3% to 20.8%."*
>
> Source M. Li et al. (2023), PDF p. 12, print p. 12, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-constrained proposition: the fluctuating lift decreases by approximately 17.0% to 19.0% as the streamwise gust amplitude increases from 5.3% to 20.8%.

Table 12: M. Li et al. (2023) high-completeness transient condition and evidence table. Comparison ceiling: within-source L0-L4. Absent entries are uniformly recorded as not reported.

source id	condition field	source recorded content	status
M. Li et al. (2023)	wind tunnel	Closed-circuit-type wind tunnel, test section 0.5 m wide, 0.5 m high, 5.0 m long.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	gust generator	Installed 2.05 m upstream of the model leading edge; composed of five blades, each a NACA 0012 aerofoil with 0.1 m chord and 0.46 m span; driven by motors for synchronous pitching motion at prescribed oscillation frequencies and amplitudes.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	qualitative rules of gust-generator design (as stated by the authors)	Smaller blade spacing increases gust amplitude but degrades flow-field uniformity; increasing pitching amplitude improves both gust amplitude and uniformity simultaneously; lower and higher blade oscillation frequencies favour the generation of streamwise and lateral gusts, respectively.	recorded within source; not reported items remain not reported

Table 12: M. Li et al. (2023) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
M. Li et al. (2023)	model	Fibreglass square sectional model, mounted horizontally at the tunnel centre, angle of attack 0 degrees.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	gust conditions, 6 in total (Table 1, physical p. 4)	not reported	recorded within source; not reported items remain not reported
M. Li et al. (2023)	regarding f_g = 2 f_os for streamwise gusts, the original text explains	The gust generator effectively actuates the streamwise flow velocity twice per motion cycle.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	smooth-flow baseline	The background turbulence intensity of the tunnel without the gust generator installed is I_u = 0.6 %, referred to throughout as nominal smooth flow.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	mean wind speed	U = 6.0 m/s for all gust conditions.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	Reynolds number	Original text: The corresponding Reynolds number of the main flow based on the square model thickness D was Re = 2.1 x 10^4 (physical p. 4). The characteristic length is the square-cylinder depth D = 50 mm, and the velocity is the mean free-stream velocity U. Only this single Reynolds number is given throughout the paper; the instantaneous Reynolds-number range due to gust-induced velocity variation is not separately provided.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	gust velocity definition	Eq. (1) U_t(t) = U + u_hat sin(2 pi f_u t); Eq. (2) w(t) = w_hat sin(2 pi f_w t).	recorded within source; not reported items remain not reported
M. Li et al. (2023)	gust amplitude	Expressed as sigma_g/U, with streamwise denoted sigma_u/U and lateral denoted sigma_w/U.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	reduced gust frequency	K_g = 2 pi f_g D / U (inline definition on physical p. 4). Streamwise K_g = 0.021, lateral K_g = 0.131.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	gust wavelength ratio	lambda/D = 300 (streamwise) and 48 (lateral).	recorded within source; not reported items remain not reported
M. Li et al. (2023)	background turbulence intensity	I_u = 0.6 % (without gust generator).	recorded within source; not reported items remain not reported

Table 12: M. Li et al. (2023) high-completeness transient condition and evidence table (continued)

source id	condition field	source recorded content	status
M. Li et al. (2023)	inflow and Reynolds-number definitions	The reduced-frequency abscissa for force spectra is nD/U; gust peaks in spectral plots are labelled by K_u/2pi and K_w/2pi.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	dynamic-pressure definition	q_bar = rho U^2 / 2, where rho is the air density. The value of air density, temperature, and atmospheric pressure conditions are not reported.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	sampling frequency	800 Hz.	recorded within source; not reported items remain not reported
M. Li et al. (2023)	sampling duration	Approximately 60 s (original text: The pressure measurements were performed for approximately 60 s).	recorded within source; not reported items remain not reported

The closest physical neighbour source Wu et al. (2021) belongs to the streamwise sinusoidal family with a 5:1 cross-section; both the axial definitions and the cross-sectional ratio differ, limiting the comparison ceiling to L1–L2. Numerical sources can only provide hypothetical context and must not be used to interpret independent single-axis physical responses as validated synchronous multi-axis effects or numerical models.

12 Comparison Matrix of Aspect Ratio, Inflow Family, and Measurement Channel

Table 13: B/D=1 evidence-condition matrix. Comparison ceiling: L0-L2 cross-source. Gaps are uniformly recorded as not reported.

source id	tier	B over D	flow family	direct channels	limitations
Moore et al. (2019)	D	1	steady uniform	P-SP;P-PIV;P-HW	included
Yu et al. (2013)	C	1	steady uniform	N-SP;N-PIF;N-FORCE;N-FIELD	included
Le et al. (2009)	D	1	steady turbulent	P-SP;P-HW	included
Knisely (1990)	D	1	steady uniform	P-BAL;P-HW	included
Brusco et al. (2022)	A	1	acceleration and deceleration	P-SP;P-VEL	included
Oka and Ishihara (2009)	C	1	steady uniform	N-FORCE;N-FIELD	included
Morello et al. (2025a)	A	1	monotonic acceleration	P-SP;P-PIF	included

Table 13: B/D=1 evidence-condition matrix (continued)

source id	tier	B over D	flow family	direct channels	limitations
Shimada et al. (2002)	C	1	steady uniform	N-SP;N-FORCE;N-FIELD	included
Okajima (1983)	D	1	steady uniform	P-SP;P-HW	included
Guo et al. (2021)	C	1	monotonic acceleration	N-SP;N-FORCE;N-FIELD	included
Da Matha Sant'Anna et al. (1988)	B	1	steady uniform	P-SP	included
Robertson et al. (1978)	B	1	steady turbulent	P-SP	included
Lunghi et al. (2024)	C	1	monotonic acceleration	N-SP;N-FORCE;N-FIELD	included
Brusco et al. (2024)	A	1	downburst-like transient	P-SP;P-PIF;P-VEL	included
van Oudheusden et al. (2005)	B	1	steady uniform	P-PIV	included
Huot et al. (1986)	D	1	steady turbulent	P-SP;P-PIF;P-HW	included
Vickery (1966)	B	1	steady turbulent	P-SP;P-BAL	included
van Oudheusden et al. (2008)	B	1	steady uniform	P-PIV	included
Lander and Letchford (2016)	D	1	steady turbulent	P-SP;P-PIF;P-PIV	included
Yang and Mason (2019)	A	1	monotonic acceleration	P-SP;P-PIF;P-HW	included
Sohankar (2008)	C	1	steady uniform	N-FORCE;N-FIELD	included
M. Li et al. (2023)	A	1	streamwise and vertical sinusoidal (independent cases)	P-SP;P-PIF;P-VEL	included

Table 14: B/D=3 evidence-condition matrix. Comparison ceiling: L0-L2 cross-source. Gaps are uniformly recorded as not reported.

source id	tier	B over D	flow family	direct channels	limitations
Moore et al. (2019)	D	3	steady uniform	P-SP;P-PIV;P-HW	included
Yu et al. (2013)	C	3	steady uniform	N-SP;N-PIF;N-FORCE;N-FIELD	included
Shimada et al. (2002)	C	3	steady uniform	N-SP;N-FORCE;N-FIELD	included

Table 14: B/D=3 evidence-condition matrix (continued)

source id	tier	B over D	flow family	direct channels	limitations
Okajima (1983)	D	3	steady uniform	P-SP;P-HW	included
Da Matha Sant'Anna et al. (1988)	B	3	steady uniform	P-SP	included
Ohya et al. (1992)	C	3	steady uniform	N-FORCE;N-FIELD	included
Nakamura et al. (1991)	B	3	steady uniform	P-HW	included
Zhang and Liu (2017)	D	3	steady uniform	P-SP;P-PIV	included
Yang and Mason (2019)	A	3	monotonic acceleration	P-SP;P-PIF;P-HW	included
Sohankar (2008)	C	3	steady uniform	N-FORCE;N-FIELD	included
Morello et al. (2025b)	D	3	monotonic acceleration	N-FORCE;N-FIELD	numerical mechanistic evidence; does not substitute for physical measurement

Table 15: B/D=5 evidence-condition matrix. Comparison ceiling: L0-L2 cross-source. Gaps are uniformly recorded as not reported.

source id	tier	B over D	flow family	direct channels	limitations
Li et al. (2021a)	B	5	streamwise sinusoidal	P-SP;P-PIF;P-VEL	included
Moore et al. (2019)	D	5	steady uniform	P-SP;P-PIV;P-HW	included
Zhang et al. (2023)	B	5	steady turbulent	N-SP;N-FORCE;N-FIELD	included
Chiarini and Quadrio (2021)	C	5	steady uniform	N-FORCE;N-FIELD;N-PROBE	included
Li et al. (2022)	D	5	downburst-like transient	P-SP;P-VEL	included
Han et al. (2025)	B	5	steady uniform	P-SP;P-PIF;P-BAL	included
Le et al. (2009)	D	5	steady turbulent	P-SP;P-HW	included
Mannini et al. (2017)	D	5	steady turbulent	P-SP;P-PIF;P-BAL;P-HW	included
Matsumoto et al. (2003)	B	5	steady turbulent	P-SP;P-HW	included
Morello et al. (2025a)	A	5	monotonic acceleration	P-SP;P-PIF	included
Li et al. (2021b)	D	5	vertical sinusoidal	P-SP;P-PIF;P-PIV;P-VEL	included
Mariotti et al. (2024)	D	5	steady uniform	P-SP;P-HW	included

Table 15: B/D=5 evidence-condition matrix (continued)

source id	tier	B over D	flow family	direct channels	limitations
Okajima (1983)	D	5	steady uniform	P-SP;P-HW	included
Tang et al. (2023)	B	5	steady turbulent	P-SP;P-PIF;P-VEL	included
Yang et al. (2021)	B	5	steady turbulent	P-SP;P-VEL	included
Wu et al. (2020)	C	5	steady uniform	P-SP;P-PIF;N-FORCE;N-FIELD	included
Schewe (2013)	D	5	steady uniform	P-SP;P-BAL	included
Zhu et al. (2020)	C	5	steady turbulent	N-SP;N-FIELD;N-PROBE	included
Mannini et al. (2011)	C	5	steady uniform	N-FORCE;N-FIELD	included
S. Li et al. (2023)	D	5	steady turbulent	P-PIF;P-VEL	included
Zhang and Xu (2020)	D	5	steady uniform	N-SP;N-FORCE;N-FIELD	included
Ricci et al. (2017)	C	5	steady turbulent	N-SP;N-FORCE;N-FIELD	included
Zhou et al. (2025)	B	5	acceleration and deceleration	P-SP;P-PIF;P-VEL	included
S. Li et al. (2024)	D	5	downburst-like transient	P-SP;P-VEL	included
Patruno et al. (2016)	C	5	steady uniform	N-SP;N-FIELD	included
Wu et al. (2021)	A	5	streamwise sinusoidal	P-SP;P-PIF;P-VEL	included
Bruno et al. (2014)	D	5	steady uniform	Compiled	included
Y. Li et al. (2024)	D	5	steady turbulent	P-SP;P-PIF;P-BAL;P-VEL;ML	included
Guissart et al. (2022)	D	5	steady uniform	P-PIV	included
Morello et al. (2025b)	D	3,5	monotonic acceleration	N-FORCE;N-FIELD	included
Bruno et al. (2010)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	included
Wu et al. (2022)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	included
Bruno et al. (2012)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	included

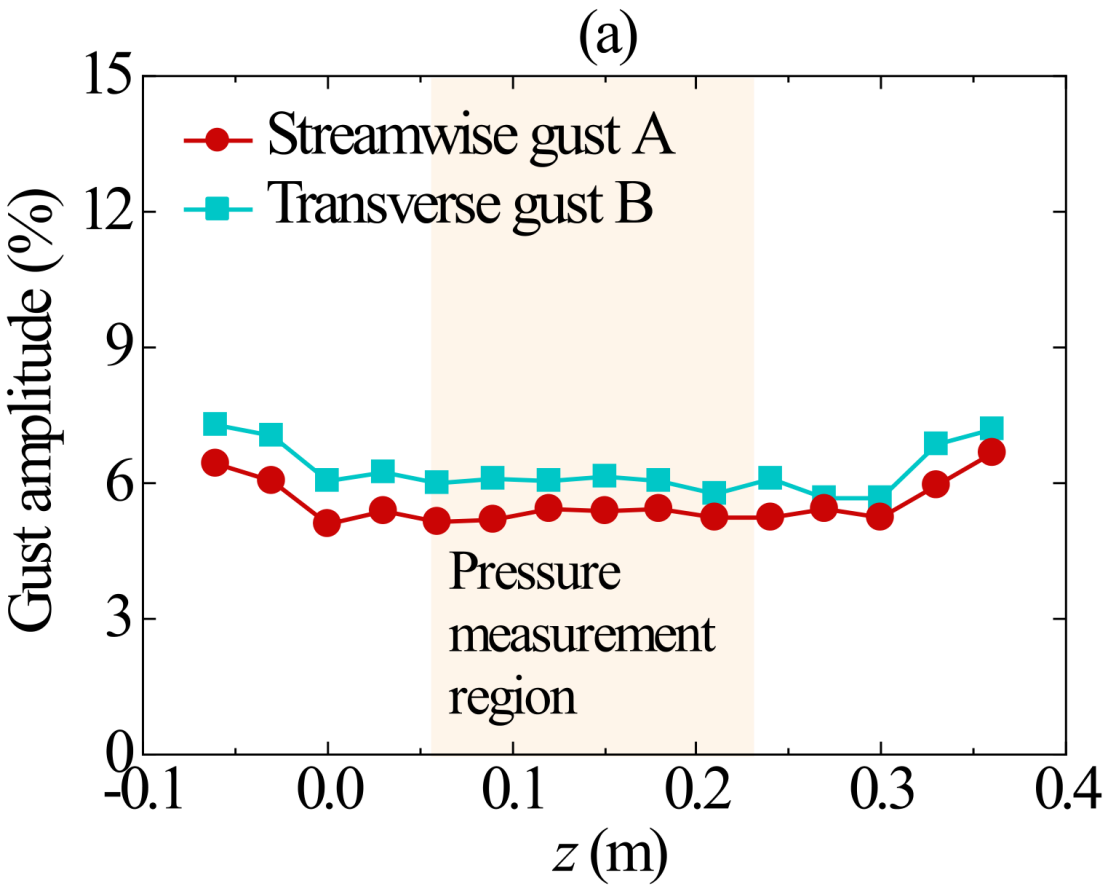


Figure 43: Source figure excerpt. Source M. Li et al. (2023), original figure/table Fig. 5, panel a, PDF p. 5, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

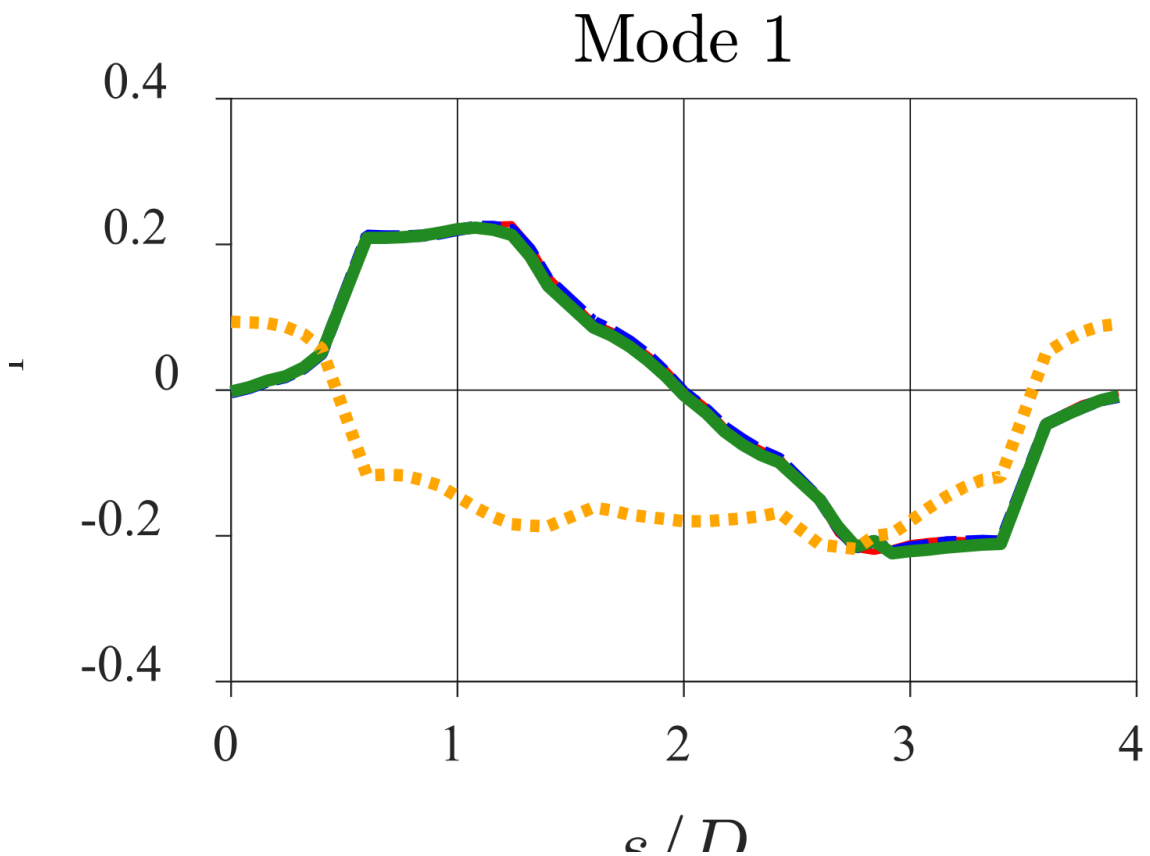


Figure 44: Source figure excerpt. Source M. Li et al. (2023), original figure/table Fig. 11, panel Mode 1, PDF p. 9, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

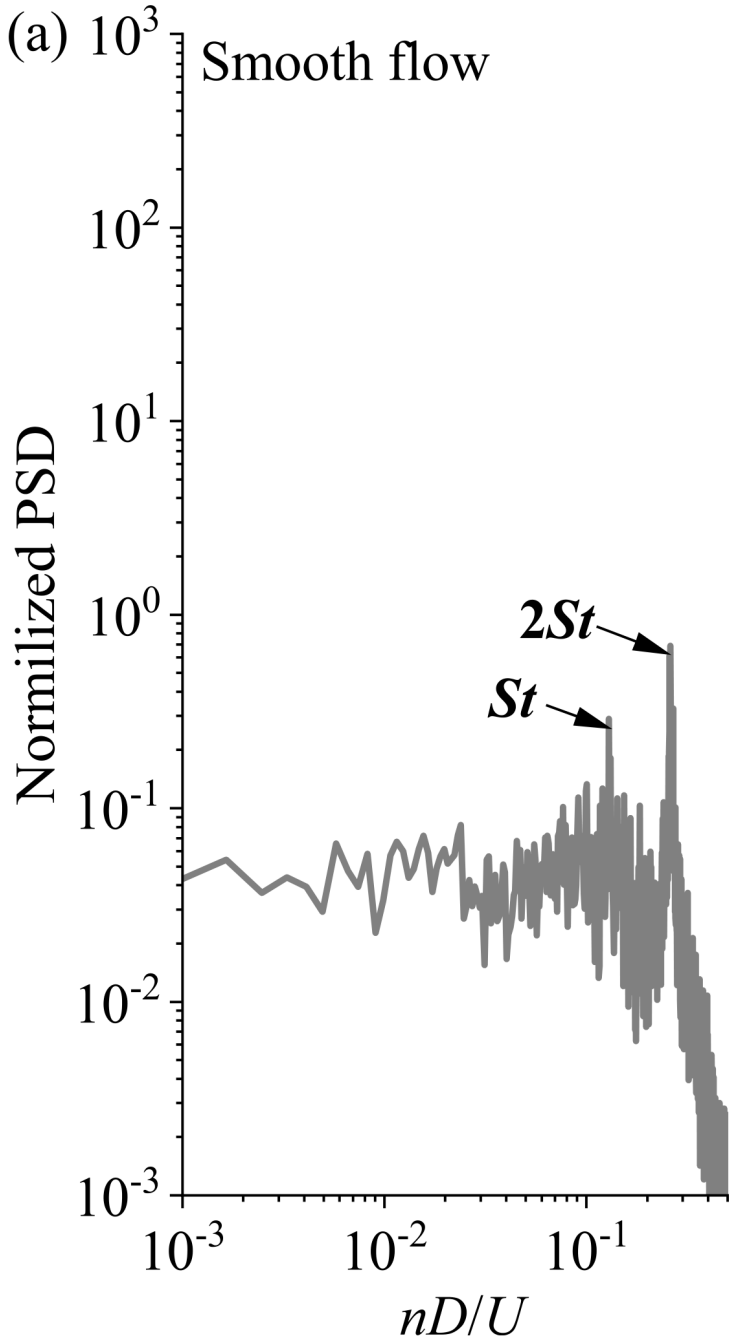


Figure 45: Source figure excerpt. Source M. Li et al. (2023), original figure/table Fig. 19, panel a, PDF p. 14, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

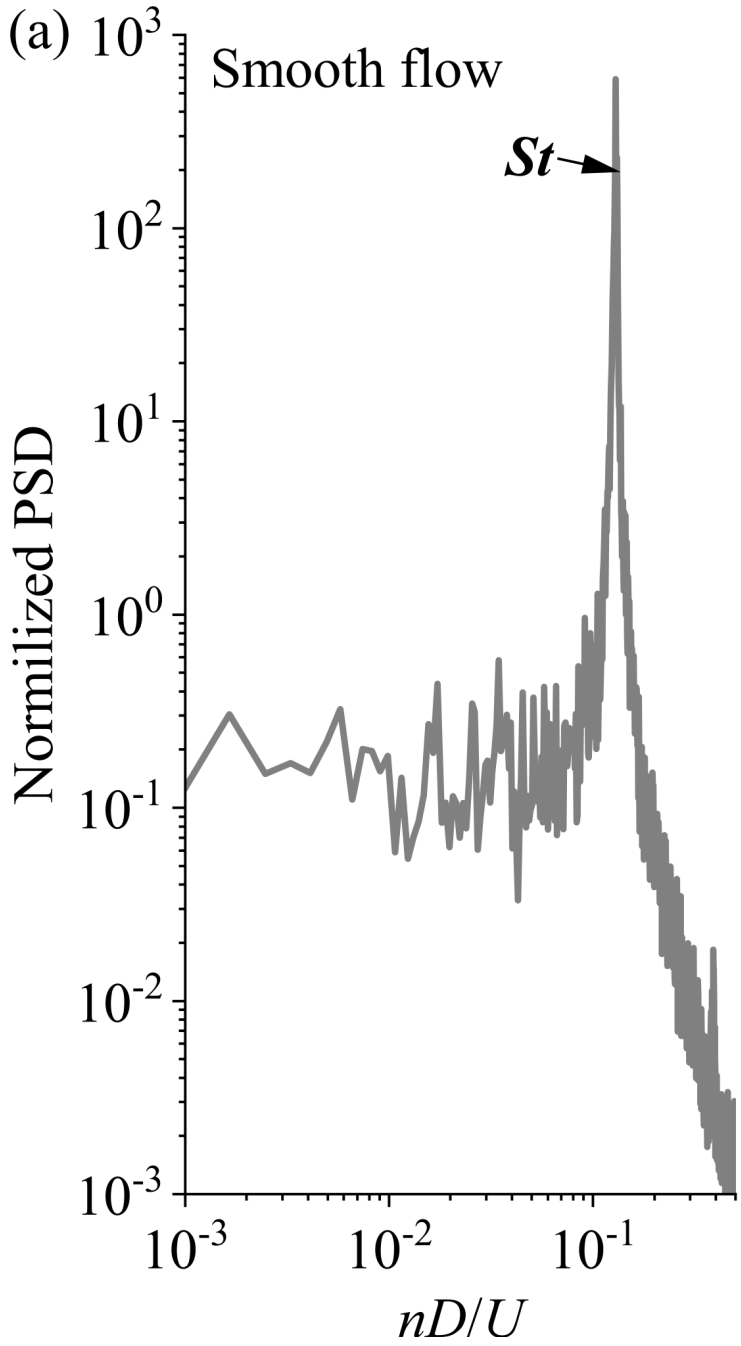
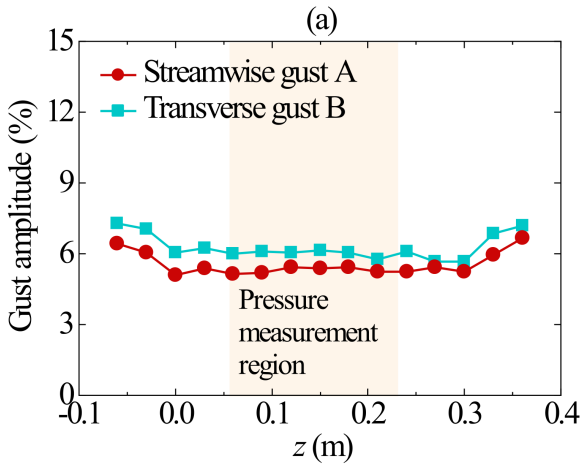
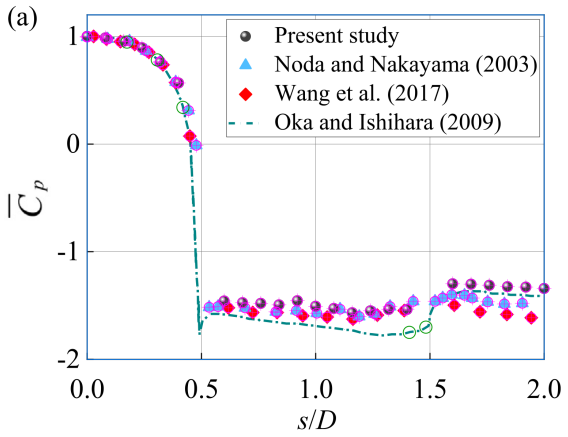


Figure 46: Source figure excerpt. Source M. Li et al. (2023), original figure/table Fig. 20, panel a, PDF p. 14, print p. not reported. Data origin: direct measurement from the source publication. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

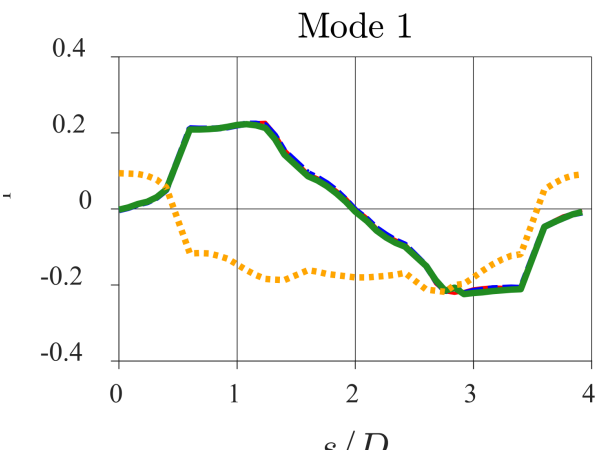


Source figure

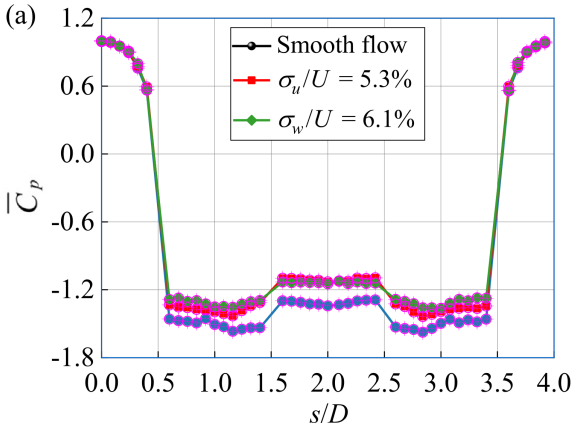


Source-constrained extraction overlay

Figure 47: Source figure and source-constrained extraction overlay shown side by side. Source M. Li et al. (2023), PDF p. 6. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.



Source figure



Source-constrained extraction overlay

Figure 48: Source figure and source-constrained extraction overlay shown side by side. Source M. Li et al. (2023), PDF p. 7. Left: source figure; right: formalised extraction overlay or mask. The overlay is used solely to verify extraction positions, series, or marker identities; it is not an independent measurement and does not elevate the comparison level. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

Table 16: Pressure-channel condition and normalization matrix. Comparison ceiling: L0-L3, no channel pooling. Gaps are uniformly recorded as not reported.

source id	tier	B over D	flow family	channel identity	normalization or reference	limitation
Li et al. (2021a)	B	5	streamwise sinusoidal	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
Moore et al. (2019)	D	1,3,5	steady uniform	P-SP;P-PIV;P-HW	as defined in source; not reported items remain not reported	included
Yu et al. (2013)	C	1,3	steady uniform	N-SP;N-PIF;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Zhang et al. (2023)	B	5	steady turbulent	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Li et al. (2022)	D	5	downburst-like transient	P-SP;P-VEL	as defined in source; not reported items remain not reported	included
Han et al. (2025)	B	5	steady uniform	P-SP;P-PIF;P-BAL	as defined in source; not reported items remain not reported	included
Le et al. (2009)	D	1,5	steady turbulent	P-SP;P-HW	as defined in source; not reported items remain not reported	included
Mannini et al. (2017)	D	5	steady turbulent	P-SP;P-PIF;P-BAL;P-HW	as defined in source; not reported items remain not reported	included
Matsumoto et al. (2003)	B	5	steady turbulent	P-SP;P-HW	as defined in source; not reported items remain not reported	included
Brusco et al. (2022)	A	1	acceleration and deceleration	P-SP;P-VEL	as defined in source; not reported items remain not reported	included
Morello et al. (2025a)	A	1,5	monotonic acceleration	P-SP;P-PIF	as defined in source; not reported items remain not reported	included
Li et al. (2021b)	D	5	vertical sinusoidal	P-SP;P-PIF;P-PIV;P-VEL	as defined in source; not reported items remain not reported	included
Mariotti et al. (2024)	D	5	steady uniform	P-SP;P-HW	as defined in source; not reported items remain not reported	included
Shimada et al. (2002)	C	1,3	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Okajima (1983)	D	1,3,5	steady uniform	P-SP;P-HW	as defined in source; not reported items remain not reported	included
Guo et al. (2021)	C	1	monotonic acceleration	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included

Table 16: Pressure-channel condition and normalization matrix (continued)

source id	tier	B over D	flow family	channel identity	normalization or reference	limitation
Tang et al. (2023)	B	5	steady turbulent	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
Da Matha Sant'Anna et al. (1988)	B	1,3	steady uniform	P-SP	as defined in source; not reported items remain not reported	included
Robertson et al. (1978)	B	1	steady turbulent	P-SP	as defined in source; not reported items remain not reported	included
Lunghi et al. (2024)	C	1	monotonic acceleration	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Brusco et al. (2024)	A	1	downburst-like transient	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
Yang et al. (2021)	B	5	steady turbulent	P-SP;P-VEL	as defined in source; not reported items remain not reported	included
Wu et al. (2020)	C	5	steady uniform	P-SP;P-PIF;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Schewe (2013)	D	5	steady uniform	P-SP;P-BAL	as defined in source; not reported items remain not reported	included
Zhu et al. (2020)	C	5	steady turbulent	N-SP;N-FIELD;N-PROBE	as defined in source; not reported items remain not reported	included
Huot et al. (1986)	D	1	steady turbulent	P-SP;P-PIF;P-HW	as defined in source; not reported items remain not reported	included
Zhang and Xu (2020)	D	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Ricci et al. (2017)	C	5	steady turbulent	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Vickery (1966)	B	1	steady turbulent	P-SP;P-BAL	as defined in source; not reported items remain not reported	included
Zhou et al. (2025)	B	5	acceleration and deceleration	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
S. Li et al. (2024)	D	5	downburst-like transient	P-SP;P-VEL	as defined in source; not reported items remain not reported	included
Zhang and Liu (2017)	D	3	steady uniform	P-SP;P-PIV	as defined in source; not reported items remain not reported	included
Patruno et al. (2016)	C	5	steady uniform	N-SP;N-FIELD	as defined in source; not reported items remain not reported	included

Table 16: Pressure-channel condition and normalization matrix (continued)

source id	tier	B over D	flow family	channel identity	normalization or reference	limitation
Wu et al. (2021)	A	5	streamwise sinusoidal	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
Lander and Letchford (2016)	D	1	steady turbulent	P-SP;P-PIF;P-PIV	as defined in source; not reported items remain not reported	included
Y. Li et al. (2024)	D	5	steady turbulent	P-SP;P-PIF;P-BAL;P-VEL;ML	as defined in source; not reported items remain not reported	included
Yang and Mason (2019)	A	1,3	monotonic acceleration	P-SP;P-PIF;P-HW	as defined in source; not reported items remain not reported	included
Bruno et al. (2010)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Wu et al. (2022)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
M. Li et al. (2023)	A	1	streamwise and vertical sinusoidal (independent cases)	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
Bruno et al. (2012)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included

Table 17: Force-channel provenance and normalization matrix. Comparison ceiling: L0-L3, no channel pooling. Gaps are uniformly recorded as not reported.

source id	tier	B over D	flow family	channel identity	normalization or reference	limitation
Li et al. (2021a)	B	5	streamwise sinusoidal	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
Yu et al. (2013)	C	1,3	steady uniform	N-SP;N-PIF;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Zhang et al. (2023)	B	5	steady turbulent	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Chiarini and Quadrio (2021)	C	5	steady uniform	N-FORCE;N-FIELD;N-PROBE	as defined in source; not reported items remain not reported	included

Table 17: Force-channel provenance and normalization matrix (continued)

source id	tier	B over D	flow family	channel identity	normalization or reference	limitation
Han et al. (2025)	B	5	steady uniform	P-SP;P-PIF;P-BAL	as defined in source; not reported items remain not reported	included
Mannini et al. (2017)	D	5	steady turbulent	P-SP;P-PIF;P-BAL;P-HW	as defined in source; not reported items remain not reported	included
Knisely (1990)	D	1	steady uniform	P-BAL;P-HW	as defined in source; not reported items remain not reported	included
Oka and Ishihara (2009)	C	1	steady uniform	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Morello et al. (2025a)	A	1,5	monotonic acceleration	P-SP;P-PIF	as defined in source; not reported items remain not reported	included
Li et al. (2021b)	D	5	vertical sinusoidal	P-SP;P-PIF;P-PIV;P-VEL	as defined in source; not reported items remain not reported	included
Shimada et al. (2002)	C	1,3	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Guo et al. (2021)	C	1	monotonic acceleration	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Tang et al. (2023)	B	5	steady turbulent	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
Lunghi et al. (2024)	C	1	monotonic acceleration	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Brusco et al. (2024)	A	1	downburst-like transient	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
Ohya et al. (1992)	C	3	steady uniform	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Wu et al. (2020)	C	5	steady uniform	P-SP;P-PIF;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Schewe (2013)	D	5	steady uniform	P-SP;P-BAL	as defined in source; not reported items remain not reported	included
Mannini et al. (2011)	C	5	steady uniform	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Huot et al. (1986)	D	1	steady turbulent	P-SP;P-PIF;P-HW	as defined in source; not reported items remain not reported	included
S. Li et al. (2023)	D	5	steady turbulent	P-PIF;P-VEL	as defined in source; not reported items remain not reported	included

Table 17: Force-channel provenance and normalization matrix (continued)

source id	tier	B over D	flow family	channel identity	normalization or reference	limitation
Zhang and Xu (2020)	D	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Ricci et al. (2017)	C	5	steady turbulent	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Vickery (1966)	B	1	steady turbulent	P-SP;P-BAL	as defined in source; not reported items remain not reported	included
Zhou et al. (2025)	B	5	acceleration and deceleration	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
Li et al. (2025)(numerical hypothetical line, B/D = not reported)	C	not reported	multi-ramp acceleration numerical	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Wu et al. (2021)	A	5	streamwise sinusoidal	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included
Lander and Letchford (2016)	D	1	steady turbulent	P-SP;P-PIF;P-PIV	as defined in source; not reported items remain not reported	included
Y. Li et al. (2024)	D	5	steady turbulent	P-SP;P-PIF;P-BAL;P-VEL;ML	as defined in source; not reported items remain not reported	included
Yang and Mason (2019)	A	1,3	monotonic acceleration	P-SP;P-PIF;P-HW	as defined in source; not reported items remain not reported	included
Morello et al. (2025b)	D	3,5	monotonic acceleration	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Bruno et al. (2010)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Wu et al. (2022)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Sohankar (2008)	C	1,3	steady uniform	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
M. Li et al. (2023)	A	1	streamwise and vertical sinusoidal (independent cases)	P-SP;P-PIF;P-VEL	as defined in source; not reported items remain not reported	included

Bruno et al. (2012)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included

Table 18: Numerical-channel condition matrix. Comparison ceiling: L0-L3, no channel pooling. Gaps are uniformly recorded as not reported.

source id	tier	B over D	flow family	channel identity	normalization or reference	limitation
Yu et al. (2013)	C	1,3	steady uniform	N-SP;N-PIF;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Zhang et al. (2023)	B	5	steady turbulent	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Chiarini and Quadrio (2021)	C	5	steady uniform	N-FORCE;N-FIELD;N-PROBE	as defined in source; not reported items remain not reported	included
Oka and Ishihara (2009)	C	1	steady uniform	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Shimada et al. (2002)	C	1,3	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Guo et al. (2021)	C	1	monotonic acceleration	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Lunghi et al. (2024)	C	1	monotonic acceleration	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Ohya et al. (1992)	C	3	steady uniform	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Zhu et al. (2020)	C	5	steady turbulent	N-SP;N-FIELD;N-PROBE	as defined in source; not reported items remain not reported	included
Mannini et al. (2011)	C	5	steady uniform	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Zhang and Xu (2020)	D	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Ricci et al. (2017)	C	5	steady turbulent	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Li et al. (2025)(numerical hypothetical line, B/D = not reported)	C	not reported	multi-ramp acceleration numerical	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Patruno et al. (2016)	C	5	steady uniform	N-SP;N-FIELD	as defined in source; not reported items remain not reported	included
Morello et al. (2025b)	D	3,5	monotonic acceleration	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included

Table 18: Numerical-channel condition matrix (continued)

source id	tier	B over D	flow family	channel identity	normalization or reference	limitation
Bruno et al. (2010)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Wu et al. (2022)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Sohankar (2008)	C	1,3	steady uniform	N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included
Bruno et al. (2012)	C	5	steady uniform	N-SP;N-FORCE;N-FIELD	as defined in source; not reported items remain not reported	included

Table 19: Monotonic acceleration, acceleration/deceleration and downburst-like family matrix. Comparison ceiling: L0-L2 cross-source. Gaps are uniformly recorded as not reported.

source id	tier	B over D	U t family	physical or numerical	channels	limitation
Li et al. (2022)	D	5	downburst-like transient	physical experiment	P-SP;P-VEL	included
Brusco et al. (2022)	A	1	acceleration and deceleration	physical experiment	P-SP;P-VEL	included
Morello et al. (2025a)	A	1,5	monotonic acceleration	physical experiment	P-SP;P-PIF	included
Guo et al. (2021)	C	1	monotonic acceleration	numerical study	N-SP;N-FORCE;N-FIELD	included
Lunghi et al. (2024)	C	1	monotonic acceleration	numerical study	N-SP;N-FORCE;N-FIELD	included
Brusco et al. (2024)	A	1	downburst-like transient	physical experiment	P-SP;P-PIF;P-VEL	included
Zhou et al. (2025)	B	5	acceleration and deceleration	physical experiment	P-SP;P-PIF;P-VEL	included
S. Li et al. (2024)	D	5	downburst-like transient	physical experiment	P-SP;P-VEL	included
Li et al. (2025)(numerical hypothetical line, B/D = not reported)	C	not reported	multi-ramp acceleration numerical	numerical study	N-FORCE;N-FIELD	included
Yang and Mason (2019)	A	1,3	monotonic acceleration	physical experiment	P-SP;P-PIF;P-HW	included
Morello et al. (2025b)	D	3,5	monotonic acceleration	numerical study	N-FORCE;N-FIELD	included

Table 20: Streamwise sinusoidal inflow condition matrix. Comparison ceiling: L0-L2 cross-source. Gaps are uniformly recorded as not reported.

source id	tier	B over D	U t family	physical or numerical	channels	limitation
Li et al. (2021a)	B	5	streamwise sinusoidal	physical experiment	P-SP;P-PIF;P-VEL	included
Wu et al. (2021)	A	5	streamwise sinusoidal	physical experiment	P-SP;P-PIF;P-VEL	included
M. Li et al. (2023)	A	1	streamwise sinusoidal (independent cases)	physical experiment	P-SP;P-PIF;P-VEL	included

Table 21: Vertical sinusoidal inflow condition matrix. Comparison ceiling: L0-L2 cross-source. Gaps are uniformly recorded as not reported.

source id	tier	B over D	U t family	physical or numerical	channels	limitation
Li et al. (2021b)	D	5	vertical sinusoidal	physical experiment	P-SP;P-PIF;P-PIV;P-VEL	included
M. Li et al. (2023)	A	1	vertical sinusoidal (independent cases)	physical experiment	P-SP;P-PIF;P-VEL	included

Table 22: Multiaxis sinusoidal inflow condition matrix. Comparison ceiling: L0-L2 cross-source. Gaps are uniformly recorded as not reported.

source id	tier	B over D	U t family	physical or numerical	channels	limitation
no synchronized multiaxis source	–	–	not observed	–	–	M. Li et al. (2023) contains only independent single-axis cases; multiaxis combined effects have not been verified

13 Source-text verification of steady-state and mechanism-anchoring studies

The role of source-text anchors is to expose definitions and limitations, not to provide paper-by-paper summaries. The short quotations below are tied respectively to end conditions, reattachment location, independent force measurement, numerical reattachment criteria, frequency branching, force-channel provenance, PIV reattachment point, sinusoidal-flow pressure recovery, pressure-integrated drag in the acceleration phase, and DNS integration-time limitations.

13.1 Source Da Matha Sant'Anna et al. (1988)

> *"The cylinders spanned the complete wind tunnel width."*
>
> Source Da Matha Sant'Anna et al. (1988), PDF p. 1, print p. 241, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the test cylinders spanned the full wind-tunnel width; the end condition corresponds to a wall-to-wall installation.

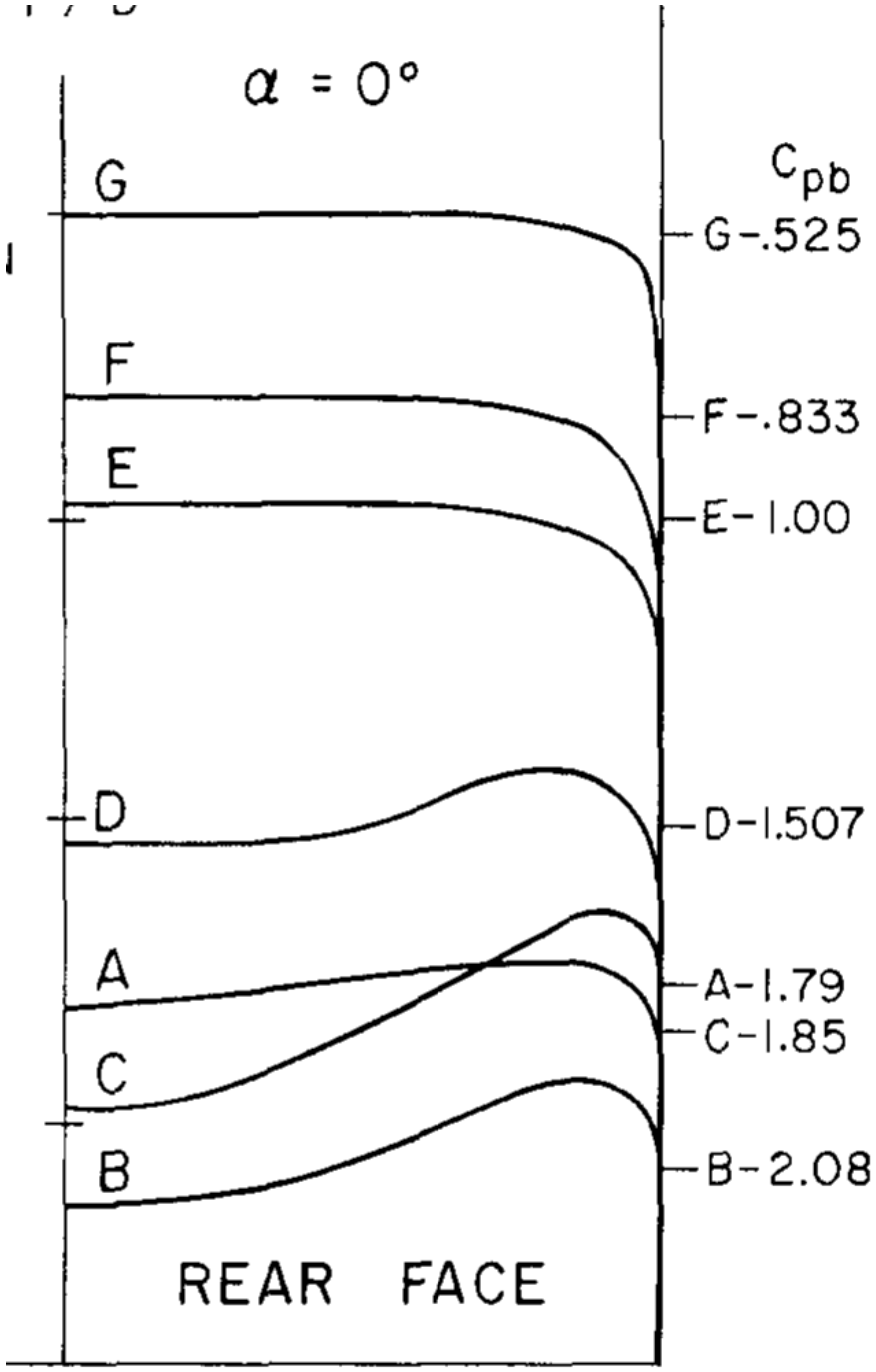


Figure 49: Source figure excerpt. Source Da Matha Sant'Anna et al. (1988), original figure/table F01d, panel as originally cropped, PDF p. not reported, print p. not reported. Data provenance: source-document figure. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

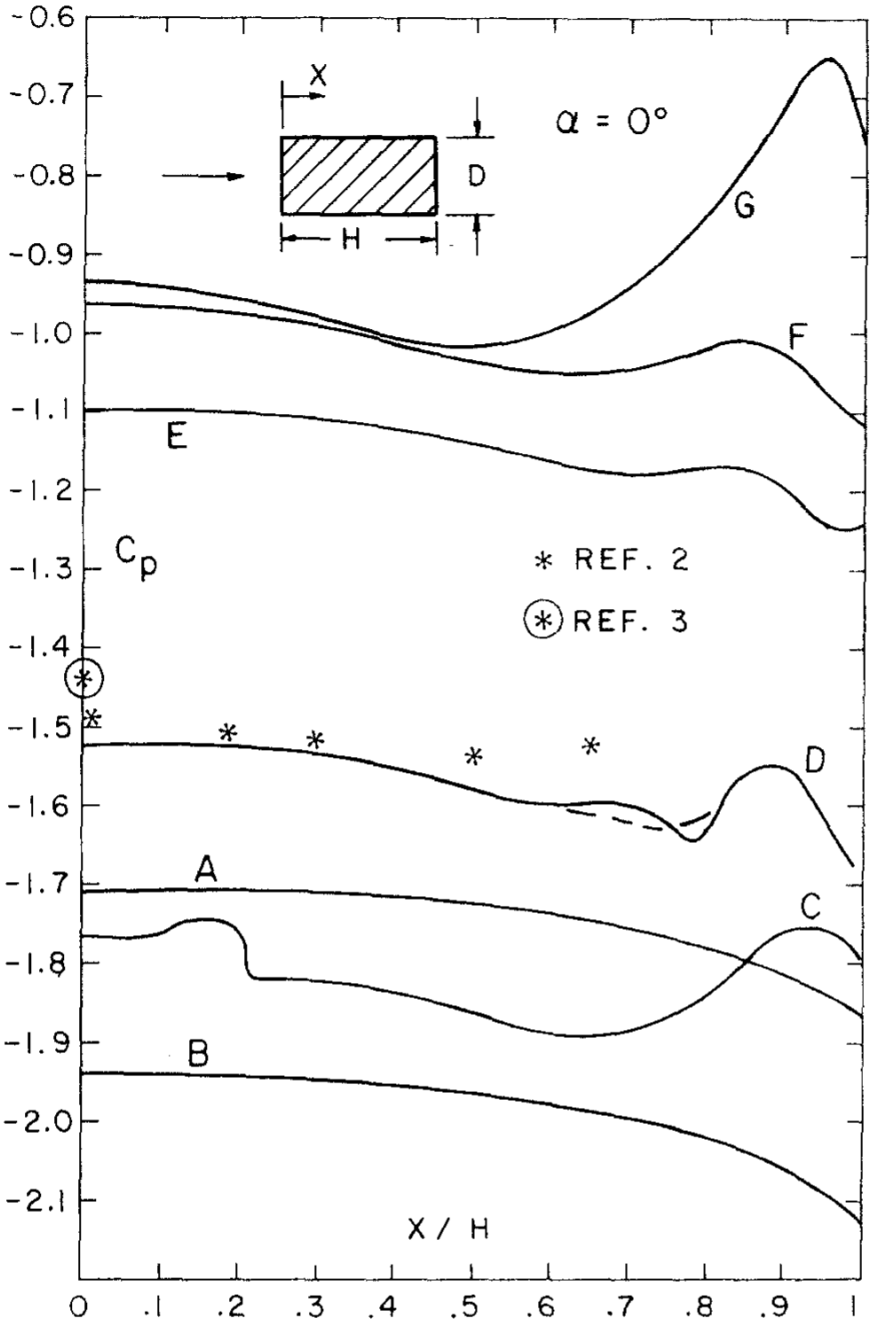


Figure 50: Source figure excerpt. Source Da Matha Sant'Anna et al. (1988), original figure/table F02a, panel as originally cropped, PDF p. not reported, print p. not reported. Data provenance: source-document figure. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

13.2 Source Robertson et al. (1978)

> *"the reattachment locale occurs somewhat down- stream of the peak in r.m.s, pressure fluctuations and upstream of the mean pressure peak."*
>
> Source Robertson et al. (1978), PDF p. 7, print p. 351, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the reattachment location lies downstream of the r.m.s. pressure-fluctuation peak and upstream of the mean pressure peak.

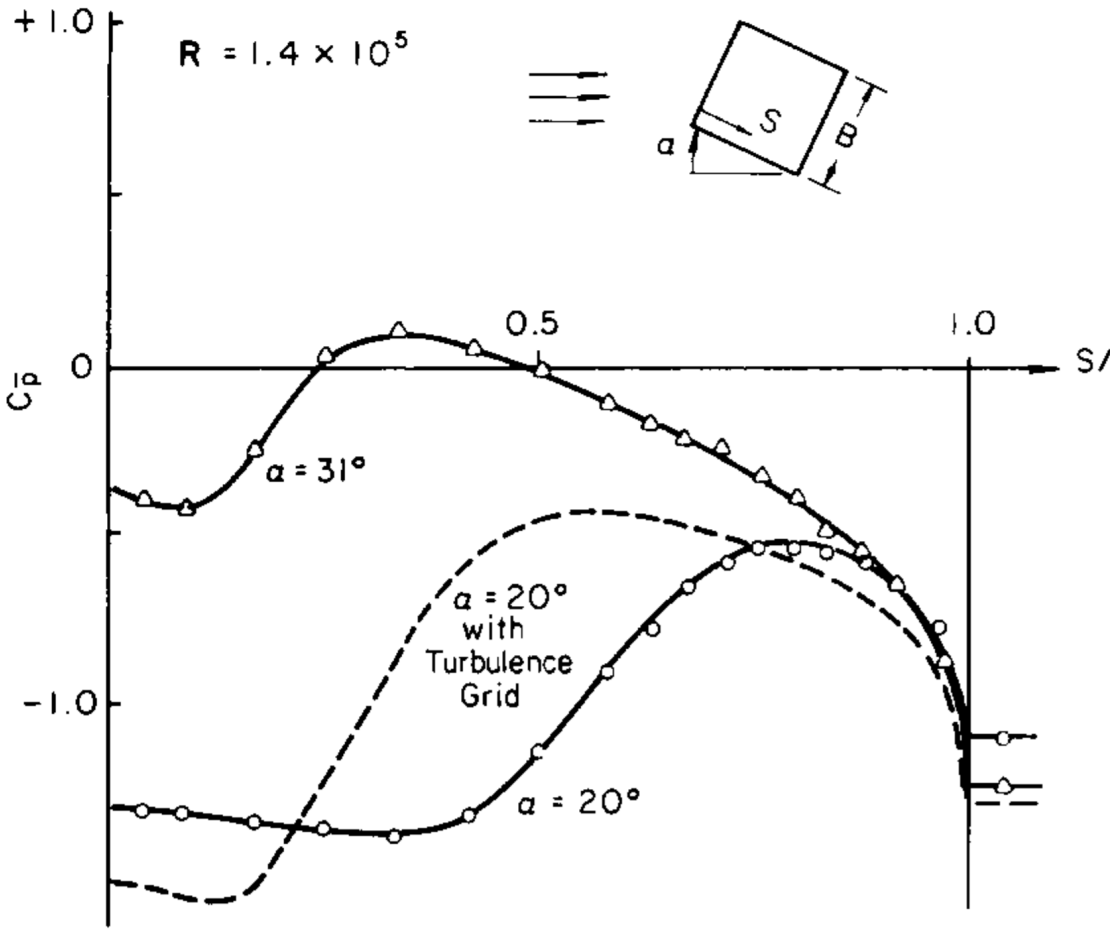


Figure 51: Source figure excerpt. Source Robertson et al. (1978), original figure/table FIG03A, panel as originally cropped, PDF p. not reported, print p. not reported. Data provenance: source-document figure. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

S/B

Fig.3. Variation in mean and r.m.s. pressure coefficient for $\alpha = 20°$ and $\alpha = 31°$.

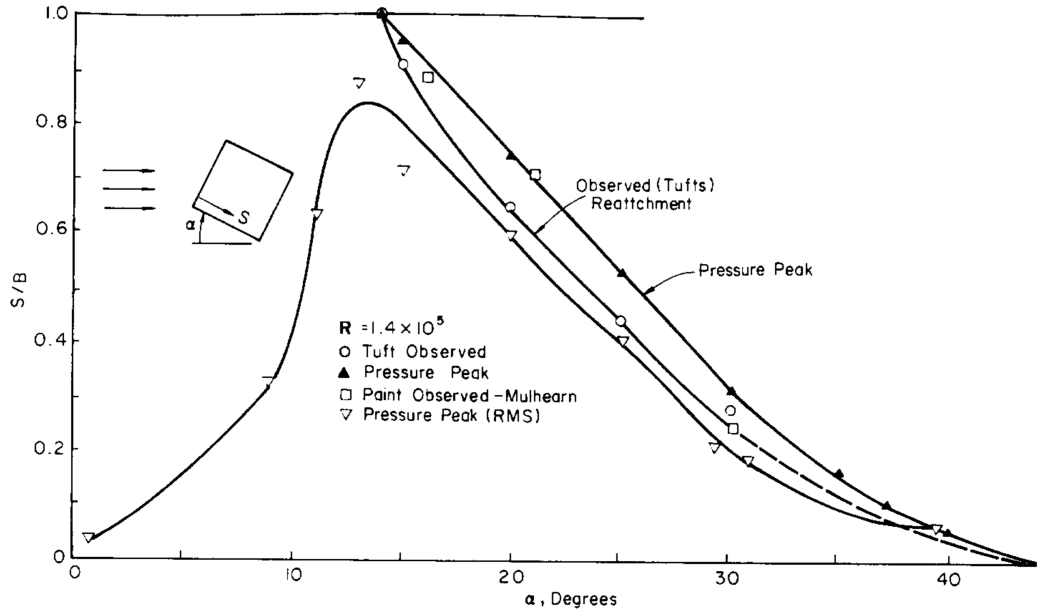


Fig.4. Reattachment and pressure peak locales for various angles.

Figure 52: Source figure excerpt. Source Robertson et al. (1978), original figure/table FIG04, panel as originally cropped, PDF p. not reported, print p. not reported. Data provenance: source-document figure. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

13.3 Source Vickery (1966)

"The root-mean-square coefficient of fluctuating drag was 0.17 in both smooth and turbulent flow"

Source Vickery (1966), PDF p. 7, print p. 487, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the root-mean-square fluctuating drag coefficient was 0.17 in both smooth and turbulent flow.

g/core/terms. https://doi.or

for tunnel blockage were quite substantial and varied from about 10 % at $\alpha = 0°$ to a maximum at $\alpha = 45°$ of 23 %. The basis on which the blockage corrections were determined is presented in Appendix 1. The correction for the average drag and the average pressure coefficient has been well substantiated by experimental observations (Maskell 1963). The corrections for the lift fluctuations and the Strouhal number have not been verified experimentally and are used on

Figure 53: Source figure excerpt. Source Vickery (1966), original figure/table FIGURE 2, panel inset, PDF p. 4, print p. not reported. Data provenance: direct measurement from the source document. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

l9 at 06:59:23, subject to the Cambridge Core terms of use, available at https://www.cambridge

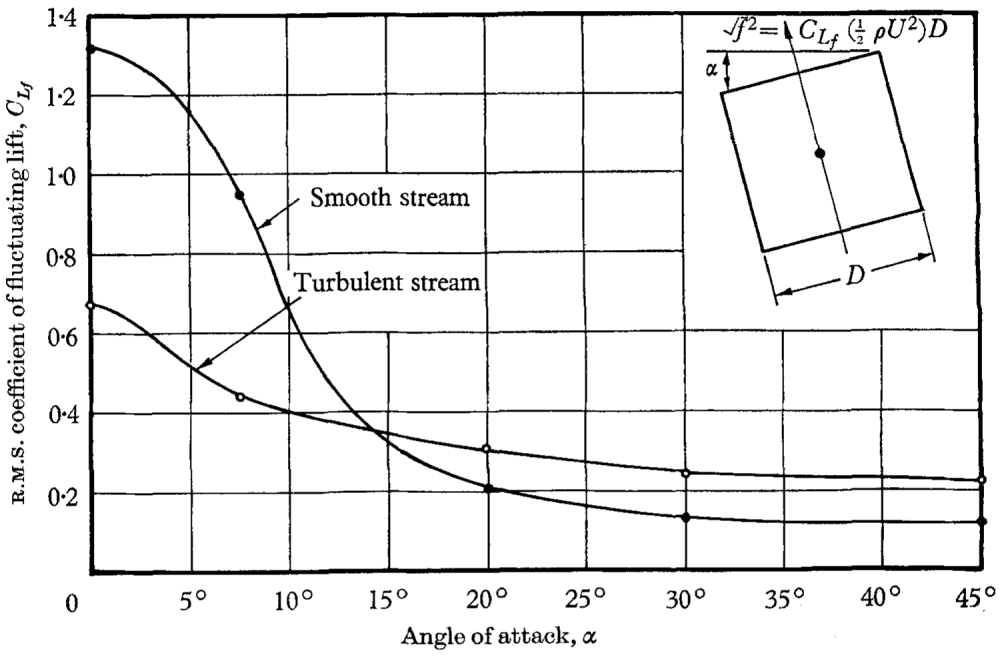


FIGURE 2. Variation of the coefficient of fluctuating lift with angle of attack.

Figure 54: Source figure excerpt. Source Vickery (1966), original figure/table FIG02-plot, panel as originally cropped, PDF p. not reported, print p. not reported. Data provenance: source-document figure. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

13.4 Source Yu et al. (2013)

"For the flow around prisms with aspect ratios of 3.0 and greater, the separated flow reattaches to the faces toward the trailing edge."

Source Yu et al. (2013), PDF p. 2, print p. 430, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the numerical results indicate that for aspect ratios of 3.0 and above, the separated flow reattaches on the side faces near the trailing edge.

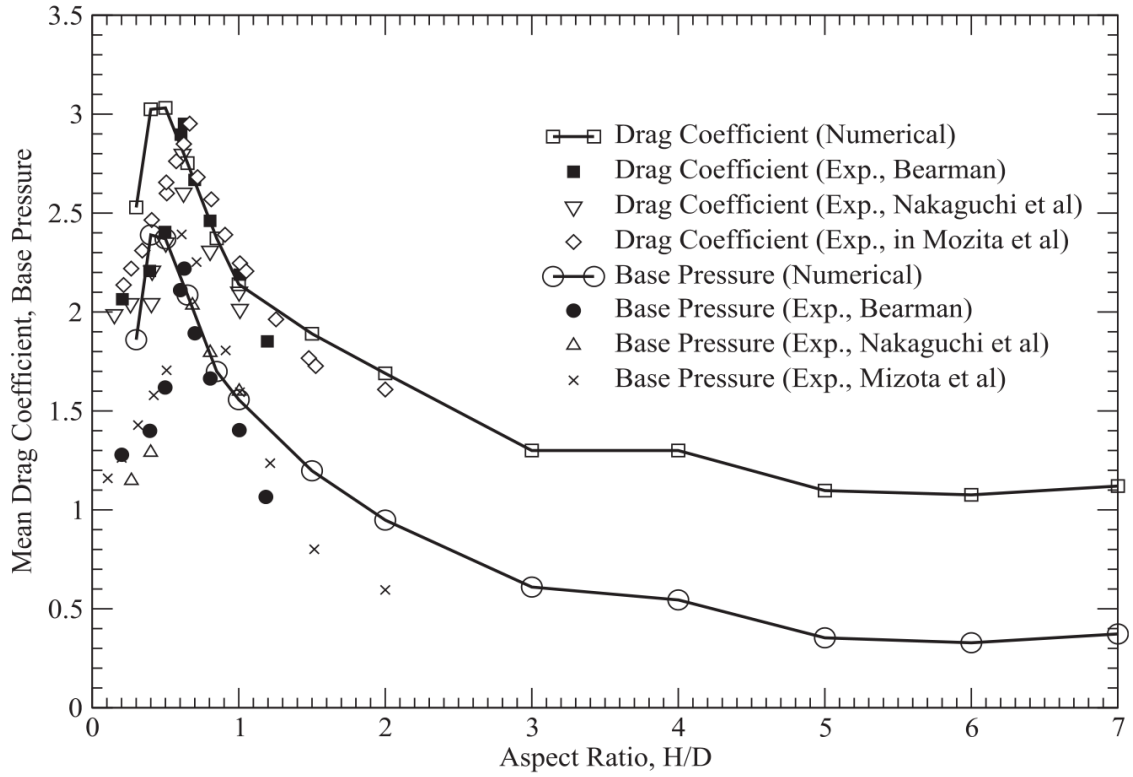


Figure 55: Source figure excerpt. Source Yu et al. (2013), original figure/table Fig. 4, panel (a), PDF p. 5, print p. not reported. Data provenance: numerical results from the source document. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

13.5 Source Nakamura et al. (1991)

"The Strouhal number which is based on the plate's chord is approximately constant and equal to 0.6 for chord-to-thickness ratios 3-5. With further increase in the ratio it increases stepwise to values that are approximately equal to integral multiples of 0.6."

Source Nakamura et al. (1991), PDF p. 1, print p. 437, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the chord-normalized Strouhal number is approximately 0.6 for chord-to-thickness ratios of 3 to 5 and increases stepwise with further increases in the ratio.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

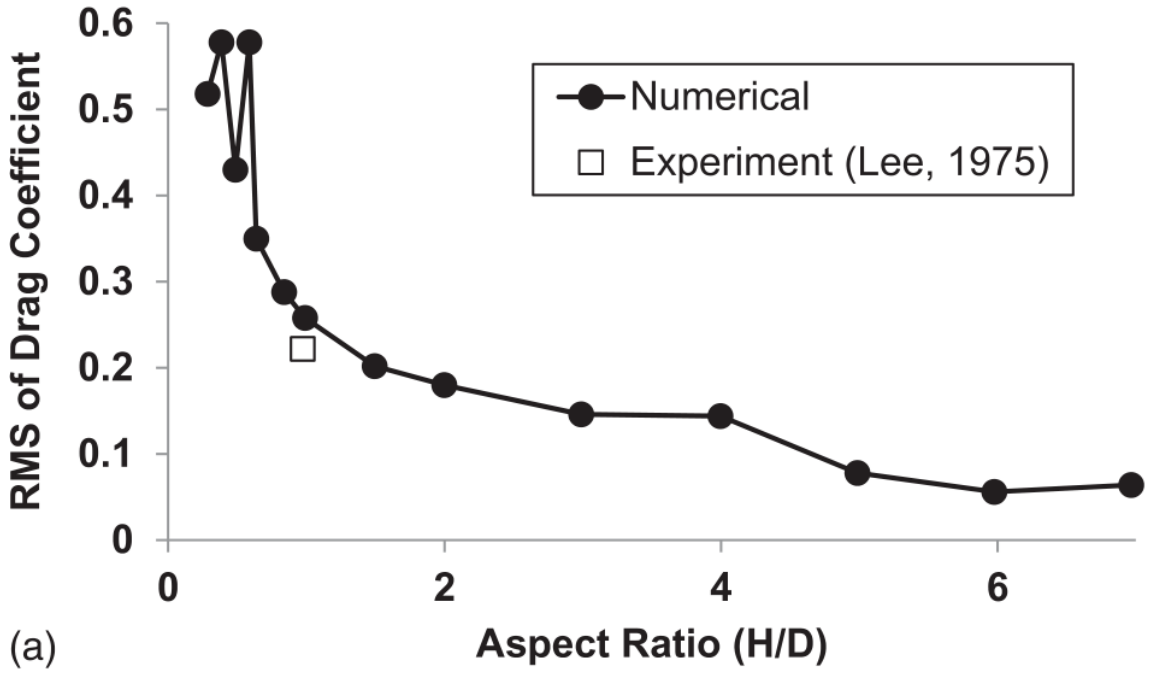


Fig. 8. RMS of the integral forces on prism

Figure 56: Source figure excerpt. Source Yu et al. (2013), original figure/table Fig. 8, panel (a), PDF p. 8, print p. not reported. Data provenance: numerical results from the source document. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

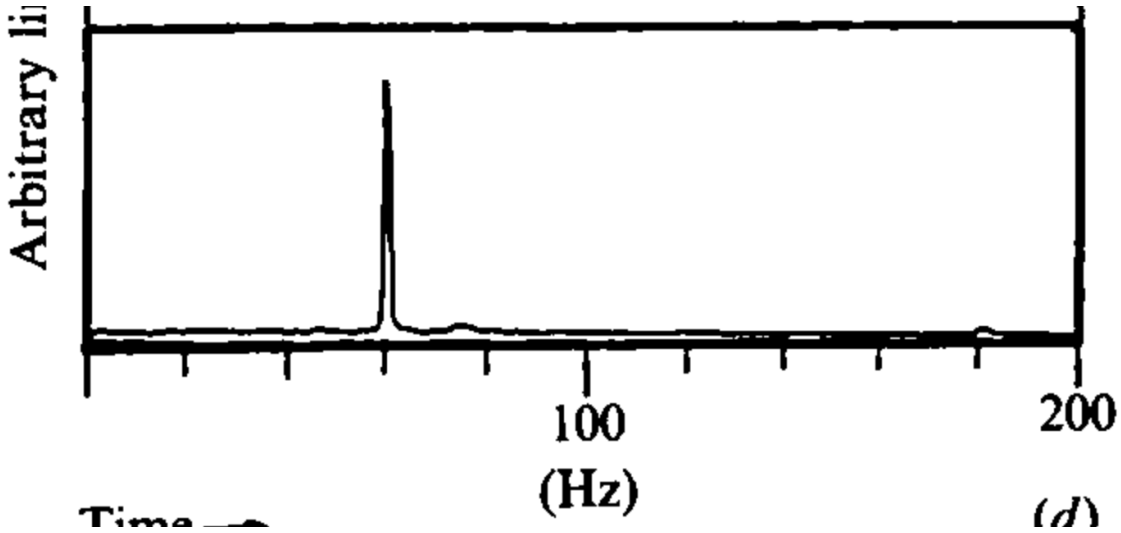


Figure 57: Source figure excerpt. Source Nakamura et al. (1991), original figure/table Figure 3, panel (a) upper box, time history, PDF p. 4, print p. not reported. Data provenance: source-document figure. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

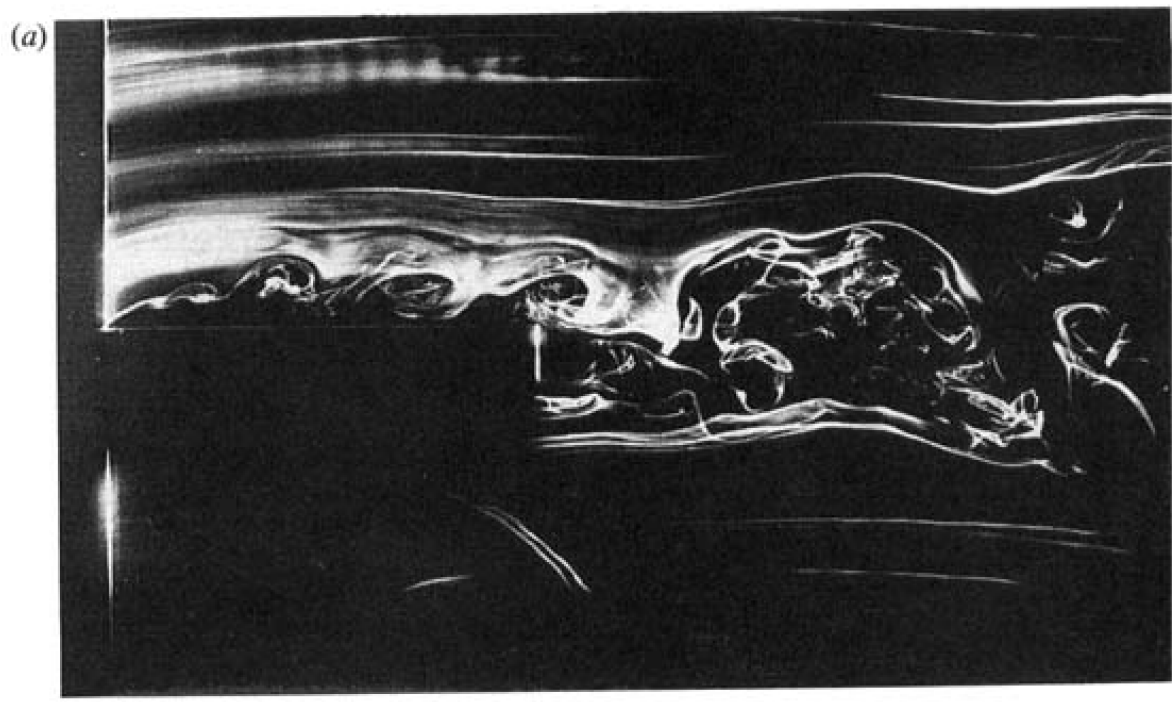


Figure 58: Source figure excerpt. Source Nakamura et al. (1991), original figure/table Figure 6, panel (a), PDF p. 7, print p. not reported. Data provenance: source-document figure. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

13.6 Source Zhang and Liu (2017)

> *"the separated shear layer reattaches onto the plate's surface near x/D = 4.7"*
>
> Source Zhang and Liu (2017), PDF p. 5, print p. 203, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the separated shear layer of the L/D=6 plate reattaches near x/D = 4.7.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

13.7 Source Ohya et al. (1992)

> *"the ratio of these values is nearly 1:2:3"*
>
> Source Ohya et al. (1992), PDF p. 6, print p. 450, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the chord-normalized vortex shedding frequencies for d/h=3, 6, and 9 are approximately in the ratio 1:2:3.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

13.8 Source Han et al. (2025)

> *"The aerodynamic coefficient time history is derived through pres- sure integration."*
>
> Source Han et al. (2025), PDF p. 7, print p. 045121-6, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the aerodynamic coefficient time histories in this source were obtained from pressure integration, not from independent balance time histories.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

13.9 Source Guissart et al. (2022)

"the reattachment point illustrated by the yellow square marker located at x/D = 1.64"

Source Guissart et al. (2022), PDF p. 4, print p. 18, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: at zero angle of attack and Re=2×10^4, the PIV-identified reattachment point is located at x/D=1.64.

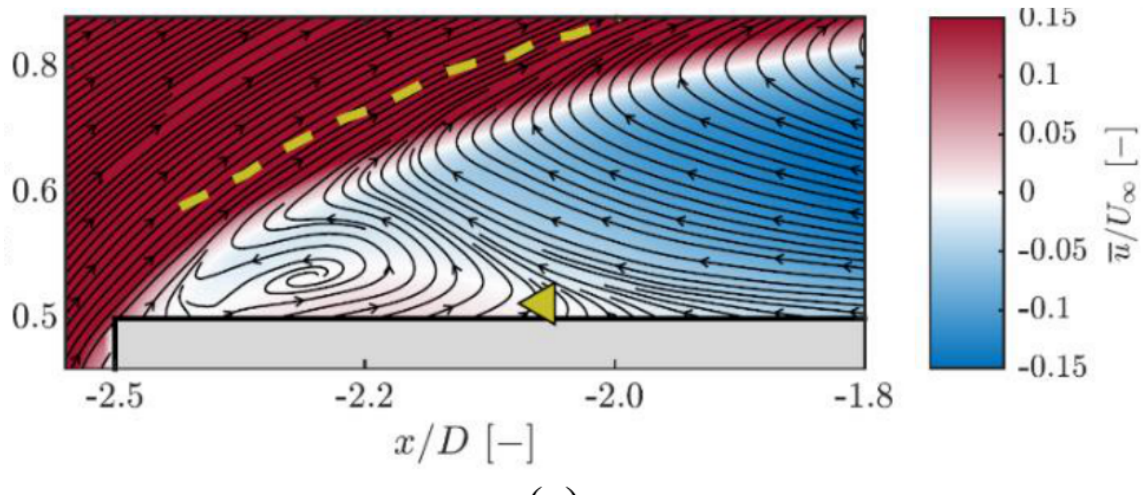


Figure 59: Source figure excerpt. Source Guissart et al. (2022), original figure/table OBJ004, panel as originally cropped, PDF p. not reported, print p. not reported. Data provenance: source-document figure. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

13.10 Source Li et al. (2021a)

"The pressure recovery starts around 2.0D in smooth flow, while the starting points are much closer to the separation point, i.e., around 0.5D in SSOFs."

Source Li et al. (2021a), PDF p. 6, print p. 6, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: in streamwise sinusoidal oscillatory flow the pressure-recovery onset is approximately 0.5D, substantially earlier than the approximately 2.0D observed in smooth flow.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

13.11 Source Zhou et al. (2025)

"CD is increased to 1.169, about 13.0% higher than that for the stationary flow"

Source Zhou et al. (2025), PDF p. 12, print p. 12, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the mean drag coefficient in the acceleration phase with af=4.30 m/s2 increased to 1.169, approximately 13.0% above the stationary-flow value.

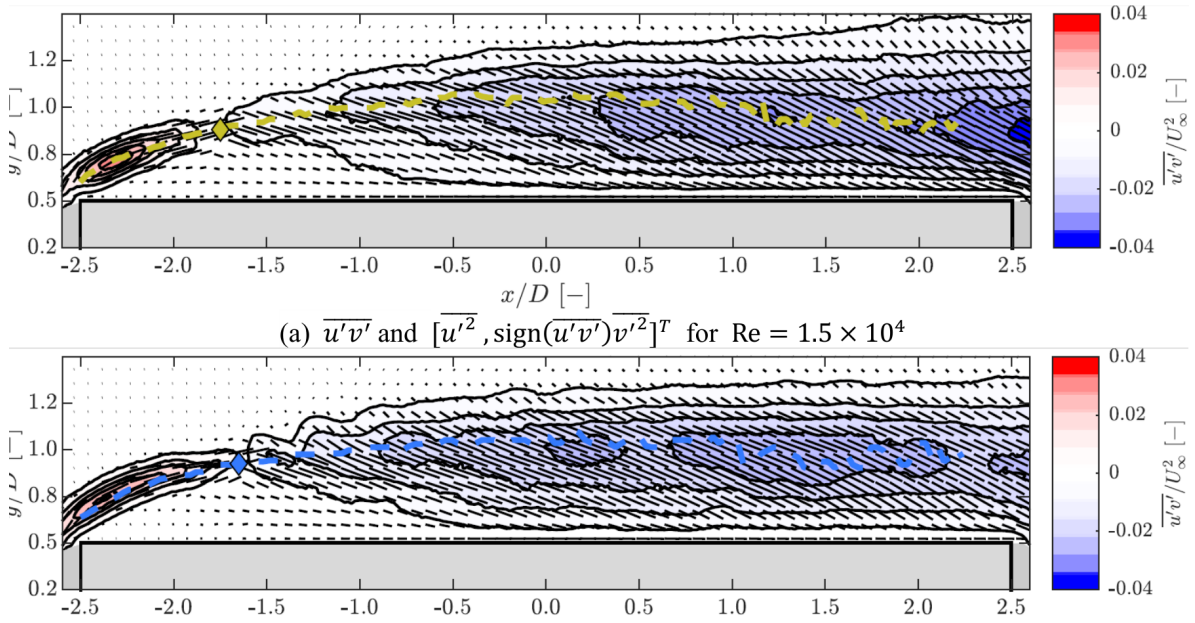


Figure 60: Source figure excerpt. Source Guissart et al. (2022), original figure/table OBJ013-014, panel as originally cropped, PDF p. not reported, print p. not reported. Data provenance: source-document figure. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

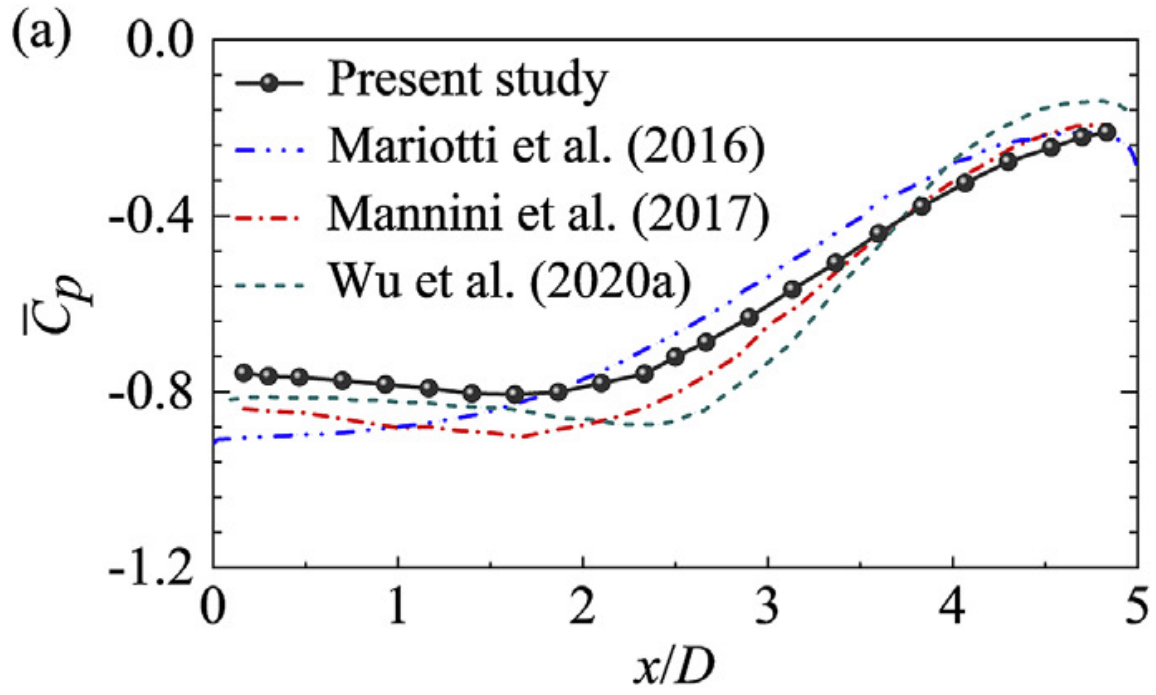


Figure 61: Source figure excerpt. Source Li et al. (2021a), original figure/table Fig. 6, panel a, PDF p. 5, print p. not reported. Data provenance: direct measurement from the source document. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

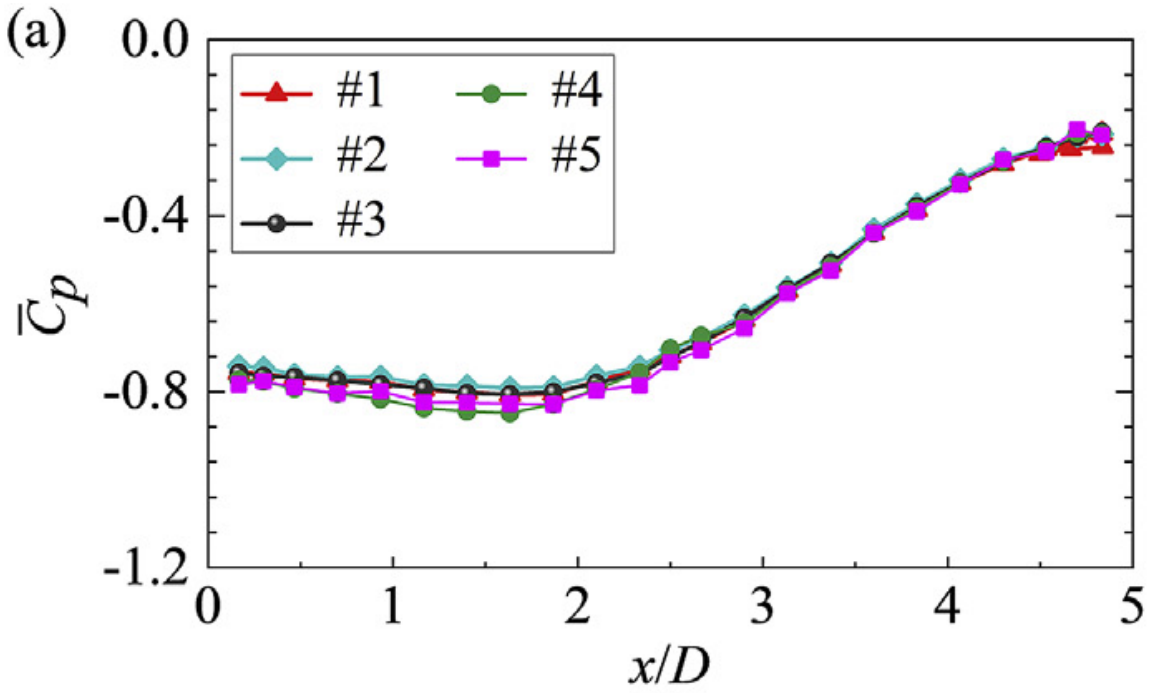


Figure 62: Source figure excerpt. Source Li et al. (2021a), original figure/table Fig. 7, panel a, PDF p. 5, print p. not reported. Data provenance: direct measurement from the source document. Permitted interpretation: only within the source conditions, methods, responses, or mechanisms stated in this caption. Prohibited interpretation: cross-source numerical merging, derivation of design corrections, or presumption of public-release permission. Source figure excerpt, internal academic review; copyright clearance required before public dissemination.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

13.12 Source Chiarini and Quadrio (2021)

> *"However, the low Reynolds number and the amount of integration time preclude any firm statement."*
>
> Source Chiarini and Quadrio (2021), PDF p. 9, print p. 883, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the low Reynolds number and limited integration time prevented the authors from drawing a definitive conclusion on low-frequency unsteadiness.

This anchor is used only within the geometry, inflow, measurement channel, normalization, and page range it records. Cross-source statements remain faceted; visual similarity does not elevate the comparison level.

14 Numerical assumptions, physical gaps, and experimental research directions

> *"For R3, R4, and R5, the maximum drag coefficients exceed the quasi-steady values by 22.4%, 32.6%, and 46.1%, respectively, while for R8, R10, R12, and R16, the increases are 74.0%, 82.0%, 94.3%, and 109.7%, respectively."*
>
> Source Li et al. (2025) (numerical-assumption line, B/D=not reported), PDF p. 14, print p. 37, 065135-13, verification: verified_in_frozen_source-only_card; authority_pdf_not_reopened. Evidence-bound proposition: the numerically reported maximum drag coefficients show overshoot relative to quasi-steady values that grows substantially with the cross-section ratio; these values serve only as numerical-assumption leads.

Li et al. (2025) (numerical-assumption line, B/D=not reported) retains this label at every occurrence. Its overshoot values cannot serve as physical validation, design coefficients, or facility input. Physical transient evidence for B/D=3 remains sparse, and numerical sources cannot fill this gap. The planned three 60 mm cross-sections with 300 Hz synchronized PIV plus pressure represent PI experimental requirements only and do not constitute a facility-executability conclusion.

Table 23: Planned three-section program evidence and gap map. Comparison ceiling: planning comparison only. Missing entries are uniformly recorded as not reported.

planned section	B over D	source evidence	required measurements	gap	status
60 x 60 mm	1	largest controlled square-section pool; high-completeness transient group and steady pressure/force/PIV anchors	surface pressure plus 300 Hz PIV or equivalent; synchronized trigger; independent velocity reference	matched accelerating-flow pressure and PIV under one protocol remains incomplete	PI planned condition; not facility-ready
60 x 180 mm	3	thin physical pool, supplemented by numerical mechanism context without substitution	surface pressure plus 300 Hz PIV; reattachment tracking; synchronization and repeatability	B/D=3 transient physical evidence is sparse and cannot be filled by CFD	PI planned condition; not facility-ready
60 x 300 mm	5	broad steady and sinusoidal pool with distinct channel and normalization definitions	surface pressure plus 300 Hz PIV; pressure/PIV synchronization; force provenance declared	matched monotonic acceleration pressure and PIV under identical reference definitions remains incomplete	PI planned condition; not facility-ready

15 Concluding remarks

The 55 sources reviewed here show that the transient aerodynamic response cannot be characterized by a single acceleration parameter or a single cross-section dimension. Multiple physical experiments on square cylinders report a decrease in C_D or St during the acceleration phase, with magnitudes of approximately 5% to 10%; changes in lift fluctuations are generally larger. Results for the 5:1 rectangular cylinder are governed by the separation-reattachment process: Morello et al. (2025a) reported that St rose from approximately 0.12 to 0.16, Zhou et al. (2025) reported a 13.1% increase in C_D at the strongest acceleration together with a pronounced increase in drag fluctuations, and streamwise and vertical sinusoidal inflows respectively altered the reattachment length, pressure fluctuations, and POD energy distribution. Physical transient evidence for $B/D = 3$ remains scarce, and numerical studies cannot substitute for this gap. The overshoot values of Li et al. (2025) continue to be classified as mechanism hypotheses only.

The methods available from the existing literature can be applied directly to the next-phase condition definition, synchronization scheme, and comparison baseline, but they are not yet sufficient to determine the final test matrix or facility operating conditions. The minimum data-quality checks should include separate archiving of measured $U(t)$ and target $U(t)$, co-triggering of pressure and PIV, dynamic calibration of pressure tubing, verification of velocity-pressure time offsets, repeat-convergence for each test condition, an independent steady-state baseline, channel-by-channel recording of pressure-integrated forces and independent balance forces, and non-overwritable archiving of raw data and processing scripts. The completion criterion is the acquisition of repeatable measured $U(t)$, pressure, PIV, and uncertainty records for all three cross-sections at selected angles of attack and inflow families; stopping criteria include synchronization drift, calibration failure, sensor saturation, model motion, facility limitations, or data-archiving anomalies. Once these measurements are completed, PI will need to decide which cross-source comparisons may enter scientific conclusions and which are to be retained solely as methodological or mechanistic context.

A Direct-quotation and page-number index

Table 24: Direct quotations, page numbers, and evidence-bound propositions in the main text

Quote ID	Source	Exact quotation	PDF p.	Governing claim
DQ-001	Brusco et al. (2022)	the black dots in Fig. 9a and b estimate the instantaneous frequency to be locked on ~ 20 Hz (τ < 13.8 s), before abruptly swapping to ~ 25.7 Hz (τ > 13.8 s)	10	An abrupt lock-on frequency transition from approximately 20 Hz to approximately 25.7 Hz was observed within a single repeat of UF1.
DQ-002	Brusco et al. (2022)	its maximum drop, around 8%, is observed in correspondence of the sixth reference wind speed U6, when up-crossed, for an acceleration < a>W = 7.11 m/s^2	14	The ensemble-averaged Strouhal number exhibits a maximum drop of approximately 8% at up-crossing of U6.
DQ-003	Morello et al. (2025a)	The cells do not have uniform duration in time; their temporal length tends to decrease as the rate of velocity variation increases.	10	The duration of constant-frequency time cells is non-uniform and tends to shorten as the rate of velocity variation increases.
DQ-004	Morello et al. (2025a)	Almost all unsteady cases exhibit a slight downstream shift in the position where pressure fluctuations reach their maximum, and this result indicates a downstream movement of the flow reattachment point along the lateral surface.	23	In most unsteady cases for the 5:1 rectangular cylinder, the peak pressure-fluctuation location shifts slightly downstream, which the authors interpret as a downstream migration of the reattachment point.
DQ-005	Brusco et al. (2024)	the ensemble mean of the values gathered from the 90 repeats under the ramp-up condition is 187.7 Pa, with a standard deviation of 0.60 Pa, while the corresponding steady reference pressure difference is 204.3 Pa	14	The ensemble-mean pressure difference during the acceleration phase of UF1 at U8 is lower than the corresponding steady-state reference pressure difference.
DQ-006	Brusco et al. (2024)	the maximum increase of the time-varying mean pressure coefficient <c>P29,W (14% in the ramp-up, and 7% in the ramp-down) is almost double than the corresponding reductions of the aerodynamic drag	18	The maximum increase in the time-varying mean pressure coefficient at C29 is 14% in the acceleration phase and 7% in the deceleration phase, approximately double the corresponding drag reduction.
DQ-007	Wu et al. (2021)	The ratio can reach about 90% when Delta u/U is higher than 0.15.	9	When the streamwise sinusoidal fluctuation amplitude exceeds 0.15, the inflow-induced drag component can account for approximately 90% of the total.
DQ-008	Wu et al. (2021)	the three components of 2D AAF do not decrease monotonously with the frequency, which is different from Davenport's, Sears' or Horlock's functions	12	The three components of the two-dimensional aerodynamic admittance function do not decrease monotonically with frequency, differing from the three classical functions.
DQ-009	Yang and Mason (2019)	Coefficient values, in both flow cases, are reduced by up to approximately 10%, 30%, 25% and 5% for <CD>, <CL'>, <absCLhat> and <St>, respectively.	12	In both accelerating-flow cases for B/D=1, the drag, lift fluctuations, peak lift, and Strouhal number decrease by different magnitudes relative to their steady-state values.
DQ-010	Yang and Mason (2019)	Coupled with the drop in lift coefficients at the initiation of flow acceleration is an overshoot of up to 10% in <CD>.	13	For B/D=3, the drop in lift coefficients at the onset of flow acceleration is accompanied by a drag-coefficient overshoot of up to 10%.
DQ-011	M. Li et al. (2023)	This is reduced by approximately 15.2% and 23.5% when sigma_u/U = 20.8% and sigma_w/U = 21.4%, respectively.	11	Under high-amplitude streamwise and vertical sinusoidal gusts the mean drag is reduced by approximately 15.2% and 23.5%, respectively.
DQ-012	M. Li et al. (2023)	The value of C_L tilde is reduced from approximately 17.0% to 19.0% when sigma_u/U increases from 5.3% to 20.8%.	12	As the streamwise gust amplitude increases from 5.3% to 20.8%, the fluctuating lift coefficient decreases by approximately 17.0% to 19.0%.
DQ-013	Da Matha Sant'Anna et al. (1988)	The cylinders spanned the complete wind tunnel width.	1	The test cylinders spanned the full wind-tunnel width; the end condition corresponds to a wall-to-wall installation.

Quote ID	Source	Exact quotation	PDF p.	Governing claim
DQ-014	Robertson et al. (1978)	the reattachment locale occurs somewhat down- stream of the peak in r.m.s, pressure fluctuations and upstream of the mean pressure peak.	7	The reattachment location lies downstream of the r.m.s. pressure-fluctuation peak and upstream of the mean pressure peak.
DQ-015	Vickery (1966)	The root-mean-square coefficient of fluctuating drag was 0.17 in both smooth and turbulent flow	7	The root-mean-square fluctuating drag coefficient was 0.17 in both smooth and turbulent flow.
DQ-016	Yu et al. (2013)	For the flow around prisms with aspect ratios of 3.0 and greater, the separated flow reattaches to the faces toward the trailing edge.	2	The numerical results indicate that for aspect ratios of 3.0 and above, the separated flow reattaches on the side faces near the trailing edge.
DQ-017	Nakamura et al. (1991)	The Strouhal number which is based on the plate's chord is approximately constant and equal to 0.6 for chord-to-thickness ratios 3-5. With further increase in the ratio it increases stepwise to values that are approximately equal to integral multiples of 0.6.	1	The chord-normalized Strouhal number is approximately 0.6 for chord-to-thickness ratios of 3 to 5 and increases stepwise with further increases in the ratio.
DQ-018	Zhang and Liu (2017)	the separated shear layer reattaches onto the plate's surface near x/D = 4.7	5	The separated shear layer of the L/D=6 plate reattaches near x/D = 4.7.
DQ-019	Ohya et al. (1992)	the ratio of these values is nearly 1:2:3	6	The chord-normalized vortex shedding frequencies for d/h=3, 6, and 9 are approximately in the ratio 1:2:3.
DQ-020	Han et al. (2025)	The aerodynamic coefficient time history is derived through pres- sure integration.	7	The aerodynamic coefficient time histories in this source were obtained from pressure integration, not from independent balance time histories.
DQ-021	Guissart et al. (2022)	the reattachment point illustrated by the yellow square marker located at x/D = 1.64	4	At zero angle of attack and Re=2×10^4, the PIV-identified reattachment point is located at x/D=1.64.
DQ-022	Li et al. (2021a)	The pressure recovery starts around 2.0D in smooth flow, while the starting points are much closer to the separation point, i.e., around 0.5D in SSOFs.	6	In streamwise sinusoidal oscillatory flow the pressure-recovery onset is approximately 0.5D, substantially earlier than the approximately 2.0D observed in smooth flow.
DQ-023	Zhou et al. (2025)	CD is increased to 1.169, about 13.0% higher than that for the stationary flow	12	The mean drag coefficient in the acceleration phase with af=4.30 m/s2 increased to 1.169, approximately 13.0% above the stationary-flow value.
DQ-024	Chiarini and Quadrio (2021)	However, the low Reynolds number and the amount of integration time preclude any firm statement.	9	The low Reynolds number and limited integration time prevented the authors from drawing a definitive conclusion on low-frequency unsteadiness.
DQ-025	Lunghi et al. (2024)	the St number decreases within the time cells in the range $0.10 \leq \mathrm{St} \leq 0.14$, whereas the expected Strouhal number for constant inflow conditions and this Reynolds number range is $\mathrm{St} \simeq 0.12$	5	In the numerically simulated accelerating-flow cases for the square cylinder, St varies between 0.10 and 0.14 within the time cells; the steady-state expected value is approximately 0.12.
DQ-026	Morello et al. (2025b)	from $\mathrm{St} \simeq 0.066$ to $\mathrm{St} \simeq 0.178$ for the 5:1 cylinder and $\mathrm{St} \simeq 0.145$ to $\mathrm{St} \simeq 0.195$ for the 3:1 cylinder	20	Across all acceleration intensities, the instantaneous St ranges differ between the 5:1 and 3:1 cylinders.
DQ-027	Li et al. (2025) (numerical-assumption line, B/D=not reported)	For R3, R4, and R5, the maximum drag coefficients exceed the quasi-steady values by 22.4%, 32.6%, and 46.1%, respectively, while for R8, R10, R12, and R16, the increases are 74.0%, 82.0%, 94.3%, and 109.7%, respectively.	14	The numerically reported maximum drag coefficients show overshoot relative to quasi-steady values that grows substantially with the cross-section ratio; these values serve only as numerical-assumption leads.

References

Luca Bruno, Davide Fransos, Nicolas Coste, Arianna Bosco. 2010. 3D flow around a rectangular cylinder: A computational study. *Journal of Wind Engineering and Industrial Aerodynamics 98(6-7), 263-276* https://doi.org/10.1016/j.jweia.2009.10.005.

Luca Bruno, Nicolas Coste, Davide Fransos. 2012. Simulated flow around a rectangular 5:1 cylinder: Spanwise discretisation effects and emerging

flow features. *Journal of Wind Engineering and Industrial Aerodynamics 104-106 (2012) 203-215* https://doi.org/10.1016/j.jweia.2012.03.018.

Luca Bruno, Maria Vittoria Salvetti, Francesco Ricciardelli. 2014. Benchmark on the Aerodynamics of a Rectangular 5:1 Cylinder: An overview after the first four years of activity. *Journal of Wind Engineering and Industrial Aerodynamics 126, 87-106* https://doi.org/10.1016/j.jweia.2014.01.005.

Stefano Brusco, Guido Buresti, Yuan-Lung Lo, Giuseppe Piccardo. 2022. Constant-frequency time cells in the vortex-shedding from a square cylinder in accelerating flows. *Journal of Wind Engineering & Industrial Aerodynamics, 230 (2022) 105182* https://doi.org/10.1016/j.jweia.2022.105182.

Stefano Brusco, Hao-Yu Bin, Yuan-Lung Lo, Giuseppe Piccardo. 2024. Transient aerodynamics of a square cylinder under downburst-like accelerating flows reproduced in a multiple-fan wind tunnel. *Journal of Fluids and Structures, 124 (2024) 104038* https://doi.org/10.1016/j.jfluidstructs.2023.104038.

Alessandro Chiarini, Maurizio Quadrio. 2021. The Turbulent Flow over the BARC Rectangular Cylinder: A DNS Study. *Flow, Turbulence and Combustion (2021) 107:875-899* https://doi.org/10.1007/s10494-021-00254-1.

F. A. Da Matha Sant'Anna, A. Laneville, J. Y. Trépanier, Z. Y. Lü. 1988. Detailed pressure field measurements for some 2-D rectangular cylinders. *Journal of Wind Engineering and Industrial Aerodynamics 28(1-3), 241-250* https://doi.org/10.1016/0167-6105(88)90120-1.

Amandine Guissart, Erik Elbæk, Jeanette Hussong. 2022. PIV study of the flow around a 5:1 rectangular cylinder at moderate Reynolds numbers and small incidence angles. *Wind and Structures 34(1), 15-27* https://doi.org/10.12989/was.2022.34.1.015.

Fangjuan Guo, Gefei Wu, Xiaoqing Du, Matthew S. Mason. 2021. Numerical investigation of flow around a square cylinder in accelerated flow. *Physics of Fluids 33(10), 104105* https://doi.org/10.1063/5.0062282.

Yuan Han, Yifei Sun, Qingkuan Liu, Hongmiao Jing, Hanjie Huang, Zhen Li, Yangxue Wang. 2025. Aerodynamic characteristics of a 5:1 rectangular cylinder at Reynolds numbers of 3.12 × 105 to 1.01 × 106. *Physics of Fluids 37, 045121 (2025)* https://doi.org/10.1063/5.0256739.

J. P. Huot, C. Rey, H. Arbey. 1986. Experimental analysis of the pressure field induced on a square cylinder by a turbulent flow. *J. Fluid Mech. (1986), vol. 162, pp. 283-298* https://doi.org/10.1017/S0022112086002057.

Charles W. Knisely. 1990. Strouhal numbers of rectangular cylinders at incidence: A review and new data. *Journal of Fluids and Structures 4(4), 371-393* https://doi.org/10.1016/0889-9746(90)90137-T.

D. C. Lander, C. W. Letchford. 2016. Influence of the bluff body shear layers on the wake of a square prism in a turbulent flow. *Physical Review Fluids 1, 044406*. https://doi.org/10.1103/PhysRevFluids.1.044406.

Thai-Hoa Le, Masaru Matsumoto, Hiromichi Shirato. 2009. Spanwise coherent structure of wind turbulence and induced pressure on rectangular cylinders. *Wind and Structures 12(5), 441-455* https://doi.org/10.12989/was.2009.12.5.441.

Ming Li, Qiusheng Li, Haoyun Shi. 2021a. Aerodynamic pressures on a 5:1 rectangular cylinder in sinusoidal streamwise oscillatory flows with non-zero mean velocities. *Journal of Wind Engineering & Industrial Aerodynamics 208, 104440*. https://doi.org/10.1016/j.jweia.2020.104440.

Ming Li, Qiusheng Li, Haoyun Shi. 2021b. Effect of sinusoidal vertical gust on the pressure distributions on and flow structures around a rectangular cylinder. *Experiments in Fluids 62, 148*. https://doi.org/10.1007/s00348-021-03192-w.

Xin Li, Shaopeng Li, Yi Su, Liuliu Peng, Shuyang Cao, Min Liu. 2022. Study on the time-varying extreme value characteristic of the transient loads on a 5:1 rectangular cylinder subjected to a thunderstorm-like wind. *Journal of Wind Engineering and Industrial Aerodynamics 229, 105161* https://doi.org/10.1016/j.jweia.2022.105161.

Ming Li, Qiu-Sheng Li, Hao-Yun Shi, Ming-Shui Li. 2023. Large-scale sinusoidal gust effect on aerodynamic pressures and forces on a square cylinder. *Journal of Fluids and Structures 120, 103917* https://doi.org/10.1016/j.jfluidstructs.2023.103917.

Shaopeng Li, Xin Li, Jingyang Li, Yi Su, Bo Wu. 2023. Measurements of the unsteady lift force on a 5:1 rectangular cylinder based on active-passive hybrid wind tunnel tests. *Measurement, 213 (2023) 112674* https://doi.org/10.1016/j.measurement.2023.112674.

Yuerong Li, Lei Yan, Huanxiang Gao, Gang Hu. 2024. A machine learning-augmented aerodynamic database of rectangular cylinders. *Physics of Fluids 36, 075127 (2024)* https://doi.org/10.1063/5.0211387.

Shaopeng Li, Liuliu Peng, Qingshan Yang, Xin Li, Jinxin Cao, Shuyang Cao, Yan Jiang. 2024. Transient surface pressure of a rectangular cylinder subjected to downburst-like winds. *Physics of Fluids 36(8), 085196* https://doi.org/10.1063/5.0222124.

Weilin Li, Yunkai Yao, Lin Zhao, Cong Chen, Luca Patruno. 2025. Aerodynamic overshoot of rectangular cylinders under accelerating inflow conditions. *Physics of Fluids 37, 065135 (2025)* https://doi.org/10.1063/5.0272163.

G. Lunghi□S. Brusco□A. Mariotti□G. Piccardo□M.V. Salvetti. 2024. Influence of inflow acceleration on the aerodynamic characteristics of a square cylinder. *Journal of Wind Engineering & Industrial Aerodynamics 252 (2024) 105814* https://doi.org/10.1016/j.jweia.2024.105814.

Claudio Mannini, Ante Šoda, Günter Schewe. 2011. Numerical investigation on the three-dimensional unsteady flow past a 5:1 rectangular cylinder. *Journal of Wind Engineering and Industrial Aerodynamics 99, 469–482*. https://doi.org/10.1016/j.jweia.2010.12.016.

Claudio Mannini, Antonino Maria Marra, Luca Pigolotti, Gianni Bartoli. 2017. The effects of free-stream turbulence and angle of attack on the aerodynamics of a cylinder with rectangular 5:1 cross section. *Journal of Wind Engineering and Industrial Aerodynamics 161, 42-58* https://doi.org/10.1016/j.jweia.2016.12.001.

A. Mariotti, G. Lunghi, M.V. Salvetti. 2024. Experimental investigations on the effect of upstream-edge rounding on the BARC configuration. *Journal*

of Wind Engineering & Industrial Aerodynamics, 251 (2024) 105787 https://doi.org/10.1016/j.jweia.2024.105787.

Masaru Matsumoto, Hiromichi Shirato, Kenji Araki, Takashi Haramura, Tomoaki Hashimoto. 2003. Spanwise coherence characteristics of surface pressure field on 2-D bluff bodies. *Journal of Wind Engineering and Industrial Aerodynamics 91(1-2), 155-163* https://doi.org/10.1016/S0167-6105(02)00342-2.

D. M. Moore, C. W. Letchford, M. Amitay. 2019. Energetic scales in a bluff body shear layer. *Journal of Fluid Mechanics 875, 543-575* https://doi.org/10.1017/jfm.2019.480.

M. Morello, H.-Y. Bin, G. Piccardo, M. V. Salvetti, A. Mariotti. 2025a. Experiments on the aerodynamic behavior of square and rectangular cylinders under accelerating inflow conditions. *Acta Mechanica*. https://doi.org/10.1007/s00707-025-04600-6.

Mario Morello, Gianmarco Lunghi, Alessandro Mariotti, Maria Vittoria Salvetti. 2025b. Influence of time-varying freestream velocity on the flow characteristics of elongated rectangular cylinders. *Flow, Turbulence and Combustion 115(4), 1585-1611*. https://doi.org/10.1007/s10494-025-00678-z.

Yasuharu Nakamura, Yuji Ohya, Hideki Tsuruta. 1991. Experiments on vortex shedding from flat plates with square leading and trailing edges. *Journal of Fluid Mechanics 222, 437-447* https://doi.org/10.1017/S0022112091001167.

Yuji Ohya, Yasuharu Nakamura, Shigehira Ozono, Hideki Tsuruta, Ryuzo Nakayama. 1992. A numerical study of vortex shedding from flat plates with square leading and trailing edges. *J. Fluid Mech. (1992), vol. 236, pp. 445-460* https://doi.org/10.1017/S0022112092001472.

Shinichi Oka, Takeshi Ishihara. 2009. Numerical study of aerodynamic characteristics of a square prism in a uniform flow. *Journal of Wind Engineering and Industrial Aerodynamics 97, 548-559*. https://doi.org/10.1016/j.jweia.2009.08.006.

Atsushi Okajima. 1983. Flow Around a Rectangular Cylinder with a Section of Various Width/Height Ratios. *Wind Engineers, JAWE 1983(17), 1-19* https://doi.org/10.5359/jawe.1983.17_1.

L. Patruno, M. Ricci, S. de Miranda, F. Ubertini. 2016. Numerical simulation of a 5:1 rectangular cylinder at non-null angles of attack. *J. Wind Eng. Ind. Aerodyn. 151 (2016) 146-157* https://doi.org/10.1016/j.jweia.2016.01.008.

M. Ricci, L. Patruno, S. de Miranda, F. Ubertini. 2017. Flow field around a 5:1 rectangular cylinder using LES: Influence of inflow turbulence conditions, spanwise domain size and their interaction. *Computers and Fluids 149 (2017) 181-193* https://doi.org/10.1016/j.compfluid.2017.03.010.

J. M. Robertson, J. B. Wedding, J. A. Peterka, J. E. Cermak. 1978. Wall pressures of separation and reattachment flow on a square prism in uniform flow. *Journal of Wind Engineering and Industrial Aerodynamics 2(4), 345-359*. https://doi.org/10.1016/0167-6105(78)90019-3.

Günter Schewe. 2013. Reynolds-number-effects in flow around a rectangular cylinder with aspect ratio 1:5. *Journal of Fluids and Structures 39, 15-26* https://doi.org/10.1016/j.jfluidstructs.2013.02.013.

K. SHIMADA, T. Ishihara. 2002. APPLICATION OF A Modified k–ε MODEL TO THE PREDICTION OF AERODYNAMIC CHARACTERISTICS OF RECTANGULAR CROSS-SECTION CYLINDERS. *Journal of Fluids and Structures (2002) 16(4), 465-485* https://doi.org/10.1006/jfls.2001.0433.

A. Sohankar. 2008. Large eddy simulation of flow past rectangular-section cylinders: side ratio effects. *Journal of Wind Engineering and Industrial Aerodynamics 96 (2008) 640-655* https://doi.org/10.1016/j.jweia.2008.02.009.

Jia Tang, Yang Yang, Mingshui Li. 2023. Experimental investigation on two-dimensional aerodynamic admittances of rectangular cylinders in various turbulent flows. *Physics of Fluids 35, 085117 (2023)* https://doi.org/10.1063/5.0160527.

B. W. van Oudheusden, F. Scarano, N. P. van Hinsberg, D. W. Watt. 2005. Phase-resolved characterization of vortex shedding in the near wake of a square-section cylinder at incidence. *Experiments in Fluids (2005) 39: 86-98* https://doi.org/10.1007/s00348-005-0985-5.

B.W. van Oudheusden, F. Scarano, N.P. van Hinsberg, E.W.M. Roosenboom. 2008. Quantitative visualization of the flow around a square-section cylinder at incidence. *Journal of Wind Engineering and Industrial Aerodynamics 96, 913–922*. https://doi.org/10.1016/j.jweia.2007.06.030.

B. J. Vickery. 1966. Fluctuating lift and drag on a long cylinder of square cross-section in a smooth and in a turbulent stream. *J. Fluid Mech. (1966), vol. 25, part 3, pp. 481-494* https://doi.org/10.1017/S002211206600020X.

Bo Wu, Shaopeng Li, Ke Li, Liangliang Zhang. 2020. Numerical and experimental studies on the aerodynamics of a 5:1 rectangular cylinder at angles of attack. *Journal of Wind Engineering & Industrial Aerodynamics, 199 (2020) 104097* https://doi.org/10.1016/j.jweia.2020.104097.

Bo Wu, Shaopeng Li, Liangliang Zhang, Ke Li. 2021. Experimental determination of the two-dimensional aerodynamic admittances of a 5:1 rectangular cylinder in streamwise sinusoidal flows. *Journal of Wind Engineering & Industrial Aerodynamics 210 (2021) 104525* https://doi.org/10.1016/j.jweia.2021.104525.

Jian Wu, Yakun Liu, Di Zhang, Ze Cao, Zijun Guo. 2022. Numerical Investigation of Vortex Shedding from a 5:1 Rectangular Cylinder at Different Angles of Attack. *Journal of Marine Science and Engineering 10(12), 1913* https://doi.org/10.3390/jmse10121913.

Ting Yang, Matthew S. Mason. 2019. Aerodynamic characteristics of rectangular cylinders in steady and accelerating wind flow. *Journal of Fluids and Structures 90, 246-262*. https://doi.org/10.1016/j.jfluidstructs.2019.07.004.

Xiongwei Yang, Yang Yang, Mingshui Li, Peiyuan Wang. 2021. Effects of free-stream turbulence on non-Gaussian characteristics of fluctuating wind pressures on a 5:1 rectangular cylinder. *Journal of Wind Engineering & Industrial Aerodynamics, 217 (2021) 104759* https://doi.org/10.1016/j.jweia.2021.104759.

Dahai Yu, Kyle Butler, Ahsan Kareem, James Glimm, Jiangang Sun. 2013. Simulation of the influence of aspect ratio on the aerodynamics of

rectangular prisms. *Journal of Engineering Mechanics 139(4), 429-438*. https://doi.org/10.1061/(ASCE)EM.1943-7889.0000494.

Qingshan Zhang, Yingzheng Liu. 2017. Separated flow over blunt plates with different chord-to-thickness ratios: Unsteady behaviors and wall-pressure fluctuations. *Experimental Thermal and Fluid Science 84, 199-216*. https://doi.org/10.1016/j.expthermflusci.2017.02.007.

Zhanbiao Zhang, Fuyou Xu. 2020. Spanwise length and mesh resolution effects on simulated flow around a 5:1 rectangular cylinder. *Journal of Wind Engineering & Industrial Aerodynamics, 202 (2020) 104186* https://doi.org/10.1016/j.jweia.2020.104186.

Yuxin Zhang, Shuyang Cao, Jinxin Cao, Jin Wang. 2023. Effects of turbulence intensity and integral length scale on the surface pressure on a rectangular 5:1 cylinder. *Journal of Wind Engineering & Industrial Aerodynamics 236 (2023) 105406* https://doi.org/10.1016/j.jweia.2023.105406.

Qiang Zhou, Shuyang Cao, Md Mahbub Alam, Bing Han. 2025. Aerodynamics of a 5:1 rectangular cylinder under transient accelerated and decelerated winds. *Journal of Wind Engineering and Industrial Aerodynamics 258, 106023* https://doi.org/10.1016/j.jweia.2025.106023.

Guowang Zhu, Shenghong Huang, Q.S. Li. 2020. Large-eddy simulation of the inflow turbulence transport and aerodynamics of a rectangular 5:1 cylinder with high-order numerical methods. *Journal of Wind Engineering & Industrial Aerodynamics 207 (2020) 104366* https://doi.org/10.1016/j.jweia.2020.104366.